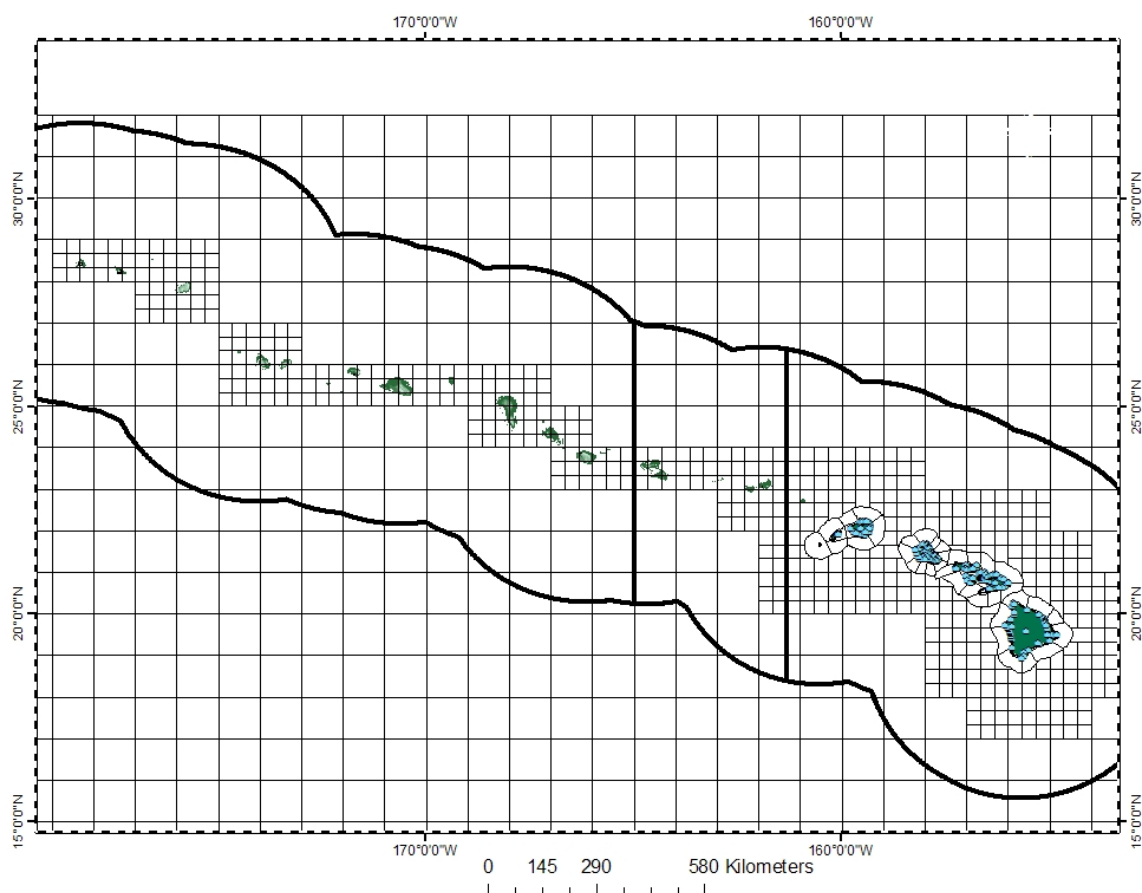


Standardization of the Hawaiian Fisherman Reporting System Database for Bottomfish: 1948-2005

Jerald S. Ault and Steven G. Smith

University of Miami
Rosenstiel School of Marine and Atmospheric Science
4600 Rickenbacker Causeway
Miami, FL 33149



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1.0 Introduction to Analysis of the Hawaiian Archipelago Bottomfish Database

Beginning in 2004, a series of stock assessment workshops were convened to evaluate the status of bottomfish and coral reef fish monitoring and assessment methodologies, and to develop a comprehensive plan to improve statistical design and evaluation methodologies for data collection to a point where reliable resource assessments could be obtained (WPRFMC 2004a, 2004b). At the workshops a number of key accomplishments were achieved, including:

- (1) current sources and state of statistical analyses of available data and design of monitoring programs were reviewed;
- (2) weaknesses in current data time-series and assessment methods were delineated;
- (3) alternative statistical assessment and modeling methodologies were reviewed and identified; and
- (4) proposed courses of action to advance the reliability of stock assessment methods and associated data collection and statistical analysis procedures were outlined.

While there was consistency among many of the recommendations stemming from the workshops, the critical review, synthesis, and comprehensive statistical analysis of existing data was identified as the highest priority by workshop participants.

In April 2005, an Ecosystem Science and Management Planning Workshop was held in Honolulu and re-emphasized the urgent need to develop robust statistical procedures for the collation, integration and synthesis of the available long term time-series data on the Hawaiian Archipelago bottomfish fishery, and further, to develop statistical models that could be readily used by resource managers to build sustainable fisheries. It was further recognized that this work would require significant expertise in the integration, statistical analysis and management of disparate fisheries data sets, with a view to subsequent implementation of formal stock assessment procedures and application of ecosystem approaches to fisheries management (EAF).

Goals and Objectives of Research Study

This research study forms the basis of a cooperative and collaborative program by the University of Miami to support the Pacific Islands Fisheries Science Center (PIFSC) bottomfish stock assessment efforts and to provide critical analyses for bottomfish resource data inventory, assimilation, and development of statistical sampling designs for fishery-dependent monitoring, ultimately supporting stock assessment efforts by NOAA in the US Pacific Islands.

The principal objective of this study was to use advanced statistical methods and procedures to analyze the Hawaii Fisherman Reporting System database for the period of 1948 to 2005 to create an improved time-series of standardized fishery catch-and-effort data by gear-type for the Hawaiian Archipelago fishery. To achieve that goal, the particular objectives were in three principal areas: (1) data inventory; (2) data assimilation; and, (3) statistical analysis and sampling design.

(1) Data Inventory. - Conduct an inventory of the existing fishery-dependent bottomfish data for consistency, completeness, accuracy, and precision, including spatial and temporal resolution. These basic analyses will identify outliers or errors, completeness over the expected

temporal and spatial domains, gaps and estimation properties (*e.g.*, precision) of primary and derived variables.

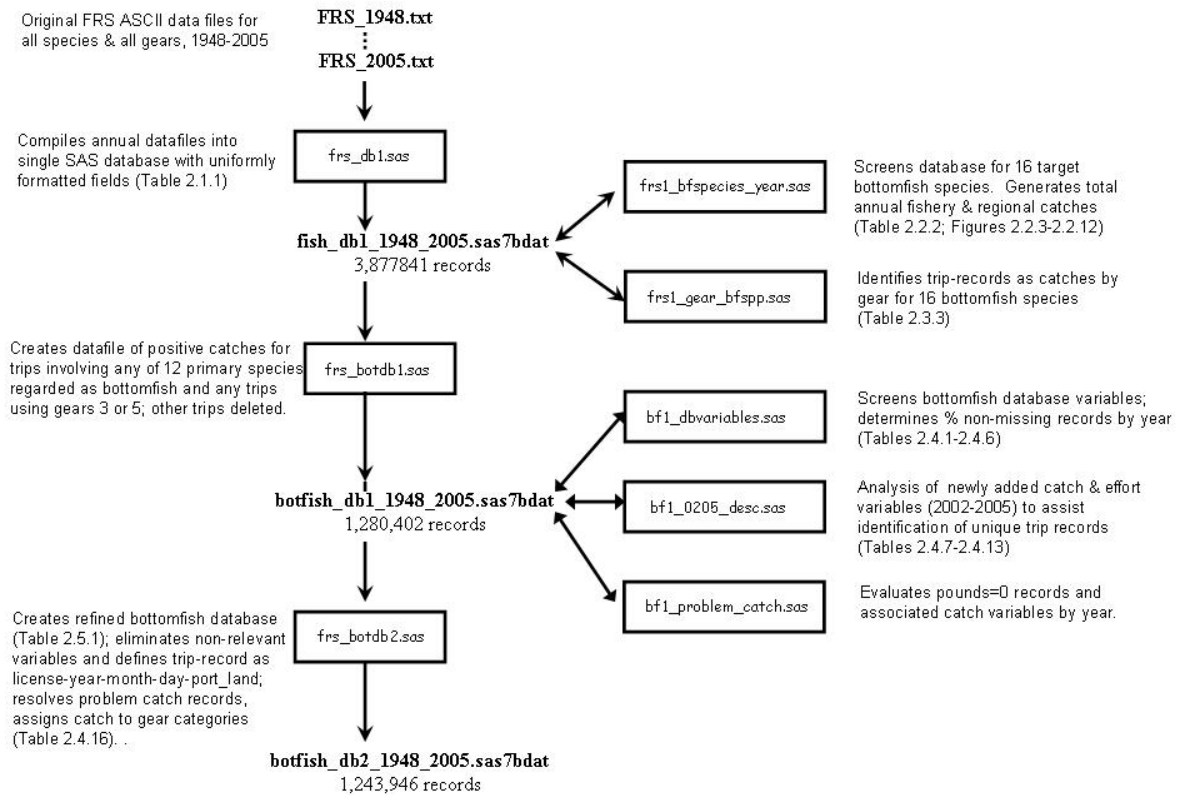
(2) Data Assimilation. - Conduct a statistical standardization of the existing data by spatial and temporal references and, if feasible, adjustment for collection biases (such as those arising from use of different sampling gears or methods) to maximize the capacity to link data sets to each other and have confidence in inferences drawn from them. These analyses will include identification of shortcomings in parameter estimation needed for formal assessment models and EAF (*e.g.*, gaps in time-series), and exploration of options for compensating for such shortcomings by use of alternative or ‘quasi’ estimates (*e.g.*, filling gaps in time-series by interpolation).

(3) Statistical Sampling Design .- Finally, evaluate the current sampling and monitoring program for the Fishermen Reporting System database and conduct a formal evaluation of these data. These analyses were conducted to support potential future applications that may include methods to identify cost-effective ways to improve the accuracy and precision for future sampling and monitoring designs such that those sampling regimes are responsive to the needs of decision makers (*e.g.*, track changes in abundance).

This report is broken into four principal sections. The first introduces the issues and discusses potential paths to solution. The second outlines the various statistical processing steps necessary to provide an overview of the compilation of a full analyses-ready database from the range of datafiles available for the Hawaiian Archipelago for the period 1948 to 2005. These analyses provided a first-order look at the historical catch records by species, gears and areas fished. This process allowed identification of gaps in the data for key classes of variables, principal gears involved in the bottomfish fishery, and what specifically represents a fishing trip and the days-fished on any particular trip for the various fleet components. Ultimately, these analyses led to a “refined” database that winnowed the dataset to focus strictly on those species and gears involved in exploitation of the Hawaii bottomfish complex. The third major section of the report involved analyses of the 2002-2005 component of the database, where essentially complete records were kept, to estimate what constituted a unit of nominal fishing effort. A unique application of a distance metric was applied using GIS to evaluate trip length from the relationship between area-fished and port-of-landing. These findings were then applied to the historical database where the duration of any given trip was poorly (if at all) recorded. Additionally, there were a number of trip records that lacked key information on what constituted a trip (*i.e.*, date, area-fished, port-of-landing, and license). From the results of the previous analyses, these data were “recovered” using algorithms that developed statistical relationships between the key variables. From these analyses, an analyses-ready database was constructed to allow for the determination of year and gear specific catch-and-effort timeseries. In the fourth major section of the report the spatio-temporal distributions of fleet effort for the entire history of the database were determined and mapped in GIS to demonstrate the substantive spatial effects present when viewing the space-compressed catch-and-effort timeseries data. Finally, a summary of recommendations for follow-on study and analysis is provided.

2.0 Processing Steps for Initial Descriptive Analysis

Figure 2.0.1.- Key steps in the initial statistical processing of the FRB database for the Hawaiian Archipelago bottomfish fishery, 1948-2005.



2.1 Compilation of a Full Database

Table 2.1.1.- Variables used in the Hawaii FRS database, 1948-2005, broken into 7 categories.

#	Variable	Type	Len	Format	Var_Category	Notes
48	species	Num	8		Catch	n/r=0 or unknown=999
50	lbs	Num	8		Catch	
49	caught	Num	8		Catch	
52	lbs_sold	Num	8	9.2	Catch	
51	num_sold	Num	8	8.2	Catch	
53	value	Num	8	10.2	Catch	
25	nocatch	Char	1		Catch	logical flag, F or T
28	num_stolen	Num	8		Catch	
55	lost	Num	8		Catch	
56	release	Num	8		Catch	
57	damage	Num	8		Catch	
64	avg_price	Num	8	7.4	Catch	missing=-9
3	year	Num	8		Date	
4	month	Num	8		Date	
5	day	Num	8		Date	
31	r_month	Num	8		Date	n/r=0
32	r_year	Num	8		Date	n/r=0
2	trip_end	Num	8		Date	missing=-9
47	trip_beg	Num	8		Date	missing=-9
34	timelink	Num	8		Date	missing=-9
18	depth_beg	Num	8		Environmental	
19	depth_end	Num	8		Environmental	
20	wind_speed	Num	8		Environmental	
21	wind_dir	Char	2		Environmental	missing=-9
22	wave_ht	Num	8		Environmental	
23	cur_speed	Char	1		Environmental	missing=Z
24	cur_dir	Char	2		Environmental	missing=-9
9	gear	Num	8		Gear	n/r=0
10	sets	Num	8		Gear	not reported or n/a=0
11	hours	Num	8		Gear	not reported or n/a=0
12	no_gears	Num	8		Gear	not reported or n/a=0
13	hooks	Num	8		Gear	not reported or n/a=0
14	net_length	Num	8		Gear	not reported or n/a=0
15	net_height	Num	8		Gear	not reported or n/a=0

Table 2.1.1.- (Continued)

#	Variable	Type	Len	Format	Var_Category	Notes
7	area	Num	8		Location	not reported=0
8	subarea	Char	1		Location	missing=Z
6	bouy	Char	2		Location	missing=-9
16	bank	Char	18		Location	missing=-9
17	bank_quad	Char	2		Location	missing=-9
26	port_land	Num	8		Location	n/r=0
45	port_land2	Num	8		Location	n/r=0
46	port_dprr	Num	8		Location	n/r=0
65	from_file	Char	2		Record Flag	missing=-9 or unknown=UN
35	f_area	Char	1		Record Flag	
36	f_buoy	Char	1		Record Flag	
38	f_ar_gr	Char	1		Record Flag	
39	f_by_pr	Char	1		Record Flag	
58	f_spec_gr	Char	1		Record Flag	
59	f_spec_ar	Char	1		Record Flag	
60	f_season	Char	1		Record Flag	
61	f_pieces	Char	1		Record Flag	
62	f_lbs	Char	1		Record Flag	
63	f_ctchlim	Char	1		Record Flag	
1	license	Num	8		Trip	missing=-9
40	lname	Char	24		Trip	missing=-9
41	fname	Char	18		Trip	missing=-9
42	vessel	Char	20		Trip	missing=-9
43	uscg	Char	7		Trip	missing=-9
44	ha_no	Char	8		Trip	missing=-9
27	logbook	Num	8		Trip	n/r=0
33	formtype	Char	8		Trip	missing=-9
30	chartered	Char	1		Trip	

2.2 Bottomfish Species Landings, 1948-2005

Table 2.2.1.- List of Hawaiian bottomfish management unit species used in the analysis. BF is a species of the bottomfish complex, Deep7 is one of the primary bottomfish species. See Appendix C for a complete list of species (from WPRFMC 2001 Annual Report).

#	Species	Species name	BF	Deep7	Family	Scientific name
1	15	HAPUUPUU (Hawaiian grouper)	1	1	Serranidae	Epinephelus quernus
2	17	KALEKALE (Pink snapper)	1	1	Lutjanidae	Pristipomoides sieboldii
3	19	OPAKAPAKA (Pink snapper)	1	1	Lutjanidae	Pristipomoides filamentosus
4	21	EHU (Red snapper)	1	1	Lutjanidae	Etelis marshi or carbunculus (KL)
5	22	ONAGA (Red snapper)	1	1	Lutjanidae	Etelis coruscans
6	58	LEHI (Red snapper/silvermouth)	1	1	Lutjanidae	Aphareus rutilans
7	97	GINDAI (snapper)	1	1	Lutjanidae	Pristipomoides zonatus
8	16	KAHALA (Greater anberjack)	1		Carangidae	Seriola dumerili
9	20	UKU (Green Jobfish)	1		Lutjanidae	Aprion virescens
10	114	TAAPE (Bluestripe snapper)	1		Lutjanidae	Lutjanus kasmira
11	140	ARMORHEAD	1		Pentacerotidae	Pseudopentaceros pectoralis
12	141	ALFONSIN	1		Berycidae	Beryx decadactylus
13	200	BUTAGUCHI (Thicklip trevally)	1		Carangidae	Caranx cheilio
14	202	GUNKAN	1		Carangidae	Caranx lugubris
15	205	WHITE PAPIO/ULUA	1		Carangidae	Caranx ignobilis
16	208	YELLOW-TAIL KALI	1		Lutjanidae	Pristipomoides auricilla

Figure 2.2.1- Map of the fishing regions in the Hawaiian Archipelago overlain on the Hawaii Division of Aquatic Resources DLNR commercial fisheries statistical chart. Region 5 is loosely defined as offshore areas in the main Hawaiian Islands.

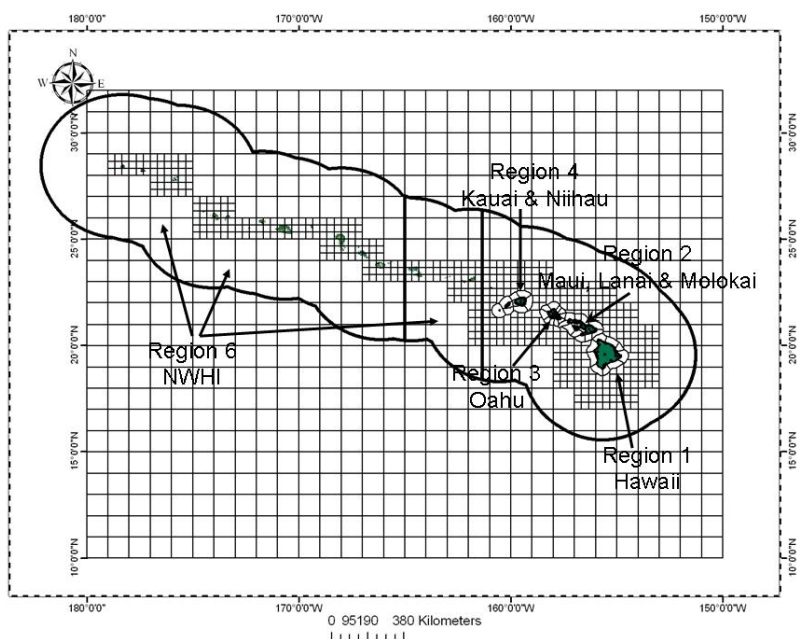


Figure 2.2.2.- Distribution map of bottomfish fishing effort from 1948-2005 in the Hawaiian Archipelago. Green squares show map cells that have received at least some fishing effort during the extent of the time-series.

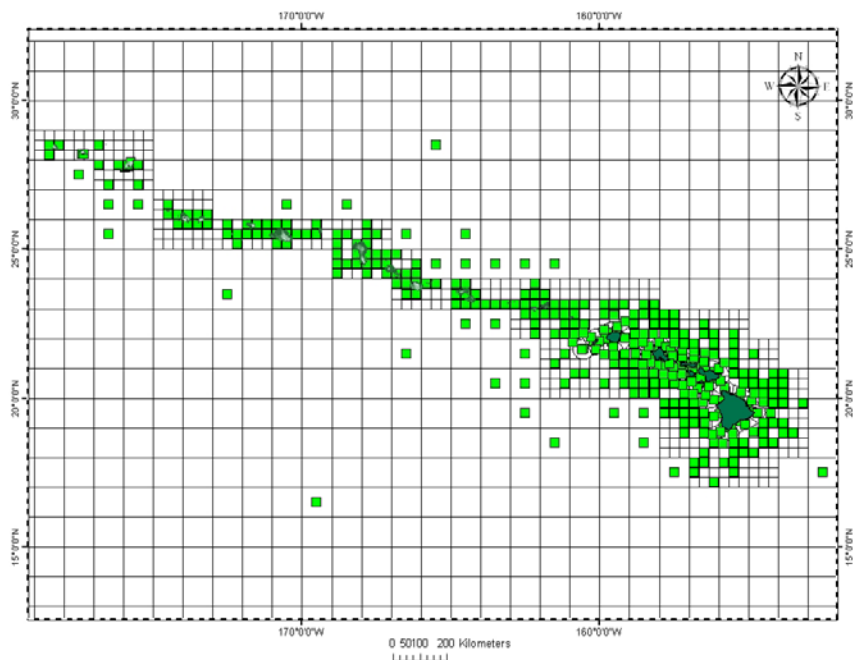
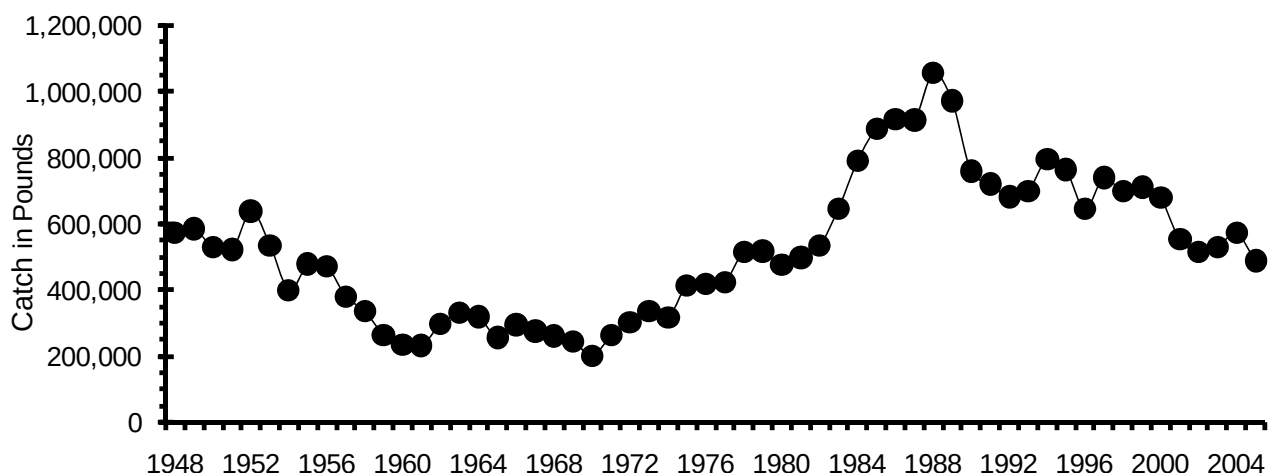


Figure 2.2.3.- Catch in pounds from 1948-2005 for the “fine 9” species (deep7 plus uku and taape): (A) combined fine9 species of the Hawaiian bottomfish community; and, (B) fine9 catches for the five fishing regions of the Hawaiian archipelago.

(A)



(B)

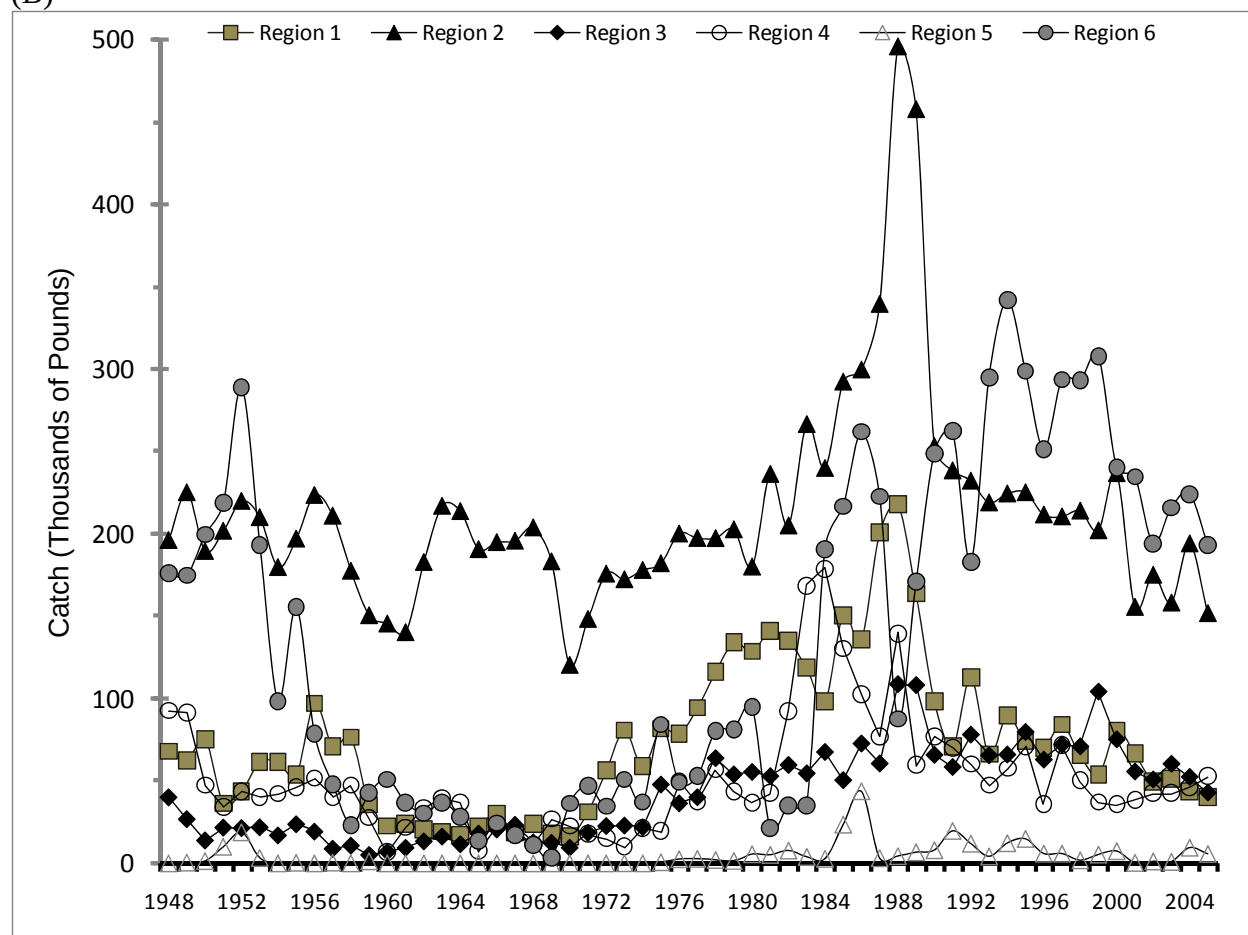


Figure 2.2.4.- Hapuupuu catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

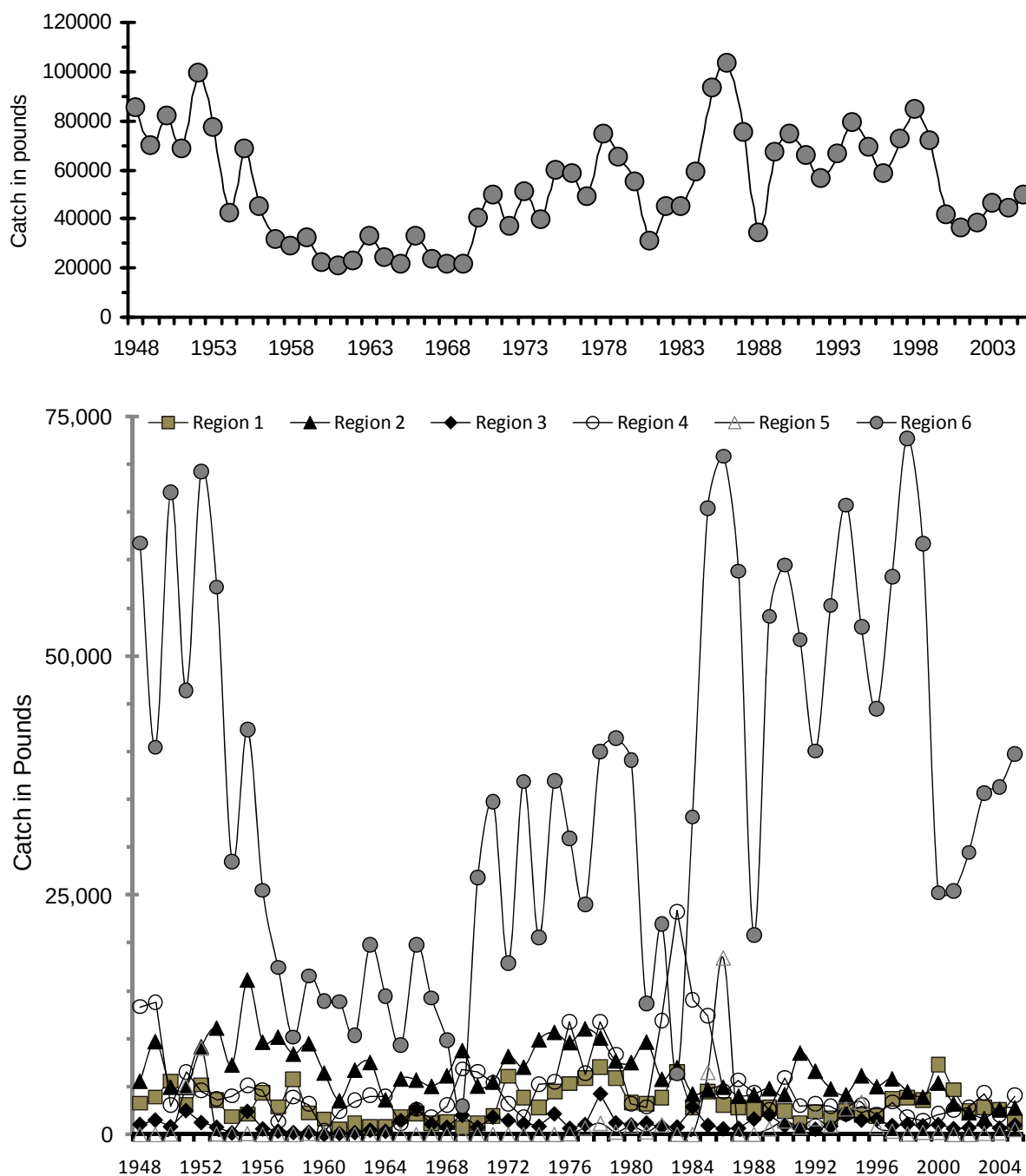


Figure 2.2.5.- Kalekale catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

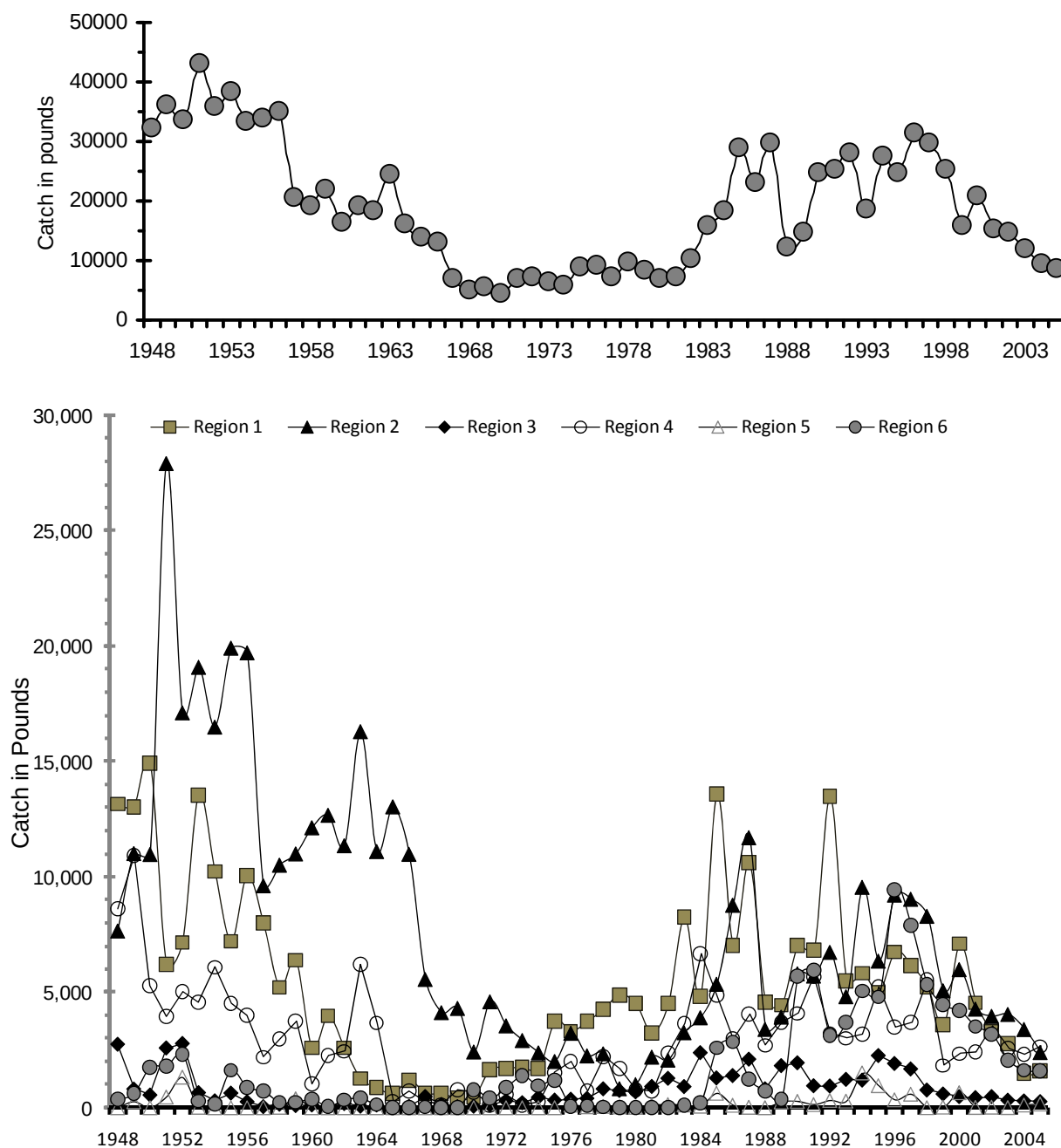


Figure 2.2.6.- Opakapaka catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

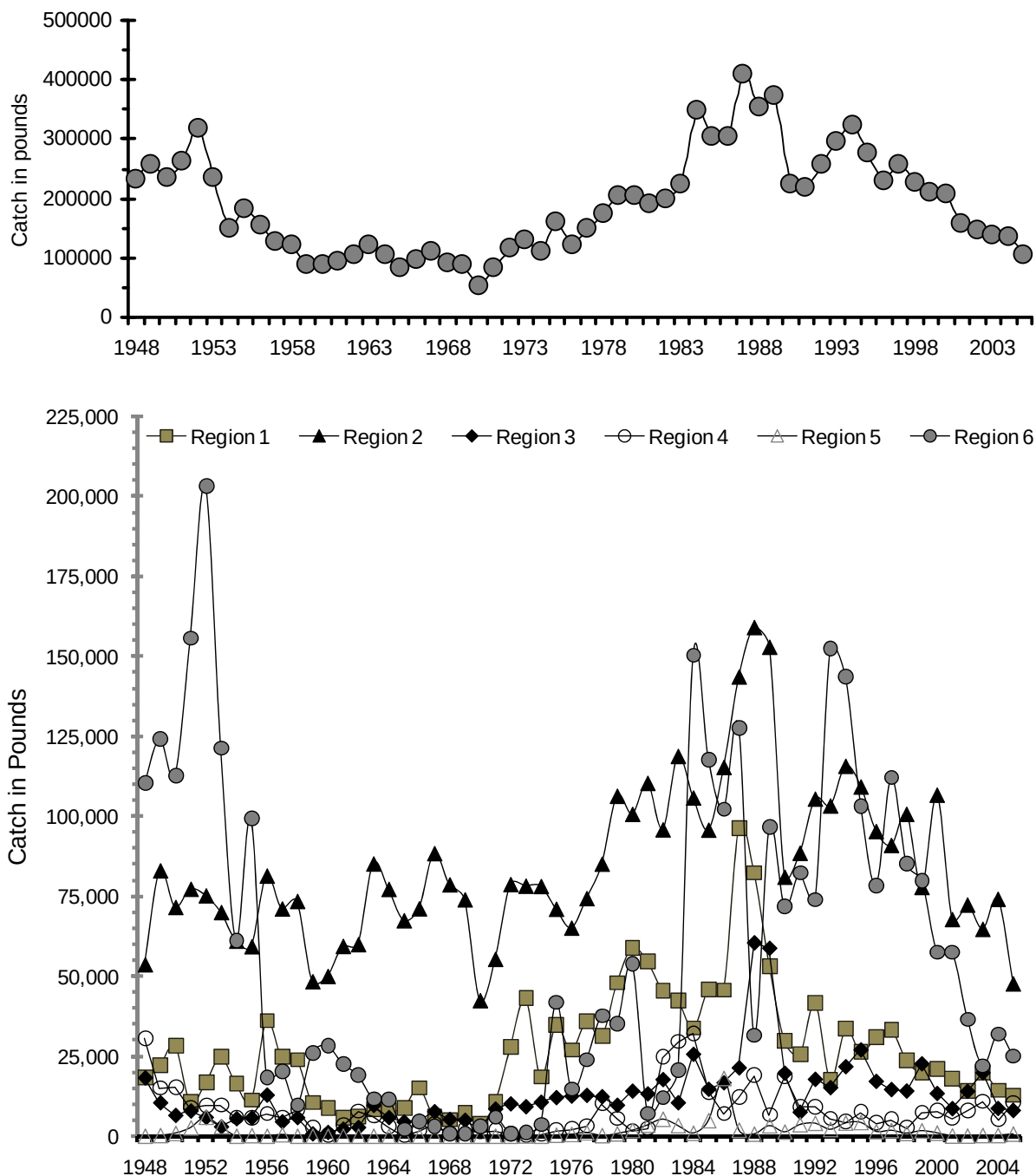


Figure 2.2.7.- Uku catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

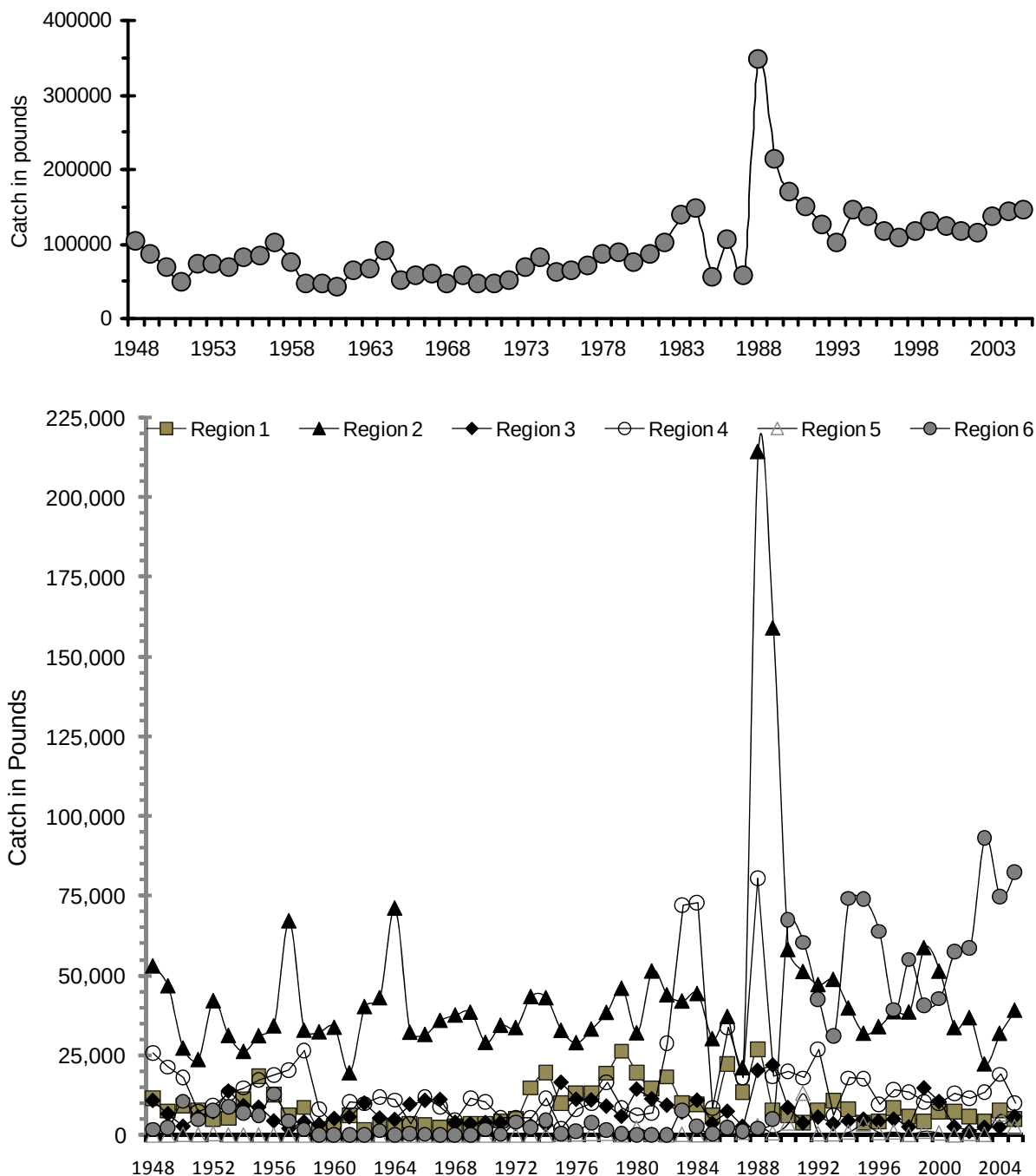


Figure 2.2.8.- Ehu catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

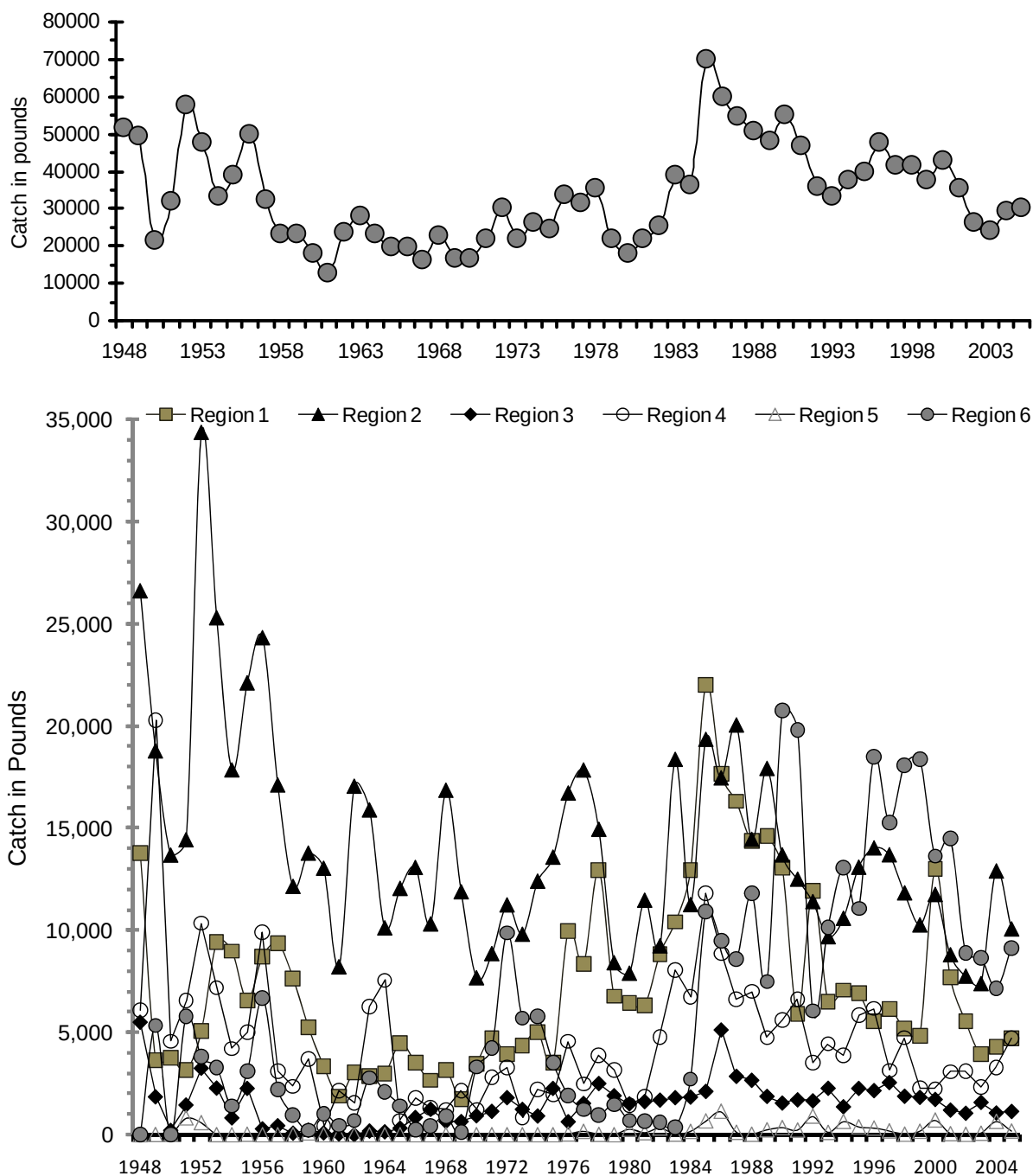


Figure 2.2.9.- Onaga catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

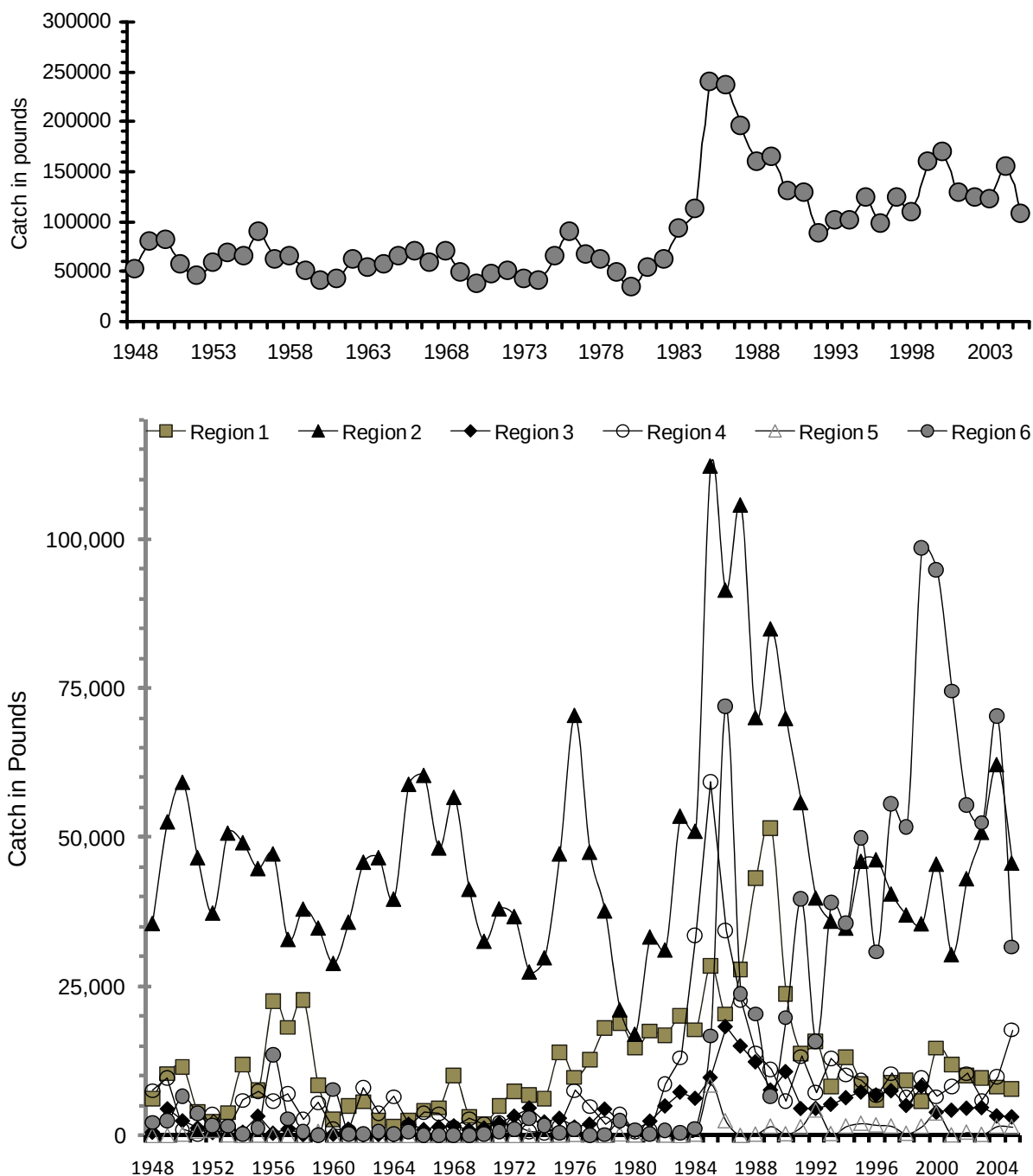


Figure 2.2.10.- Lehi catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

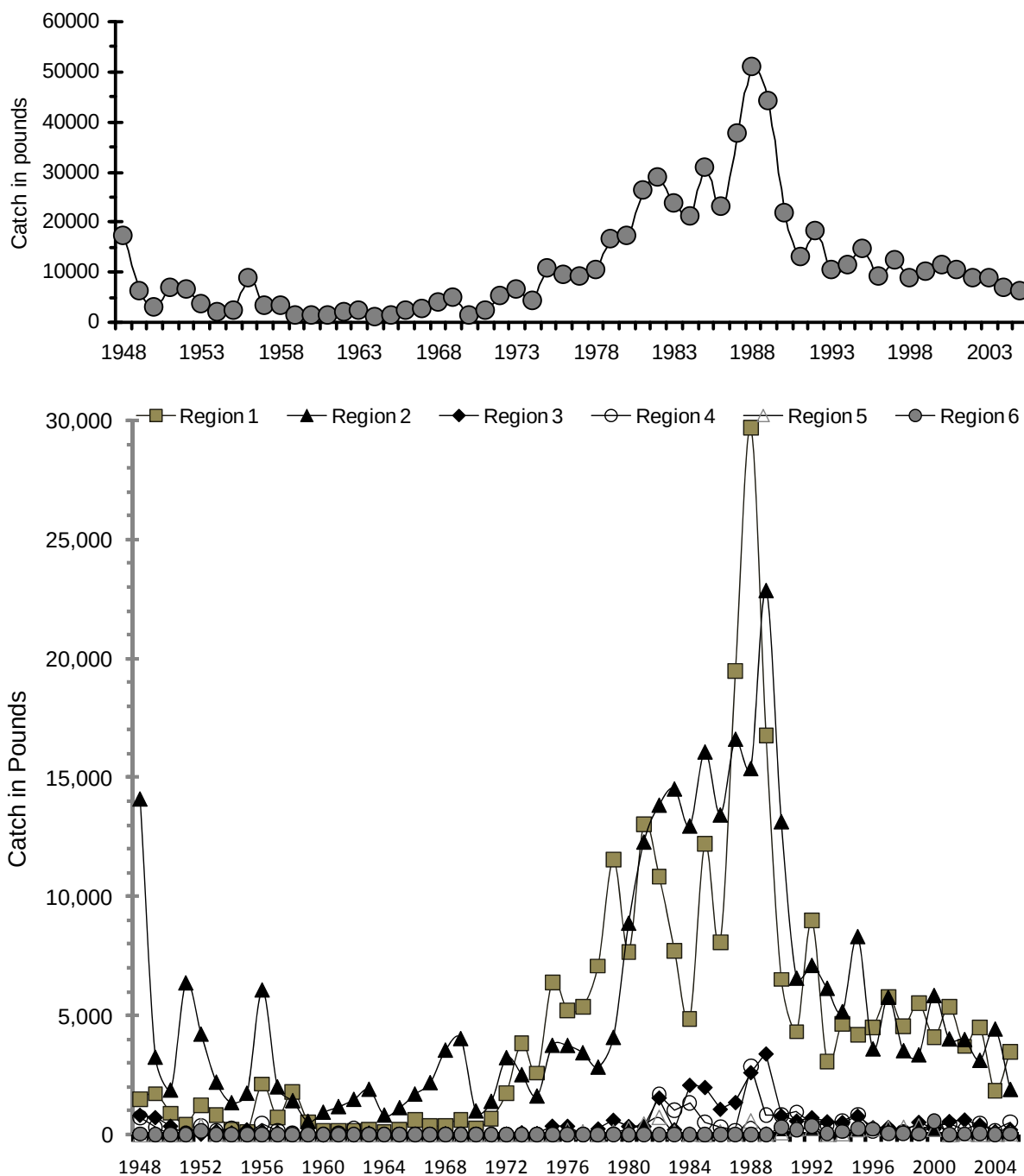


Figure 2.2.11.- Gindai catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.

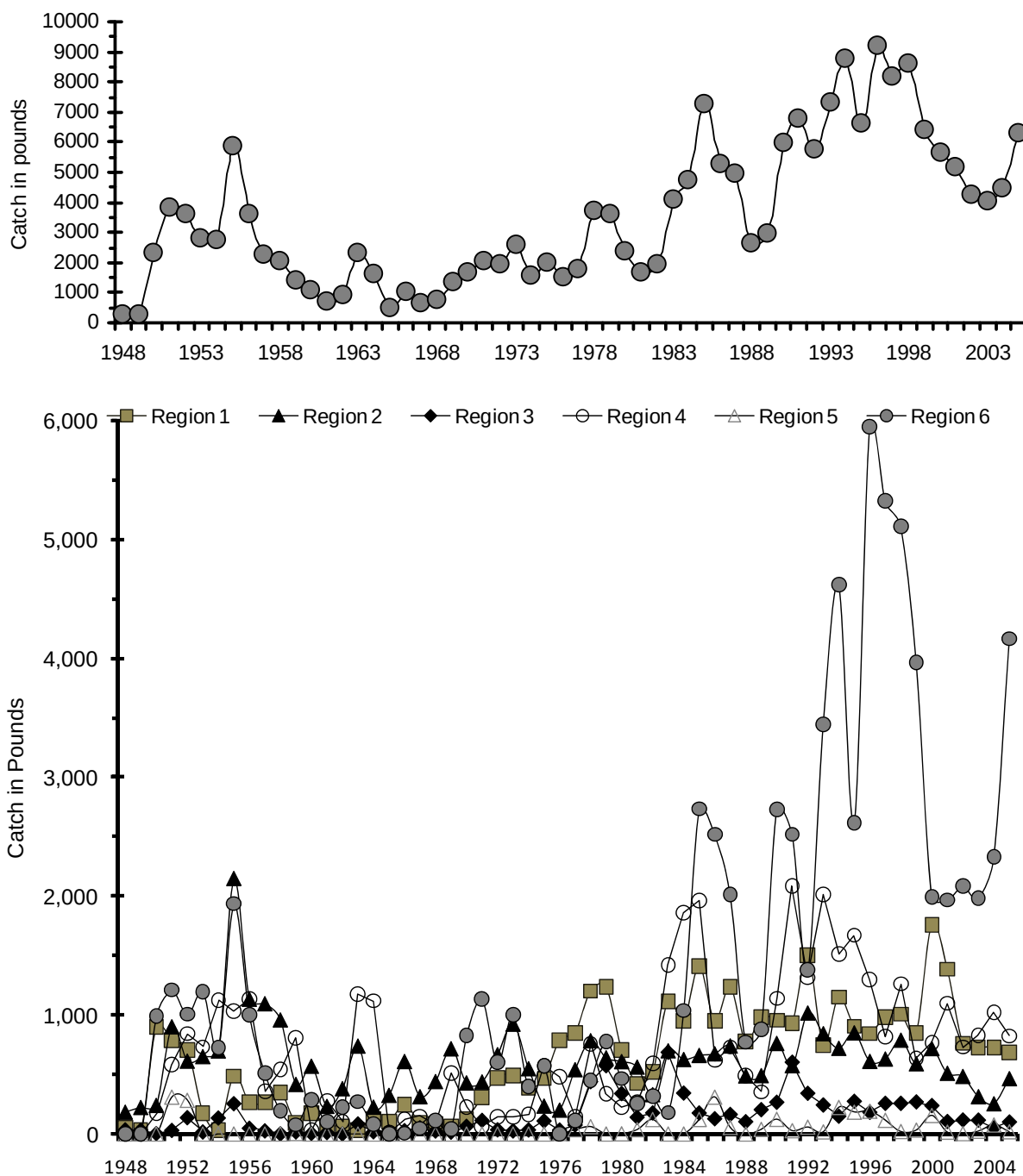
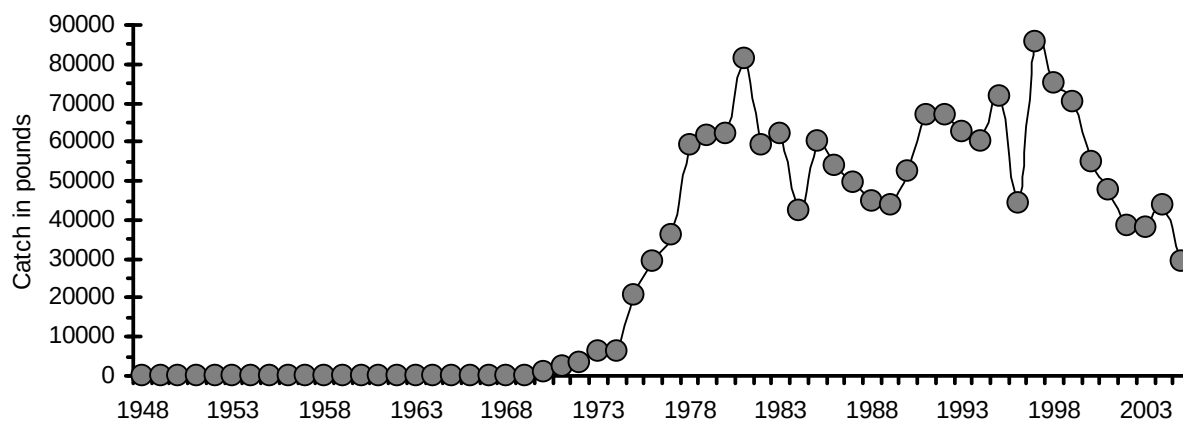
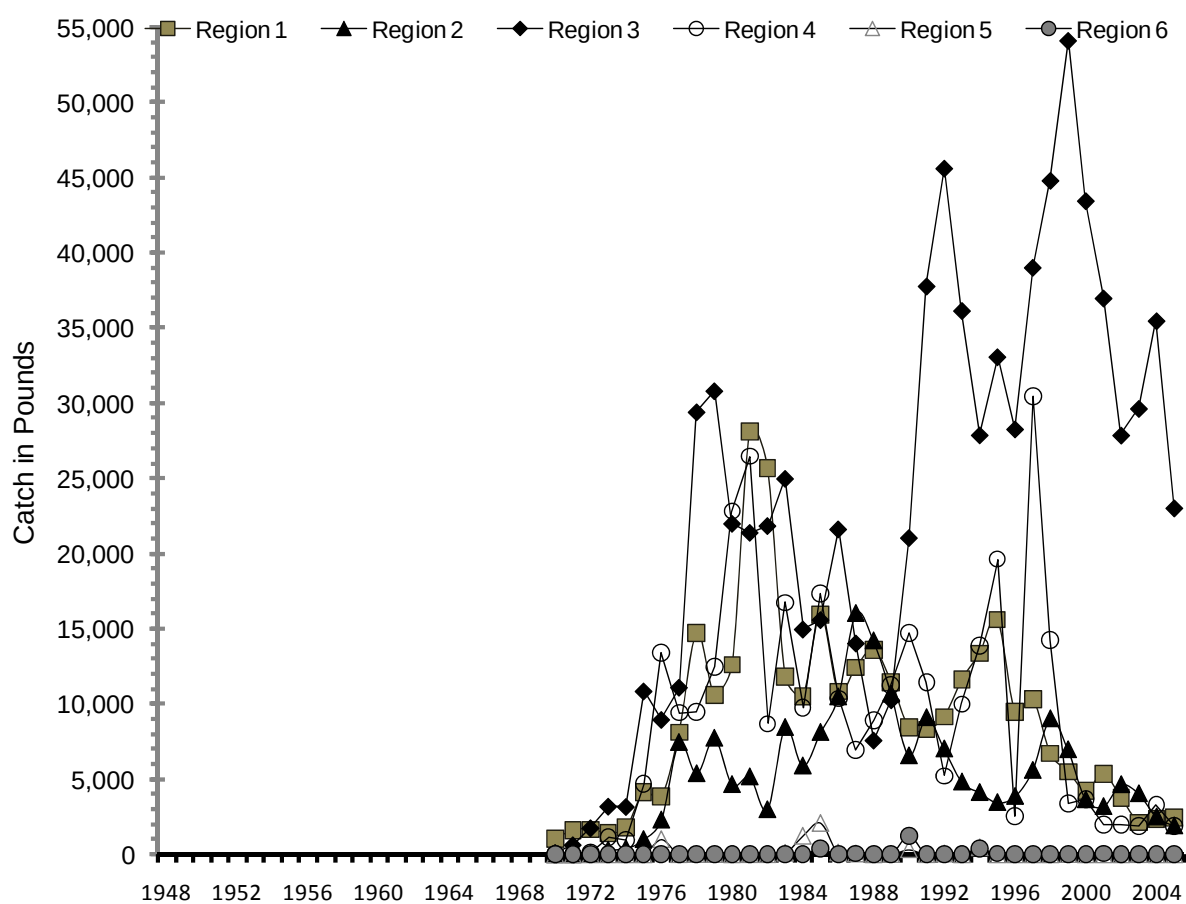


Figure 2.2.12.- Taape catches (in pounds) for the years 1948-2005: (A) Hawaiian archipelago; (B) five fishing regions.



5



2.3 Initial Bottomfish-Specific Database

Table 2.3.1- List of commercial marine fishing gears used to capture bottomfish species in the Hawaiian archipelago found in the Fisherman Reporting System database.

Gear	Description
0	Unknown, not reported
1	AKUBOAT, POLE & LINE
2	LOGLINE, AHI BOAT, FLAGLINE
3	DEEP-SEA HANDLINE, BOTTOM HANDLINE
4	INSHORE HANDLINE, COWRIE SHELL (TAKO)
5	KAKA LINE, SET LINE
6	TROLLING (MISC.)
7	Shark Liver
8	IKA-SHIBI
9	PALU AHI, DROP STONE, MAKE DOG
10	CASTING, LIGHT TACKLE, SPINNER, WHIPPING
11	TRAP (MISC.)
12	Unknown
13	SPEARFISHING, DIVE, SQUIDING (TAKO)
14	DIVING, DIVE FOR LOBSTER OR NAMAKO, ETC.
15	BLACK CORAL DIVE
20	NET (MISC.)
21	LIFT NET, OPELU
22	GILL NET, FENCE NET, LAY NET, CROSS NET
23	SEINE NET
24	BULLPEN TRAP
25	LOBSTER NET
26	CRAB NET
27	THROW NET
28	Bag net (discontinued 2002)
29	Turtle net
30	BAIT NET
31	Surround net (discontinued 2002)
32	SHRIMP TRAWL NET
33	PURSE SEINE NET (PELAGIC)
35	Tuna handline (misc.), SURFACE PELAGIC HANDLINE
40	KONA CRAB NET, LOOPS
45	AQUARIUM COLLECTING NET
51	CRAB TRAP (CAN BE DEEP SEA)
52	FISH TRAP
53	LOBSTER TRAP, LOBSTER POT
54	SHRIMP TRAP
60	HANDPICKED (LIMU, OPIHI, WANA, AAMA CRAB, NAMAKO, INA)
61	TROLLING - LURES
62	TROLLING - BAIT
63	TROLLING - STICK
70	ALBACORE TROLLING
90	FISH POND
91	FLOATLINE
92	SHORTLINE
93	VERTICAL LOGLINE
97	HYBRID
98	SUBMERSIBLE, PRECIOUS CORAL
99	OTHER

Table 2.3.2.- Total trip-records in the FRS database, 1948-2005: by gear for all trip records; positive catch trips where at least 1 bottomfish species (of the 16 species in the complex) was caught; and, positive catch trips where at least 1 deep7 species was caught. Gears 3 and 5 appear to be directed bottomfish gears.

Gear	Gear description	Total	Positive Catch for Bottomfish (16 spp)		Positive Catch for Deep 7	
		Trip Records	Trips	% of Total Trips	Trips	% of Total Trips
3	DEEP-SEA HANDLINE, BOTTOM HANDLINE	279688	200754	71.78	161259	57.66
4	INSHORE HANDLINE, COWRIE SHELL (TAKO)	348220	43724	12.56	8737	2.51
11	TRAP (MISC.)	43169	6648	15.4	1328	3.08
6	TROLLING (MISC.)	620394	4275	0.69	854	0.14
22	GILL NET, FENCE NET, LAY NET, CROSS NET	46299	2505	5.41	164	0.35
10	CASTING, LIGHT TACKLE, SPINNER	8595	2491	28.98	15	0.17
13	SPEARFISHING, DIVE, SQUIDING (TAKO)	39307	2399	6.1	227	0.58
52	FISH TRAP	3866	1795	46.43	69	1.78
20	NET (MISC.)	24265	1596	6.58	91	0.38
9	PALU AHI, DROP STONE, MAKE DOG	23251	1178	5.07	537	2.31
31		3224	857	26.58	16	0.5
5	KAKA LINE, SET LINE	1804	569	31.54	400	22.17
62	TROLLING - BAIT	7854	472	6.01	4	0.05
99	OTHER	17642	390	2.21	64	0.36
61	TROLLING - LURES	77763	346	0.44	8	0.01
23	SEINE NET	3880	343	8.84	11	0.28
24	BULLPEN TRAP	1797	321	17.86	4	0.22
35	SURFACE PELAGIC HANDLINE	39841	260	0.65	74	0.19
21	LIFT NET, OPELU	34228	258	0.75	28	0.08
0	NOT RECORDED	1421	250	17.59	197	13.86
2	LONGLINE, AHI BOAT, FLAGLINE	51468	206	0.4	64	0.12
8	IKA-SHIBI	53035	146	0.28	25	0.05
27	THROW NET	10128	55	0.54	25	0.25
28		6769	51	0.75	3	0.04
14	DIVING, DIVE FOR LOBSTER OR NAMAOKO	4374	47	1.07	8	0.18
90	FISH POND	2330	41	1.76	1	0.04
51	CRAB TRAP (CAN BE DEEP SEA)	2484	38	1.53	0	0
33	PURSE SEINE NET (PELAGIC)	119	26	21.85	0	0
1	AKUBOAT, POLE & LINE	70997	17	0.02	12	0.02
92	SHORTLINE	147	5	3.4	0	0
93	VERTICAL LONGLINE	230	4	1.74	1	0.43
7		6	3	50	2	33.33
12		14602	2	0.01	1	0.01
25	LOBSTER NET	3644	2	0.05	2	0.05
30	BAIT NET	58	2	3.45	0	0
53	LOBSTER TRAP, LOBSTER POT	7221	2	0.03	1	0.01
40	KONA CRAB NET, LOOPS	9940	1	0.01	0	0
54	SHRIMP TRAP	1409	1	0.07	0	0
60	HANDPICKED	18643	1	0.01	0	0
97	HYBRID	476	1	0.21	0	0
98	SUBMERSIBLE, PRECIOUS CORAL	47	1	2.13	0	0

Table 2.3.3.- Ranked total catches (pounds) for the period 1948-2005 by individual gear-types for the 16 species in the Hawaiian bottomfish complex. Highlighted gears represent those thought to be “directed” at bottomfish. Gears 3,4 & 5 represent 99.4% of the catches of the Deep7 species.

Species # Name	DEEP7															208 YT Kali	BF16 TOTALS
	15 Hapuupuu	17 Kalekale	19 Opakapaka	21 Ehu	22 Onaga	58 Lehi	97 Gindai	20 Uku**	114 Taape**	16 Kahala	200 Butaguchi	202 Gunkan	205 Ulua	140 Armorhead	141 Alfonsin		
GEAR																	
3	3,022,977	1,035,220	10,740,473	1,883,363	5,188,272	626,526	198,370	4,843,645	168,234	3,278,516	1,068,031	15,058	239,436	72	1,268	196	32,309,655
4	22,934	51,413	183,080	63,540	31,948	10,806	5,189	420,587	435,144	253,810	835	766	19,123	.	.	.	1,499,174
22	723	953	678	242	3,083	136	87	1,894	304,736	834	40	.	15,574	.	.	.	328,980
6	5,808	1,910	28,469	4,931	9,243	1,398	361	156,034	2,067	42,765	569	114	39,671	.	.	.	293,340
20	511	201	7,445	369	532	31	92	4,923	228,909	3,754	.	.	14,670	.	.	.	261,437
31	463	75	.	.	.	.	11	207	246,724	1,055	17	.	5,469	.	.	.	254,021
11	2,744	1,937	17,252	692	407	87	85	28,970	64,696	50,852	56	.	2,985	.	.	.	170,763
52	711	60	874	31	126	16	4	4,018	90,001	1,061	432	29	12,394	.	.	.	109,757
23	.	286	6	46	.	12	.	4,815	58,085	460	7	.	2,045	.	.	.	65,762
5	2,084	2,090	12,032	2,812	1,212	972	280	22,256	639	3,282	23	1,171	9,995	.	.	.	58,846
13	1,670	394	115	419	416	67	409	12,226	18,850	4,727	120	36	5,309	.	.	.	44,758
24	.	2	.	6	.	.	537	302	27,844	45	.	.	12,306	.	.	.	41,042
10	15	106	9	22	.	.	7	11,116	21,862	1,292	60	28	4,431	.	.	.	38,947
9	305	107	2,491	193	51	5,783	58	9,729	361	12,932	166	554	2,797	.	.	.	35,525
62	35	.	.	1	11	12	.	18,884	14	535	45	52	3,765	.	.	.	23,353
1	.	119	592	40	5,567	.	.	14,344	.	296	.	.	44	.	.	.	21,002
0	2,954	2,838	3,915	1,505	565	321	469	1,601	.	693	1,785	615	878	.	1	.	18,140
35	43	109	728	95	221	448	.	10,345	295	2,182	.	.	2,427	.	.	.	16,893
21	.	17	152	3,599	79	.	.	5,502	1	4,660	.	.	.	.	.	.	14,010
99	419	254	1,561	102	198	115	2	2,537	2,624	4,517	.	.	1,261	.	.	.	13,590
2	601	809	2,097	215	104	227	25	3,436	.	5,009	.	.	870	.	.	.	13,393
61	18	22	230	26	8	20	.	3,256	15	1,336	20	418	5,276	.	.	.	10,644
33	.	.	.	.	.	.	.	.	10,394	0	.	.	.	.	.	.	10,394
14	40	.	.	.	.	9	.	22	6,121	559	.	.	.	.	.	.	6,751
28	.	.	165	.	.	.	.	581	1,348	2,008	.	.	.	.	.	.	4,102
8	36	12	141	49	5	257	18	1,157	72	777	150	40	680	.	.	.	3,393
90	.	.	.	22	.	.	.	.	2,317	.	.	.	.	.	.	.	2,339
27	230	124	4	88	37	.	3	209	862	30	.	.	26	.	.	.	1,613
54	.	.	.	.	.	.	.	40	575	.	.	.	.	.	.	.	615
51	.	.	.	.	.	.	.	10	377	.	.	.	7	.	.	.	394
93	.	.	11	.	.	.	.	68	.	.	.	.	.	.	.	.	79
7	.	.	.	24	.	.	8	.	39	.	.	.	.	.	.	.	71
53	10	.	.	.	.	.	.	.	55	.	.	.	.	.	.	.	65
97	.	.	.	.	.	.	.	63	.	.	.	.	.	.	.	.	63
30	.	.	.	.	.	.	.	36	.	.	.	.	.	.	.	.	36
92	.	.	.	.	.	.	.	31	.	.	.	.	.	.	3	.	34
98	.	.	.	.	.	.	.	.	30	.	.	.	.	.	.	.	30
12	1	.	.	.	.	.	.	.	17	.	.	.	.	.	.	.	18
60	.	.	.	.	.	.	.	.	.	.	.	.	18	.	.	.	18
25	7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	7
40	.	.	.	.	.	.	.	.	.	.	.	.	3	.	.	.	3
	3,065,339	1,099,057	11,002,520	1,962,432	5,242,084	647,242	206,015	5,582,842	1,693,306	3,677,986	1,072,355	18,879	401,459	72	1,272	196	35,673,056

Table 2.3.4.- Catches of 16 Hawaiian bottomfish species for the period 1948-2005 by 8 ‘directed’ gears. Notice that these 8 gears were responsible for 96.7% of the 16-species bottomfish complex. Gear 3 (deep sea bottom handline) accounted for 93.7% of the total bottomfish catches. The highlighted species (Armorhead, Alfonsin, Gunkan, and Yellowtail Kali) accounted for less than 0.05% of the total catches and thus were deleted from further consideration.

Species #	Hawaiian	GEAR								Total
		3	4	5	9	10	11	31	52	
15	Hapuupuu	3022977	22934	2084	305	15	2744	463	711	3,052,234
16	Kahala	3278516	253810	3282	12932	1292	50852	1055	1061	3,602,800
17	Kalekale	1035220	51413	2090	107	106	1937	75	60	1,091,007
19	Opakapaka	10740473	183080	12032	2491	9	17252		874	10,956,211
20	Uku	4843645	420587	22256	9729	11116	28970	207	4018	5,340,527
21	Ehu	1883363	63540	2812	193	22	692		31	1,950,653
22	Onaga	5188272	31948	1212	51		407		126	5,222,016
58	Lehi	626526	10806	972	5783		87		16	644,189
97	Gindai	198370	5189	280	58	7	85	11	4	204,004
114	Taape	168234	435144	639	361	21862	64696	246724	90001	1,027,661
140	Armorhead	72								72
141	Alfonsin	1268								1,268
200	Butaguchi	1068031	835	23	166	60	56	17	432	1,069,619
202	Gunkan	15058	766	1171	554	28			29	17,605
205	Ulua	239436	19123	9995	2797	4431	2985	5469	12394	296,630
208	Yellowtail Kali	196								196
		32,309,655	1,499,174	58,846	35,525	38,947	170,763	254,021	109,757	34,476,689
% of Total		0.9371	0.0435	0.0017	0.0010	0.0011	0.0050	0.0074	0.0032	1.0000
Cum %		0.9371	0.9806	0.9823						

2.4 Analysis of Variables in Bottomfish Database

This section analyses the database by the seven categories of variables (i.e., catch, date, environmental, gear, location, record flag, and trip) to determine their frequency of presence (or non-zero values) for the period 1948 to 2005. This analysis will help to determine what variables will be retained in the final streamlined database used for generation of the bottomfish catch and effort time-series.

Table 2.4.2.- Percent non-missing (or non-zero) records by year for “Date” variables.

year	db records	year	month	day	r_month	r_year	trip_end	trip_beg	timelink
1948	16983	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1949	17263	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1950	14362	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1951	12975	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1952	14640	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1953	18181	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1954	13810	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1955	15539	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1956	16306	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1957	13643	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1958	18376	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1959	13224	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1960	10881	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1961	8312	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1962	11016	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1963	11498	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1964	9129	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1965	8238	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1966	9357	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1967	8220	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1968	9346	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1969	8076	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1970	7667	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1971	10732	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1972	10800	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1973	11383	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1974	10859	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1975	12849	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1976	14794	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1977	19553	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1978	17451	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1979	19759	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1980	23869	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1981	27431	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1982	27983	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1983	31523	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1984	35259	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1985	34191	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1986	30354	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1987	31304	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1988	37451	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1989	35336	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1990	28301	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1991	25594	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1992	30103	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1993	25661	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1994	28310	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1995	29973	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1996	29691	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1997	32209	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1998	28150	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
1999	26167	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
2000	27104	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00
2001	22674	100.00	100.00	100.00	99.92	99.92	0.08	0.08	0.08
2002	20295	100.00	100.00	100.00	84.26	84.26	15.74	15.74	36.58
2003	18320	100.00	100.00	100.00	74.91	74.91	25.09	25.09	100.00
2004	19293	100.00	100.00	100.00	76.26	76.26	23.74	23.74	100.00
2005	16238	100.00	100.00	100.00	77.55	77.55	22.45	22.45	100.00

Table 2.4.3.- Percent non-missing (or non-zero) records by year for “Environmental” variables.

year	db records	depth_beg	depth_end	wind_speed	wind_dir	wave_ht	cur_speed	cur_dir
1948	16983	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1949	17263	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1950	14362	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1951	12975	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1952	14640	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1953	18181	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1954	13810	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1955	15539	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1956	16306	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1957	13643	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1958	18376	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1959	13224	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1960	10881	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1961	8312	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1962	11016	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1963	11498	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1964	9129	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1965	8238	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1966	9357	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1967	8220	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1968	9346	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1969	8076	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1970	7667	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1971	10732	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1972	10800	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1973	11383	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1974	10859	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1975	12849	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1976	14794	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1977	19553	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1978	17451	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1979	19759	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1980	23869	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1981	27431	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1982	27983	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1983	31523	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1984	35259	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1985	34191	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1986	30354	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1987	31304	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1988	37451	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1989	35336	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1990	28301	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1991	25594	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1992	30103	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1993	25661	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1994	28310	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1995	29973	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1996	29691	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1997	32209	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1998	28150	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1999	26167	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2000	27104	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2001	22674	0.08	0.08	0.08	0.08	0.08	0.08	0.08
2002	20295	12.09	12.09	11.76	10.05	12.54	10.68	10.39
2003	18320	11.03	11.03	11.43	10.60	11.42	9.27	9.27
2004	19293	14.48	14.48	14.56	13.35	14.86	10.72	9.89
2005	16238	15.07	15.07	14.58	10.16	14.89	13.06	12.81

Table 2.4.4.- Percent non-missing (or non-zero) records by year for “Gear” variables.

year	db records	gear	sets	hours	no_gears	hooks	net_length	net_height
1948	16983	99.99	0.00	0.00	0.00	0.00	0.00	0.00
1949	17263	99.44	0.00	0.00	0.00	0.00	0.00	0.00
1950	14362	99.85	0.00	0.00	0.00	0.00	0.00	0.00
1951	12975	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1952	14640	99.99	0.00	0.00	0.00	0.00	0.00	0.00
1953	18181	93.46	0.00	0.00	0.00	0.00	0.00	0.00
1954	13810	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1955	15539	99.98	0.00	0.00	0.00	0.00	0.00	0.00
1956	16306	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1957	13643	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1958	18376	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1959	13224	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1960	10881	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1961	8312	99.98	0.00	0.00	0.00	0.00	0.00	0.00
1962	11016	99.65	0.00	0.00	0.00	0.00	0.00	0.00
1963	11498	99.95	0.00	0.00	0.00	0.00	0.00	0.00
1964	9129	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1965	8238	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1966	9357	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1967	8220	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1968	9346	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1969	8076	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1970	7667	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1971	10732	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1972	10800	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1973	11383	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1974	10859	99.98	0.00	0.00	0.00	0.00	0.00	0.00
1975	12849	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1976	14794	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1977	19553	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1978	17451	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1979	19759	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1980	23869	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1981	27431	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1982	27983	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1983	31523	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1984	35259	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1985	34191	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1986	30354	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1987	31304	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1988	37451	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1989	35336	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1990	28301	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1991	25594	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1992	30103	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1993	25661	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1994	28310	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1995	29973	100.00	0.00	0.00	0.00	0.00	0.00	0.00
1996	29691	99.76	0.00	0.00	0.00	0.00	0.00	0.00
1997	32209	99.84	0.00	0.00	0.00	0.00	0.00	0.00
1998	28150	99.81	0.00	0.00	0.00	0.00	0.00	0.00
1999	26167	99.91	0.00	0.00	0.00	0.00	0.00	0.00
2000	27104	99.94	0.00	0.00	0.00	0.00	0.00	0.00
2001	22674	99.86	0.00	0.08	0.08	0.08	0.00	0.00
2002	20295	99.84	1.00	35.71	35.35	13.98	0.66	0.66
2003	18320	99.91	6.76	98.99	92.99	15.86	5.15	5.15
2004	19293	99.92	5.19	99.25	95.58	18.38	2.78	2.78
2005	16238	100.00	4.61	99.33	97.03	17.69	1.43	1.43

Table 2.4.5.- Percent non-missing (or non-zero) records by year for “Location” variables.

year	db records	area	subarea	buoy	bank	bank_quad	port_land	port_land2	port_dprtr
1948	16983	99.99	0.00	0.00	0.00	0.00	49.56	0.00	0.00
1949	17263	99.66	0.00	0.00	0.00	0.00	99.33	0.00	0.00
1950	14362	99.69	0.00	0.00	0.00	0.00	99.97	0.00	0.00
1951	12975	99.99	0.00	0.00	0.00	0.00	99.94	0.00	0.00
1952	14640	99.24	0.00	0.00	0.00	0.00	99.54	0.00	0.00
1953	18181	97.75	0.00	0.00	0.00	0.00	99.93	0.00	0.00
1954	13810	100.00	0.00	0.00	0.00	0.00	99.76	0.00	0.00
1955	15539	100.00	0.00	0.00	0.00	0.00	98.89	0.00	0.00
1956	16306	99.75	0.00	0.00	0.00	0.00	99.02	0.00	0.00
1957	13643	100.00	0.00	0.00	0.00	0.00	99.31	0.00	0.00
1958	18376	100.00	0.00	0.00	0.00	0.00	98.98	0.00	0.00
1959	13224	99.99	0.00	0.00	0.00	0.00	99.26	0.00	0.00
1960	10881	100.00	0.00	0.00	0.00	0.00	99.79	0.00	0.00
1961	8312	100.00	0.00	0.00	0.00	0.00	99.98	0.00	0.00
1962	11016	97.27	0.00	0.00	0.00	0.00	99.64	0.00	0.00
1963	11498	100.00	0.00	0.00	0.00	0.00	99.95	0.00	0.00
1964	9129	99.61	0.00	0.00	0.00	0.00	99.34	0.00	0.00
1965	8238	99.87	0.00	0.00	0.00	0.00	99.99	0.00	0.00
1966	9357	100.00	0.00	0.00	0.00	0.00	99.97	0.00	0.00
1967	8220	100.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
1968	9346	100.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
1969	8076	100.00	0.00	0.00	0.00	0.00	99.98	0.00	0.00
1970	7667	100.00	0.00	0.00	0.00	0.00	99.62	0.00	0.00
1971	10732	100.00	0.00	0.00	0.00	0.00	99.94	0.00	0.00
1972	10800	99.99	0.00	0.00	0.00	0.00	100.00	0.00	0.00
1973	11383	100.00	0.00	0.00	0.00	0.00	99.95	0.00	0.00
1974	10859	99.95	0.00	0.00	0.00	0.00	99.65	0.00	0.00
1975	12849	100.00	0.00	0.00	0.00	0.00	99.89	0.00	0.00
1976	14794	100.00	0.00	0.00	0.00	0.00	99.85	0.00	0.00
1977	19553	100.00	0.00	0.00	0.00	0.00	99.76	0.00	0.00
1978	17451	100.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
1979	19759	99.99	0.00	0.00	0.00	0.00	99.94	0.00	0.00
1980	23869	100.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
1981	27431	99.98	0.00	7.03	0.00	0.00	99.95	0.00	0.00
1982	27983	100.00	0.00	4.72	0.00	0.00	99.89	0.00	0.00
1983	31523	99.75	0.00	2.01	0.00	0.00	99.82	0.00	0.00
1984	35259	99.71	0.00	1.52	0.00	0.00	99.95	0.00	0.00
1985	34191	98.37	0.00	1.73	0.00	0.00	99.97	0.00	0.00
1986	30354	98.87	0.00	1.53	0.00	0.00	100.00	0.00	0.00
1987	31304	98.44	0.00	2.10	0.00	0.00	99.96	0.00	0.00
1988	37451	99.41	0.00	1.87	0.00	0.00	99.99	0.00	0.00
1989	35336	98.71	0.00	0.66	0.00	0.00	100.00	0.00	0.00
1990	28301	99.74	0.68	0.95	0.00	0.00	100.00	0.00	0.00
1991	25594	99.81	1.76	1.32	0.00	0.00	100.00	0.00	0.00
1992	30103	99.86	0.81	1.40	0.00	0.00	99.96	0.00	0.00
1993	25661	99.88	2.03	1.10	0.00	0.00	99.89	0.00	0.00
1994	28310	99.80	7.94	0.82	0.00	0.00	99.98	0.00	0.00
1995	29973	99.82	5.63	0.91	0.00	0.00	100.00	0.00	0.00
1996	29691	99.84	10.86	0.44	0.00	0.00	99.75	0.00	0.00
1997	32209	99.80	8.93	0.61	0.00	0.00	99.93	0.00	0.00
1998	28150	99.88	10.76	0.76	0.00	0.00	100.00	0.00	0.00
1999	26167	99.93	11.11	0.83	0.00	0.00	100.00	0.00	0.00
2000	27104	99.96	7.43	0.84	0.00	0.00	100.00	0.00	0.00
2001	22674	99.87	9.58	0.66	0.08	0.04	100.00	0.08	0.00
2002	20295	99.81	11.24	0.79	13.19	12.21	100.00	15.74	0.00
2003	18320	99.73	11.34	1.02	12.67	11.59	99.87	24.99	0.09
2004	19293	99.55	14.01	1.10	15.82	14.75	99.96	23.72	0.00
2005	16238	99.70	13.91	1.06	15.13	15.56	99.18	21.66	16.71

Table 2.4.6.- Percent non-missing (or non-zero) records by year for “Record Flag” variables.

year	db records	from_file	f_area	f_buoy	f_ar_gr	f_by_pr	f_spec_gr	f_spec_ar	f_season	f_pieces	f_lbs	f_ctchlim
1948	16983	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1949	17263	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1950	14362	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1951	12975	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1952	14640	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1953	18181	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1954	13810	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1955	15539	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1956	16306	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1957	13643	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1958	18376	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1959	13224	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1960	10881	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1961	8312	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1962	11016	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1963	11498	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1964	9129	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1965	8238	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1966	9357	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1967	8220	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1968	9346	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1969	8076	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1970	7667	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1971	10732	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1972	10800	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1973	11383	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1974	10859	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1975	12849	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1976	14794	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1977	19553	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1978	17451	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1979	19759	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1980	23869	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1981	27431	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1982	27983	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1983	31523	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1984	35259	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1985	34191	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1986	30354	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1987	31304	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1988	37451	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1989	35336	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1990	28301	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1991	25594	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1992	30103	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1993	25661	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1994	28310	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1995	29973	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1996	29691	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1997	32209	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1998	28150	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1999	26167	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2000	27104	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2001	22674	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2002	20295	63.42	0.00	0.00	0.01	0.00	0.08	0.08	0.00	0.53	0.00	0.01
2003	18320	100.00	0.00	0.16	0.05	0.01	9.56	8.25	0.56	0.47	0.98	2.62
2004	19293	100.00	0.00	0.40	0.23	0.03	11.33	10.44	0.62	0.88	0.51	2.07
2005	16238	0.00	0.01	0.34	0.17	0.03	1.34	2.05	0.03	1.56	0.06	0.14

Table 2.4.7.- Percent non-missing (or non-zero) records by year for “Trip” variables.

year	db records	license	lname	fname	vessel	uscg	ha_no	logbook	formtype	chartered
1948	16983	99.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1949	17263	97.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1950	14362	99.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1951	12975	98.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1952	14640	93.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1953	18181	67.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1954	13810	97.55	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1955	15539	89.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1956	16306	86.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1957	13643	73.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1958	18376	58.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1959	13224	70.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1960	10881	86.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1961	8312	86.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1962	11016	66.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1963	11498	83.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1964	9129	82.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1965	8238	81.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1966	9357	80.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1967	8220	83.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1968	9346	81.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1969	8076	96.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1970	7667	95.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1971	10732	88.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1972	10800	94.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1973	11383	96.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1974	10859	90.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1975	12849	98.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1976	14794	99.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1977	19553	99.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1978	17451	99.64	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1979	19759	99.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1980	23869	98.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1981	27431	93.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1982	27983	94.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1983	31523	93.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1984	35259	90.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1985	34191	95.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1986	30354	98.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1987	31304	93.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1988	37451	96.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1989	35336	98.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1990	28301	99.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1991	25594	99.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1992	30103	99.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1993	25661	97.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1994	28310	99.84	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1995	29973	99.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1996	29691	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1997	32209	99.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1998	28150	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1999	26167	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2000	27104	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2001	22674	100.00	0.08	0.08	0.00	0.08	0.00	0.00	0.08	0.00
2002	20295	100.00	36.58	36.58	18.70	13.87	21.59	0.00	36.58	0.71
2003	18320	100.00	100.00	100.00	73.05	23.40	73.91	0.84	100.00	3.22
2004	19293	100.00	100.00	100.00	73.07	25.48	72.64	14.67	100.00	3.88
2005	16238	100.00	100.00	100.00	86.11	23.64	73.35	16.50	100.00	3.44

Table 2.4.8.- Description of values for variable ‘number of gears’. The column ‘most freq’ represents the number of units of that gear most frequently recorded in the database. These data were determined to be largely unreliable.

gear_desc	gear	most freq	dominant freq. range
DEEP-SEA HANDLINE, BOTTOM HANDLINE	3	2	1-4
INSHORE HANDLINE, COWRIE SHELL (TAKO)	4	2	1-4
KAKA LINE, SET LINE	5	1	1-4
TROLLING (MISC.)	6	4	2-9
IKA-SHIBI	8	1	1-3
PALU AHI, DROP STONE, MAKE DOG	9	2	1-4
CASTING, LIGHT TACKLE, SPINNER, WHIPPING	10	6	1-6
SPEARFISHING, DIVE, SQUIDING (TAKO)	13	1	1-3
DIVING, DIVE FOR LOBSTER OR NAMAKO, ETC.	14	2	1-2
LIFT NET, OPELU	21	1	1
GILL NET, FENCE NET, LAY NET, CROSS NET	22	1200	125,150,800,900,1200,1400,1500,1600
SEINE NET	23	1500	1,500,600,800,1000,1200,1500
BULLPEN TRAP	24	1000	100,800,1000
CRAB NET	26	6	6
THROW NET	27	1	1
PURSE SEINE NET (PELAGIC)	33	1	1,200,1200
SURFACE PELAGIC HANDLINE	35	2	1-2
KONA CRAB NET, LOOPS	40	30	6,10,15,30,60,100,120,250
CRAB TRAP (CAN BE DEEP SEA)	51	120	80,120
FISH TRAP	52	20	2-8,10,13,15,20
LOBSTER TRAP, LOBSTER POT	53	5	5
SHRIMP TRAP	54	10	10
HANDPICKED (LIMU, OPIHI, WANA, AAMA CRAB, NAMAKO, INA	60	1	1
TROLLING - LURES	61	3	1-7
TROLLING - BAIT	62	3	1-4
TROLLING - STICK	63	2	2,5
SHORTLINE	92	4	1-4
VERTICAL LONGLINE	93	2	1-4
OTHER	99	1	1-3

Table 2.4.9.- Frequency analysis for several key “Gear” variables found in the bottomfish database (c.f., Table 2.1.1) for the period 2002-2005: (A) “Sets”; (B) “Hooks”; (C) “net length”; and, (D) “net height”. Gear is the gear number, most freq is the most frequent observation in the database, and freq range is the range of observations in the database.

(A) Variable ‘sets’.

gear	most freq	freq range
21	2	1-10
22	1	1,6,8,10-11
23	1	1-4
27	1,20	1,20
33	6	1-3,6-7
40	2	1-3,5-6

(B) Variable ‘hooks’.

gear	most freq	freq range
3	8	1-6,8
4	1	1,3,5
6	1	1,2
9	1	1
61	1	1-2

(C) Variable ‘net length’.

gear	most freq	freq range	note
22	150	150,900,1200,1400,1500,2500	usually no_gears=0 for these ranges
23	1400	250,400,500,800,1000,1400	

(D) Variable ‘net height’.

gear	most freq	freq range
22	12	7-9,12,18,20
23	12	12,20

Table 2.4.10.- Proportion of trip-records containing single or multiple ports of landing, 2002-2005.

Region_Lg	Trip Type	Daily	% trips	
		Trip-Records	1 Landing Port	2 Landing Ports
MHI	Commercial	17919	99.85	0.15
MHI	Chartered	1106	100	0
NWHI	Commercial	2103	100	0
NWHI	Chartered	0		

This analysis shows that the data largely reflect usage of a single port of landing for 99.8% of the observed trip records.

Table 2.4.11.- Proportion of trip-records containing single or multiple fishing areas, 2002-2005.

Region_Lg	Trip Type	Daily	% trips			
		Trip-Records	1 Area	2 Areas	3 Areas	4+ Areas
MHI	Commercial	17945	93.04	5.26	1.46	0.24
MHI	Chartered	1106	86.71	8.41	3.53	1.35
NWHI	Commercial	2103	97.43	2.05	0.52	0

This analysis shows that a single area is generally fished for Main Hawaiian Island (MHI) and Northwestern Hawaiian Islands (NWHI) commercial trips, and for MHI chartered trips.

Table 2.4.12.- Proportion of trip-records containing single or multiple fishing gears, 2002-2005.

Region_Lg	Trip Type	Daily	% trips			
		Trip-Records	1 Gear	2 Gears	3 Gears	4 Gears
MHI	Commercial	17945	78.81	19.79	1.38	0.02
MHI	Chartered	1106	76.31	14.56	9.13	0
NWHI	Commercial	2103	85.50	14.46	0.04	0

By and large a single gear is fished on greater than 75 percent of the MHI commercial and chartered trips, and greater than 85% of the NWHI commercial trips.

These analyses show that the typical case for the bulk of the database is that vessels fish mostly 1 area per trip, use between 1-2 gears, and access 1 landing port.

Table 2.4.13.- Descriptive statistical analysis showing the minimum, mean, median, and maximum of 2000-2005 daily fishing trip-records, and ports of landing-per-month per license-holder: (A) Main Hawaiian Islands commercial; (B) Main Hawaiian Islands charter; and, (C) Northwestern Hawaiian Islands commercial.

(A) MHI Commercial

Year	Daily trip-records per month per license-holder				Different ports per month per license-holder			
	min_trips	av_trips	med_trips	max_trips	min_ports	av_ports	med_ports	max_ports
2002	1	2.23	1	23	1	1.05	1	4
2003	1	2.37	2	25	1	1.04	1	3
2004	1	2.28	2	22	1	1.04	1	3
2005	1	2.20	1	23	1	1.04	1	3

(B) MHI Chartered

Year	Daily trip-records per month per license-holder				Different ports per month per license-holder			
	min_trips	av_trips	med_trips	max_trips	min_ports	av_ports	med_ports	max_ports
2002	1	4.61	2	18	1	1.00	1	1
2003	1	4.87	2	25	1	1.00	1	1
2004	1	6.03	3	22	1	1.02	1	2
2005	1	5.00	2	19	1	1.00	1	1

(C) NWHI Commercial

Year	Daily trip-records per month per license-holder				Different ports per month per license-holder			
	min_trips	av_trips	med_trips	max_trips	min_ports	av_ports	med_ports	max_ports
2002	1	7.21	6	16	1	1.00	1	1
2003	1	6.88	6	16	1	1.00	1	1
2004	1	8.71	9	18	1	1.03	1	2
2005	1	9.45	10	19	1	1.00	1	1

Table 2.4.14.- Number of fish released by species and fishing region/sector, 2002-2005. Shading denotes the 12 target bottomfish species; ‘—’ denotes species not captured in region/sector. See Appendix A for full list of species.

Bottomfish Name	Species	Number of Fish Released, 2002-2005		
		MHI Commercial	MHI Charter	NWHI
	1	13	--	0
	2	17	0	0
	3	39	0	0
	4	0	--	--
	6	0	0	0
	7	0	0	0
	8	0	--	--
	9	0	1	0
	10	1	1	0
	11	0	--	--
	12	0	--	0
	13	6	0	0
	14	16	0	0
HAPUUPUU	15	4	0	0
KAHALA	16	5056	369	0
KALEKALE	17	18	0	0
	18	471	0	0
OPAKAPAKA	19	369	0	0
UKU	20	42	0	0
EHU	21	41	0	0
ONAGA	22	31	0	0
	23	15	0	0
	24	4	0	0
	25	43	0	0
	26	0	0	--
	27	0	--	--
	28	50	--	--
	29	0	--	--
	30	1	--	0
	31	0	--	--
	33	1	--	--
	34	203	--	0
	35	0	--	--
	37	0	--	--
	38	7	0	0
	39	--	--	0
	40	3	--	--
	41	0	--	--
	42	23	--	--
	43	301	12	--
	45	0	--	--
	46	33	34	0
	47	821	--	--
	48	5	1	0
	49	0	0	0
	51	4	--	--
	52	527	--	0
	53	0	--	--
	54	14	0	--
	55	6	0	--
	56	0	0	--
	57	0	--	--
LEHI	58	12	0	0
	59	0	--	--
	60	24	--	--
	62	0	--	--
	63	183	--	--
	64	41	--	--
	65	683	71	0
	66	4	--	--
	67	0	--	--
	68	420	0	0
	69	0	--	--
	71	9	--	--
	72	624	--	--
	74	21	--	--
	75	0	--	0
	76	0	--	--
	77	152	--	--
	78	20	--	--
	79	0	--	--
	80	6	--	--
	81	38	0	--
	84	0	--	--

Table 2.4.14.- (cont.)

Bottomfish Name	Species	Number of Fish Released, 2002-2005		
		MHI Commercial	MHI Charter	NWHI
	85	0	--	--
	86	137	--	--
	87	36	--	--
	88	0	--	--
	89	0	0	--
	92	16	--	--
	93	99	--	0
	94	1	--	--
	95	0	--	--
	96	1035	--	--
GINDAI	97	12	0	0
	98	0	--	--
	99	3	0	--
	100	1045	0	0
	101	3	--	--
	102	25	--	0
	103	6	--	--
	104	1	0	0
	107	0	1	0
	110	10	--	--
	111	0	--	--
	112	1	--	--
	113	0	--	--
TAAPE	114	760	1145	0
	115	16	--	--
	116	0	--	--
	117	0	--	--
	118	0	0	0
	119	7	--	--
	120	0	--	--
	121	0	--	--
	122	2	--	--
	124	1255	--	--
	125	23	--	--
	126	9	--	--
	127	2046	0	0
	128	51	--	--
	129	0	--	--
	140	--	--	0
	141	0	--	0
BUTAGUCHI	200	8	--	0
	201	12	--	0
	202	0	0	0
	203	0	--	0
	204	12	--	0
WHITE PAPIO/ULUA	205	192	14	0
	206	11	--	--
	207	0	--	--
	208	0	--	0
	209	4	--	--
	220	0	--	--
	221	0	--	--
	300	23	--	0
	320	2	1	--
	321	1	--	0
	322	3	--	0
	324	11	--	0
	700	0	--	--
	701	227	--	0
	702	0	--	--
	706	0	--	--
	707	0	--	--
	708	0	--	--
	709	0	--	--
	716	0	--	--
	717	0	--	--
	719	3	--	--
	721	0	--	--
	722	0	--	--
	724	0	--	--
	726	9	--	--
	727	0	--	--
	752	0	--	--
	800	0	--	--
	999	881	--	0

Table 2.4.15.- Problem catch records ('lbs'=0).

Year	lbs=0	Number of database records associated with lbs=0 values							
	records	lbs_sold	caught	num_sold	nocatch	num_stolen	lost	release	damage
1949	30	0	0	0	0	0	0	0	0
1950	37	0	2	0	0	0	0	0	0
1952	27	0	1	0	0	0	0	0	0
1953	665	493	30	0	0	0	0	0	0
1955	1	1	0	0	0	0	0	0	0
1956	57	43	8	0	0	0	0	0	0
1957	3	0	3	0	0	0	0	0	0
1958	5	0	4	0	0	0	0	0	0
1959	6	0	6	0	0	0	0	0	0
1960	2	0	2	0	0	0	0	0	0
1961	4	1	1	0	0	0	0	0	0
1962	2	0	1	0	0	0	0	0	0
1964	22	0	2	0	0	0	0	0	0
1966	3	0	0	0	0	0	0	0	0
1967	22	0	0	0	0	0	0	0	0
1968	22	0	0	0	0	0	0	0	0
1969	5	0	0	0	0	0	0	0	0
1970	5	0	0	0	0	0	0	0	0
1971	15	0	0	0	0	0	0	0	0
1972	1	0	0	0	0	0	0	0	0
1974	4	1	1	0	0	0	0	0	0
1975	2	0	1	0	0	0	0	0	0
1976	1	0	1	0	0	0	0	0	0
1982	4	1	2	0	0	0	0	0	0
1983	148	93	42	0	0	0	0	0	0
1984	32	1	22	0	0	0	0	0	0
1985	8	0	6	0	0	0	0	0	0
1986	2	0	2	0	0	0	0	0	0
1989	339	0	0	0	0	0	0	0	0
1990	341	0	0	0	0	0	0	0	0
1991	342	0	3	0	0	0	0	0	0
1992	444	0	2	0	0	0	0	0	0
1993	376	0	1	0	0	0	0	0	0
1994	316	0	0	0	0	0	0	0	0
1995	377	0	3	0	0	0	0	0	0
1996	278	0	0	0	0	0	0	0	0
1997	271	0	0	0	0	0	0	0	0
1998	293	0	0	0	0	0	0	0	0
1999	262	0	0	0	0	0	0	0	0
2000	260	0	0	0	0	0	0	0	0
2001	187	0	0	0	0	0	0	0	0
2002	870	0	0	0	162	28	40	224	8
2003	1322	0	13	0	496	1	100	509	2
2004	1555	2	13	2	526	0	85	525	4
2005	1457	2	8	2	512	7	118	414	2
Total	10425	638	180	4	1696	36	343	1672	16

Table 2.4.16.- Final designation of a reduced set of gear categories describing combinations of gears used in the Hawaiian bottomfish fishery.

Gear Category Name	gear_cat	gear
Deep-sea handline	HLD	3
Kaka line	KKL	5
Shallow handline	HLS	4,9
Other hook-line gears	HLO	1,2,6,8,10,35,61,62,63,92,93
Trap gears	TRP	11,24,51,52,53,54
Spear & diving gears	SPR	13,14
Net gears	NET	20,21,22,23,25,26,27,28,30,31,33,40
Other gears	OTH	all other gear codes

2.5 Refined Bottomfish Database

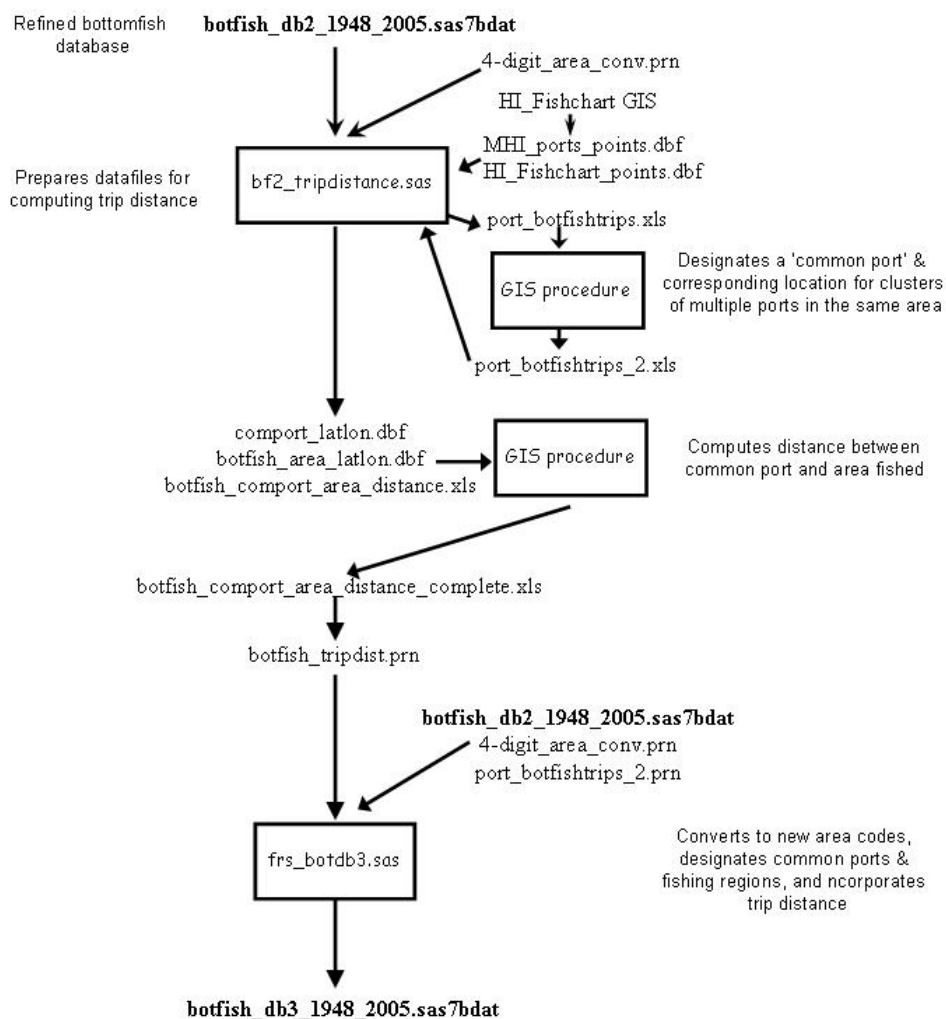
Table 2.5.1.- Variables in the refined bottomfish database. Advantages of this streamlining: it eliminates variables irrelevant to further analyses; further refines the definition of a bottomfish trip' (i.e., =license-year-month-day-port_land); and, resolves (deletes) remaining problem catch records. # refers to original variable number; *variable* is the refined variable name; *type* is the format of the variable; and, *len* is the number of character spaces allotted for the variable.

#	Variable	Type	Len
7	area	Num	8
16	bank	Char	18
17	bank_quad	Char	2
25	bf	Num	8
6	bouy	Char	2
21	chartered	Char	1
4	day	Num	8
27	deep7	Num	8
18	depth_beg	Num	8
19	depth_end	Num	8
26	fine9	Num	8
9	gear	Num	8
28	gear_cat	Char	3
13	hooks	Num	8
11	hours	Num	8
24	lbs	Num	8
1	license	Num	8
3	month	Num	8
15	net_height	Num	8
14	net_length	Num	8
12	no_gears	Num	8
20	port_land	Num	8
10	sets	Num	8
23	species	Num	8
8	subarea	Char	1
22	trip_beg	Num	8
5	trip_end	Num	8
2	year	Num	8

3.0 Data Processing and Statistical Analysis to Estimate Nominal Fishing Effort

This section details the sequential process of evaluating the database to enable computation of nominal fishing effort for the bottomfish fishery from 1948-2005. The procedures for determining the distances for unique fishing trips are given in **Figure 3.0.1**.

Fig. 3.0.1.- Steps in data processing required for computation of bottomfish fishing-trip distances.



3.1 Computation of Trip Distances

The analysis begins with the computation of trip distances between fishing areas designated on the Hawaiian Fishing Areas map (GIS format) and the specific ports-of-landing also identified in the Fishery Resources Database (GIS format).

Each trip in the fishery resources database provides information on the area fished and port-of-landing (**Figure 3.1.1**).

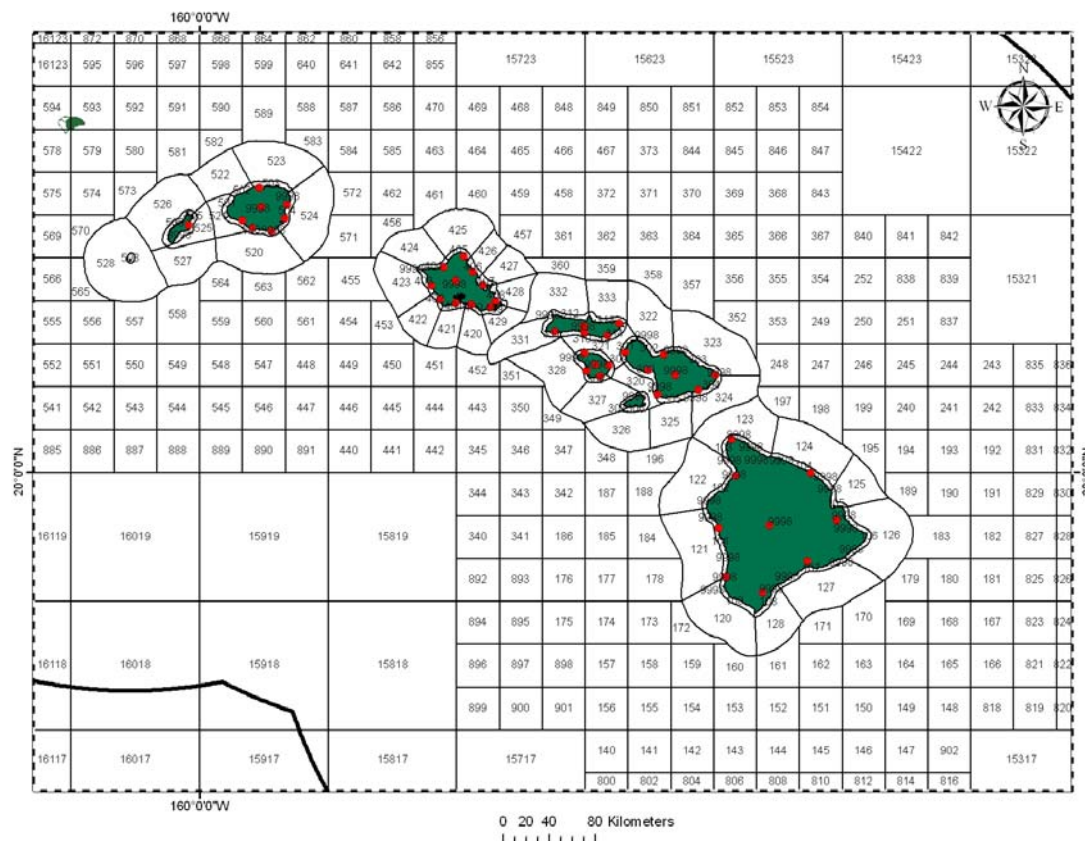


Figure 3.1.1.- Location of principal bottomfish ports (red dots) where fishing vessels either departed or landed during the period of 1948-2005 used for computation of trip distance.

The total database of ports-of-landing provided by Hawaii DLNR was reduced to a subset of common ports around each island (**Figure 3.1.2**). These ports were matched with specific fishing areas reported for the entire FRB database for the period 1948-2005, and these produced a matrix of trip-distances associated with each specific area fished (AF)-port of landing (POL) pair (**Table 3.1.1**).

To clearly evaluate potential multiple trip-days from the historical database that contained information on AF and POL and a single date of entry, we analyzed the contents of the 2002-2005 database.

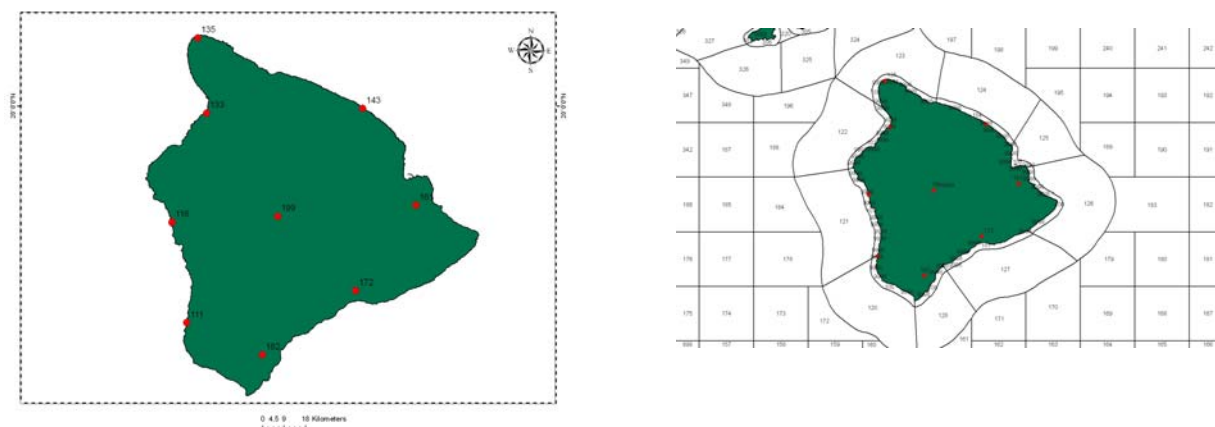


Figure 3.1.2.- Steps on the computation of trip distance: (A) location of principal common bottomfish ports (red dots) where fishing vessels landed during the period of 1948-2005; and, (B) fishing area designations used for determining where a trip took place.

For the entire database we computed trip-distance combinations for all reported combinations of AF and POL.

Table 3.1.1.- Example matrix of computed bottomfish trip distance from port on the Island of Hawai'i to specific fishing areas around the island.

Area	Common Landing Port								
	111	116	133	135	143	161	172	182	199
100	12.8	35.4	67.2	80.2	.	90.6	46.4	25.8	37.5
101	16.8	5.9	37.5	51.3	90.7	118.9	75.4	56.6	23.5
102	47.6	25.7	6.8	21.2	61.3	88.0	105.6	84.1	29.6
103	.	55.0	28.3	5.5	33.4	60.0	.	113.5	39.3
104	.	82.8	55.6	.	4.2	31.0	.	.	30.8
105	125.8	103.8	77.0	.	.	10.0	.	78.4	31.0
106	94.3	109.7	.	.	.	13.3	39.2	57.5	38.4
107	55.8	78.2	.	.	.	48.6	4.5	25.8	28.0
108	33.0	54.4	85.6	.	.	71.6	26.8	5.2	35.7
120	19.7	42.0	72.9	87.0	.	94.0	49.1	29.6	48.1
121	18.5	14.2	41.1	.	94.8	119.4	74.8	55.8	34.9
122	47.3	25.5	18.1	25.4	65.4	92.3	105.0	85.0	39.9
123	76.4	54.1	26.6	7.4	37.9	64.8	111.5	111.2	48.0
124	105.1	83.6	56.4	.	11.7	35.9	.	103.3	40.3
125	112.3	107.3	80.5	.	.	20.5	.	80.6	43.3
126	90.8	113.4	102.8	.	.	22.4	38.7	61.6	49.8
127	58.8	81.0	113.5	.	.	47.5	15.3	29.7	38.8
128	34.8	57.0	88.3	.	.	75.4	32.7	16.6	45.8

3.2 Procedures for Identifying Unique Fishing Trips

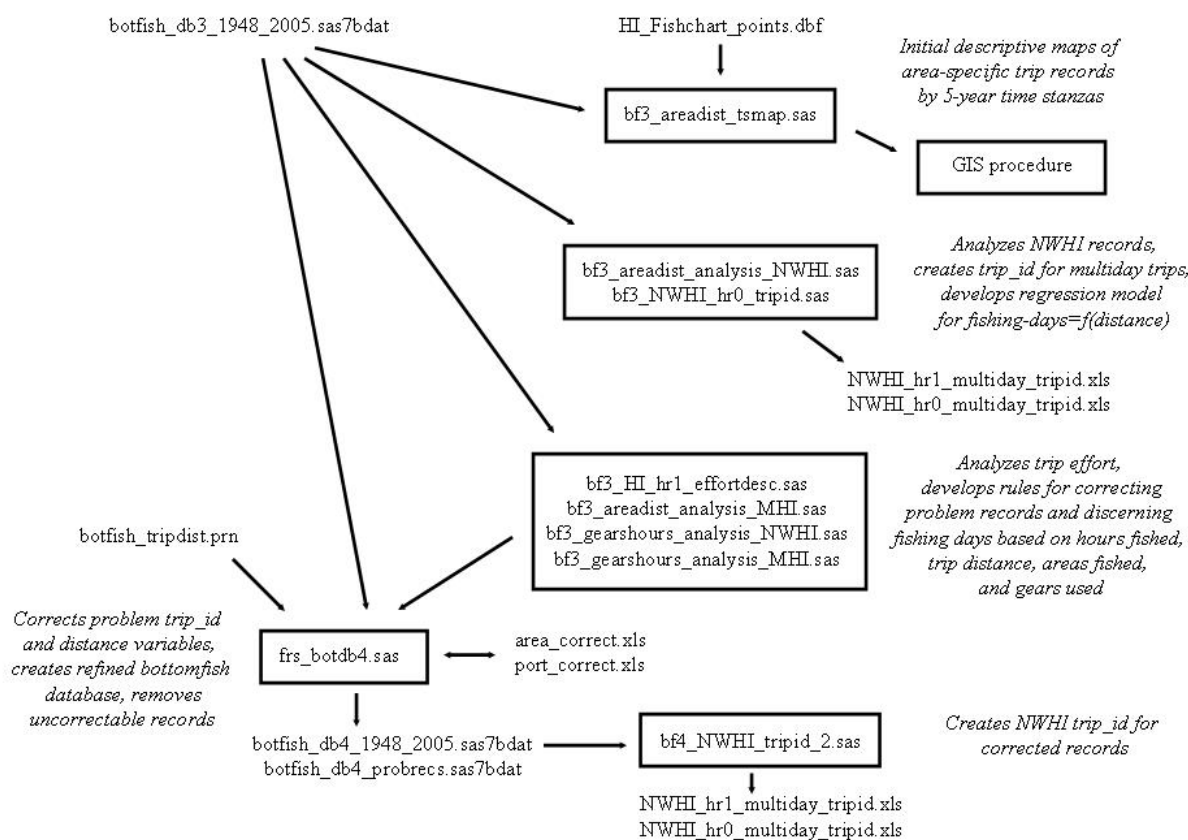


Figure 3.2.0.- Procedure flow for computing unique fishing trips.

Table 3.2.1.- Illustration of procedure for identification of complete fishing trips in the Northwestern Hawaiian Islands. A complete trip was defined as a trip with a sequential number of days fishing and a port of landing.

license	year	month	day	trip_id	port	land	area	subarea	gear	dist	gear_cat	hours
1157	2003	7	16	1	511		16423	C	3	369.517	HLD	1
1157	2003	7	17	1	511		16423	C	3	369.517	HLD	3
1157	2003	7	17	1	511		16423	C	6	369.517	HLO	4
1157	2003	7	18	1	511		16423	C	6	369.517	HLO	12
1157	2003	7	19	1	511		16423	D	6	414.639	HLO	12
1157	2003	8	8	2	511		16423	E	3	407.203	HLD	6
1157	2003	8	8	2	511		16423	E	6	407.203	HLO	5
1157	2003	8	9	2	511		16423	E	3	407.203	HLD	10
1157	2003	8	10	2	511		16423	E	6	407.203	HLO	4
1157	2003	8	11	2	511		16423	E	6	407.203	HLO	5
1157	2003	8	13	2	511		16222	I	3	257.249	HLD	10
1157	2003	8	26	3	511		16423	E	6	407.203	HLO	5
1157	2003	8	27	3	511		16423	E	6	407.203	HLO	12
1157	2003	8	28	3	511		16423	E	6	407.203	HLO	12
1157	2003	8	29	3	511		16423	E	6	407.203	HLO	12
1157	2003	8	30	3	511		16423	E	3	407.203	HLD	8
1157	2003	8	30	3	511		16423	E	6	407.203	HLO	4
1157	2003	8	31	3	511		16423	E	3	407.203	HLD	13
1157	2003	8	31	3	511		16423	E	6	407.203	HLO	1
1157	2003	10	5	4	511		875	Z	6	240.288	HLO	7
1157	2003	10	5	4	511		16223	C	3	266.462	HLD	2
1157	2003	10	6	4	511		16222	I	3	257.249	HLD	8
1157	2003	10	7	4	511		16223	A	3	299.020	HLD	5
1157	2003	10	7	4	511		16223	A	6	299.020	HLO	3
1157	2003	10	8	4	511		16223	A	3	299.020	HLD	2
1157	2003	10	8	4	511		16223	A	6	299.020	HLO	8
1157	2003	10	9	4	511		16222	I	3	257.249	HLD	9
1157	2003	10	10	4	511		16222	I	3	257.249	HLD	8
1157	2003	10	10	4	511		16222	I	6	257.249	HLO	4
1157	2003	11	15	5	511		16423	C	3	369.517	HLD	11
1157	2003	11	16	5	511		16423	E	3	407.203	HLD	8
1157	2003	11	16	5	511		16423	E	6	407.203	HLO	5
1157	2003	11	17	5	511		16423	E	3	407.203	HLD	12
1157	2003	11	18	5	511		16423	E	3	407.203	HLD	12
1157	2003	11	18	5	511		16423	E	6	407.203	HLO	2
1157	2003	11	19	5	511		16423	E	3	407.203	HLD	10
1157	2003	11	20	5	511		16423	E	3	407.203	HLD	6
1157	2003	11	20	5	511		16423	E	6	407.203	HLO	5
1157	2003	11	21	5	511		16223	A	3	299.020	HLD	2
1157	2003	11	21	5	511		16223	A	6	299.020	HLO	2

3.2.1 Relationship Between Trip Distance and Number of Trip-Days

To determine the number of “trip days” involving active fishing for bottomfish in the historical data records (1948-2002), an analysis of the data for bottomfish trips taken to the Northwestern Hawaiian Islands during the period 2002-2005 was conducted.

Descriptive analysis of these data indicated that the values increased linearly up to about 500 nautical miles (nm) and then asymptoted. To estimate the number of “trip days” from a historical data record that contained only the location fish and the port-of-landing, we estimated the distance and fit these records to an empirical function using multivariate regression analysis.

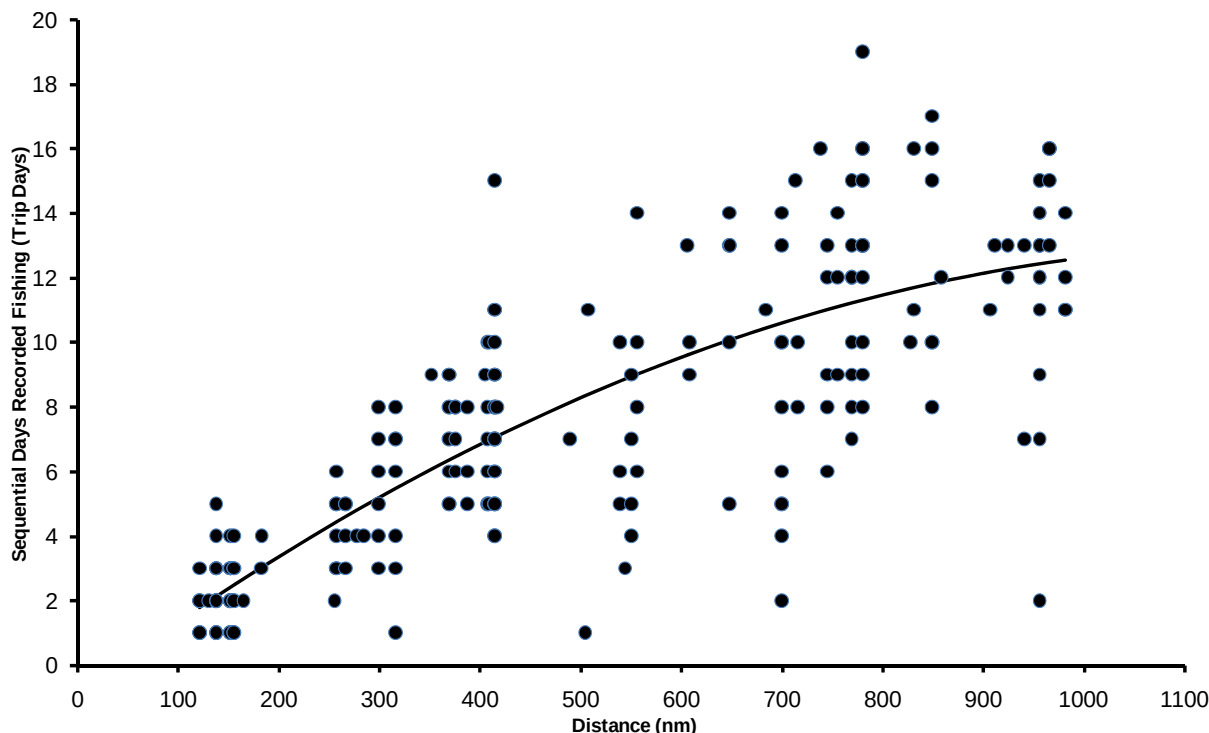


Figure 3.2.1.- Trip-days as a function of distance from landing port for NWHI commercial bottomfish trips for the period 2002-2005.

The multiple regression equation expressing trip days (Y_i) as a function of distance (X_i) was

$$Y_i = -0.87262 + 0.02314X - 0.00000965X^2$$

where, $n = 282$; $r^2 = 0.71$. More than 95% of the trips to the NWHI originated from Oahu (particularly Kewalo Basin). This trip-distance analysis for the Northwestern Hawaiian Islands for the period 2002-2005 was in excellent agreement with Kurt Kawamoto (1997) analysis for NWHI data for the 1990s.

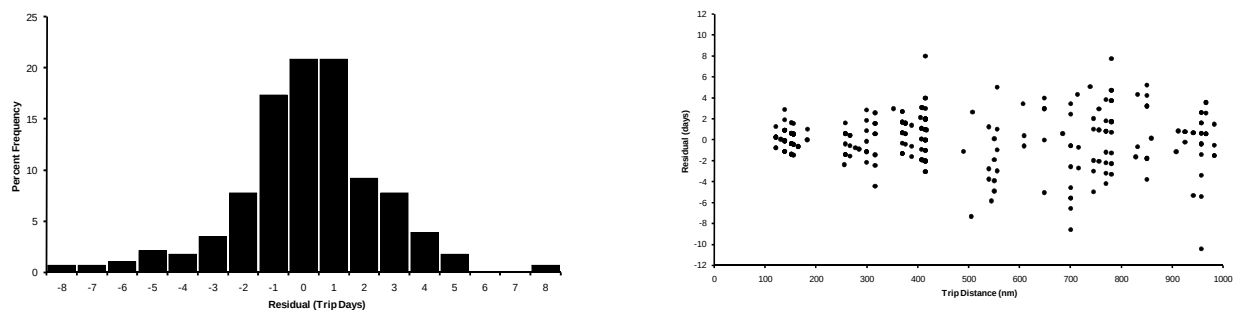


Figure 3.2.2._ Residual patterns from multivariate regression analysis of bottomfish trip length (i.e., trip-days) dependent on trip distance: (A) residual distribution of the response variate trip-days; and, (B) residuals dependent on the predictor variable (trip distance).

MHI Commercial

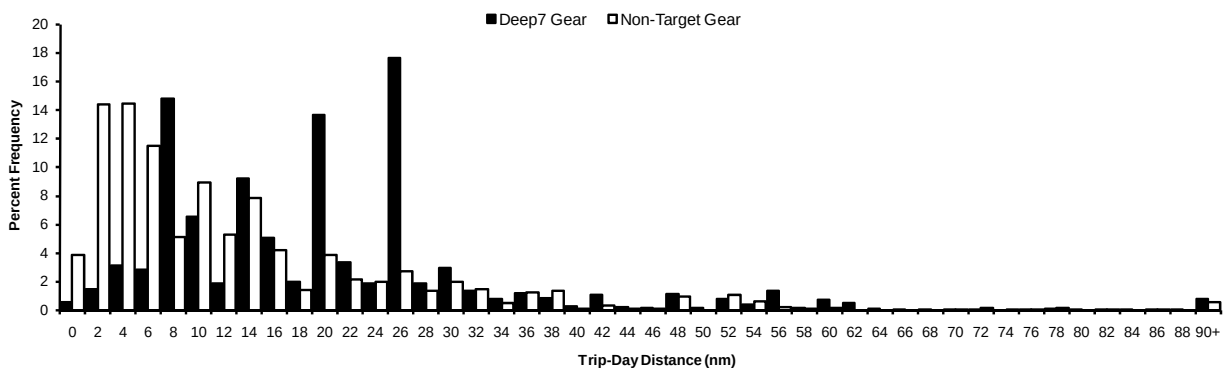


Figure 3.2.3.- Distribution of trip-day distance traveled for Main Hawaiian Islands (MHI) commercial bottomfish fleet using “Deep7” target gears (#3 and 5, dark squares) and non-target gears (open squares) from historical trips with AF and POL data.

MHI Charter

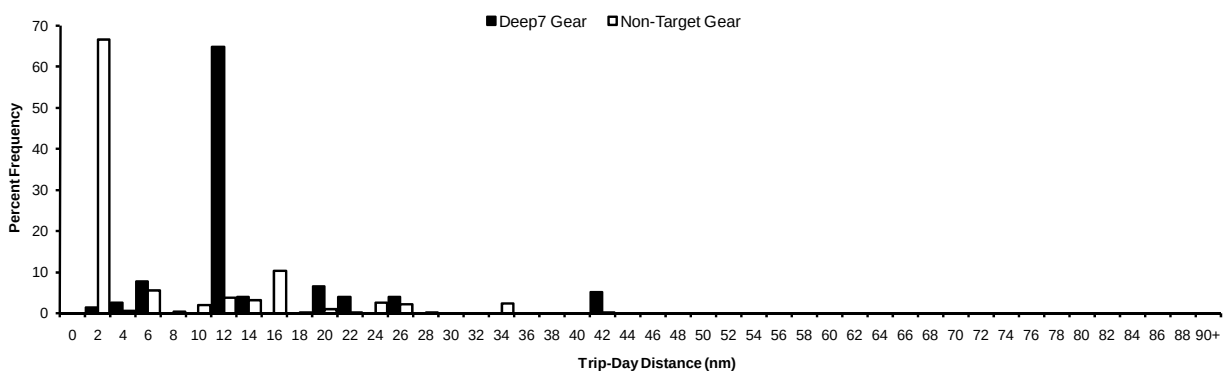


Figure 3.2.4.- Distribution of trip-day distance traveled for Main Hawaiian Islands (MHI) chartered bottomfish fleet using “Deep7” target gears (#3 and 5) and non-target gears from historical trips with distance.

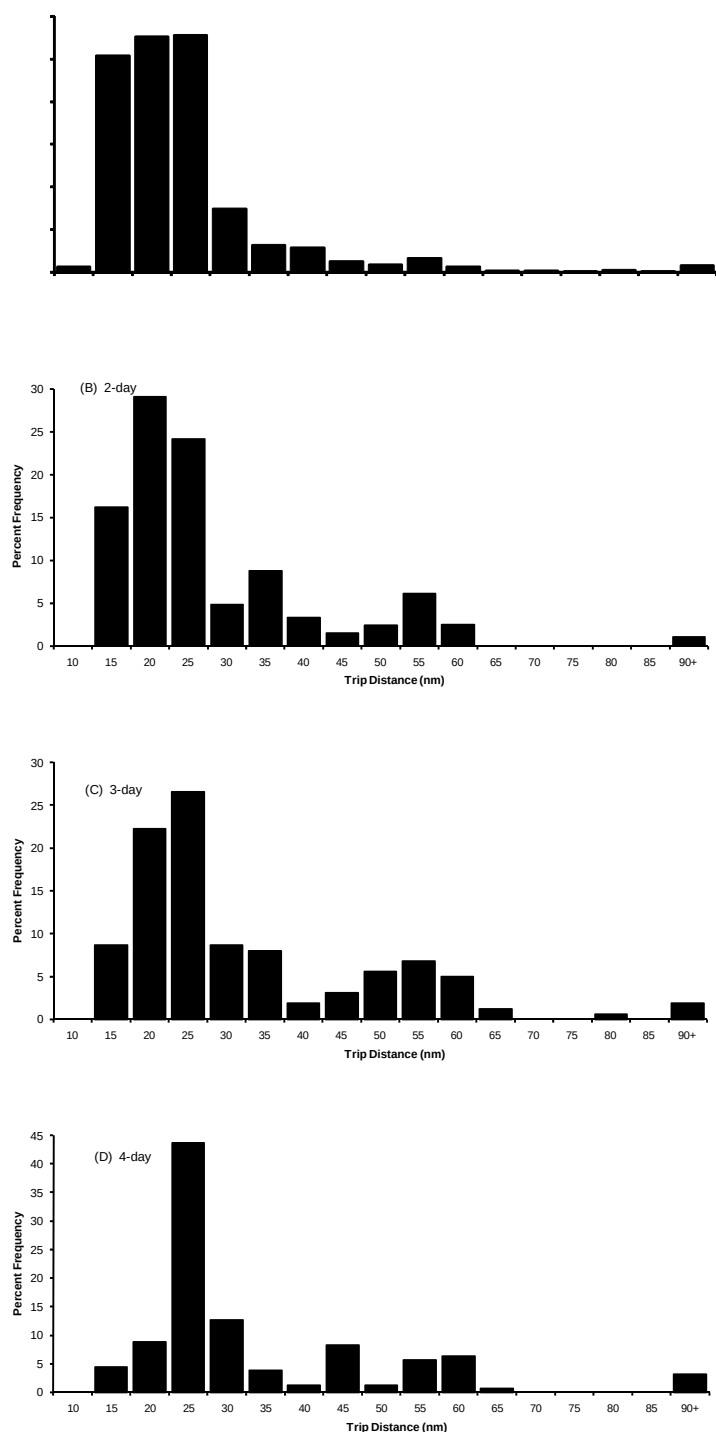


Figure 3.2.5.- Distribution of distances (nm) for commercial MHI Deep7 bottomfish trips of differing lengths from the FRS database for the period 2002-2005: (A) 1 day; (B) 2 day; (C) 3 day; and (D) 4 day. One day trips were considered to be those that travelled < 30 nm; 2 day, $30 \text{ nm} \leq T \leq 60 \text{ nm}$; 3 day, $> 60 \text{ nm}$.

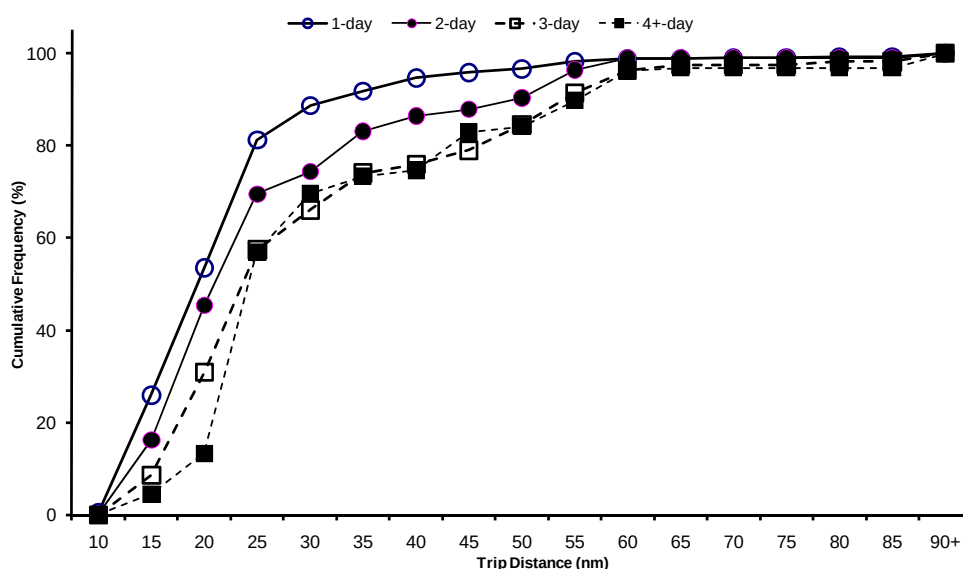


Figure 3.2.6.- Cumulative frequency of observations of distances to areas-fished from port-of-landing for varying length trips in the Main Hawaiian Islands (MHI) bottomfish fishery for the period 2002-2005. Single day trips were <30 nm; 2-day trips were ≥ 30 , but ≤ 60 nm; and, ≥ 3 day trips were > 60 nm.

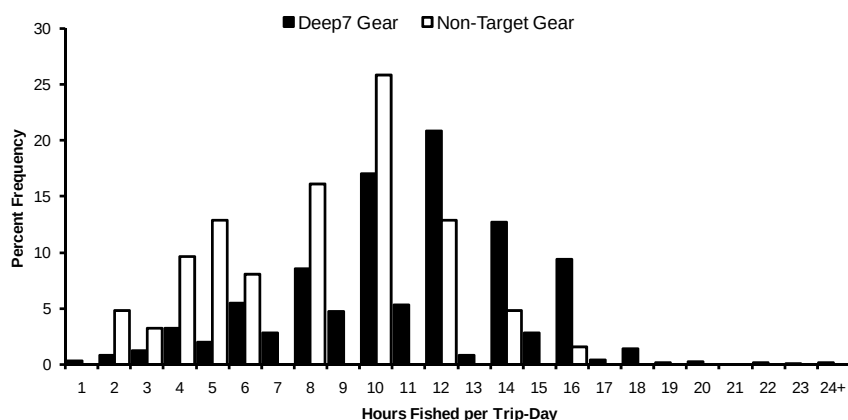
(A)		NWHI		MHI Commercial		MHI Chartered	
# Areas Fished		trip-days	% tripdays	trip-days	% tripdays	trip-days	% tripdays
1		1933	97.58	14317	95.00	825	91.36
2		38	1.92	721	4.78	56	6.20
3		10	0.50	29	0.19	21	2.33
4		0	0.00	3	0.02	1	0.11
Total		1981	100	15070	100	903	100.0

(B)		NWHI		MHI Commercial		MHI Chartered	
# gear categories		trip-days	% tripdays	trip-days	% tripdays	trip-days	% tripdays
1		1713	86.47	12536	83.19	871	96.46
2		268	13.53	2430	16.12	30	3.32
3		0	0.00	100	0.66	2	0.22
4		0	0.00	4	0.03	0	0.00
Total		1981	100	15070	100	903	100

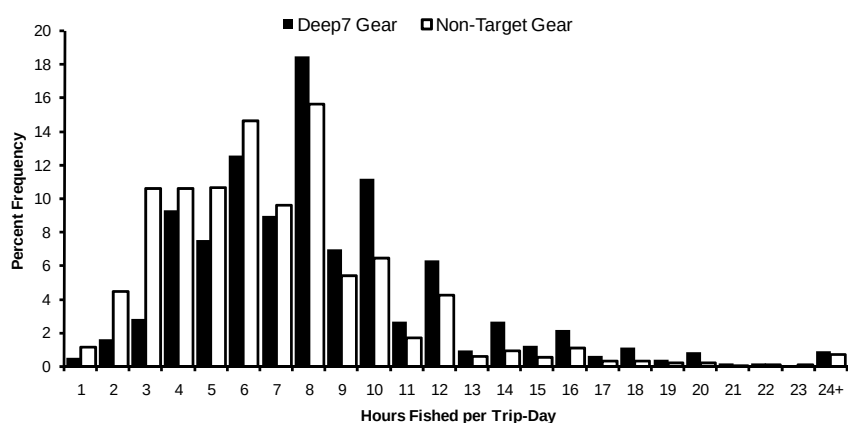
(C)		NWHI		MHI Commercial		MHI Chartered	
gear category		trip-days	min-max hrs	trip-days	min-max hrs	trip-days	min-max hr
HLD		1919	1-24	10901	0.08-48	76	0.5-16
KKL		0		49	1-16	1	3
HLS		1	11	2721	0.5-40	25	1-8
HLO		329	1-16	2946	0.05-33	849	1-18
NET		0		554	0.25-14	0	
SPR		0		295	0.5-10	0	
TRP		0		219	1-528	0	
OTH		0		13	1-12	0	

Table 3.2.1.- Frequency of observed trips for the NWHI commercial, MHI commercial, and MHI charter bottomfish fleets from 2002-2005: (A) number of areas fished per trip days; (B) number of gears fished per trip-day; and, (C) types of gears fished during trip days.

(A) NWHI



(B) MHI Commercial



(C) MHI Charter

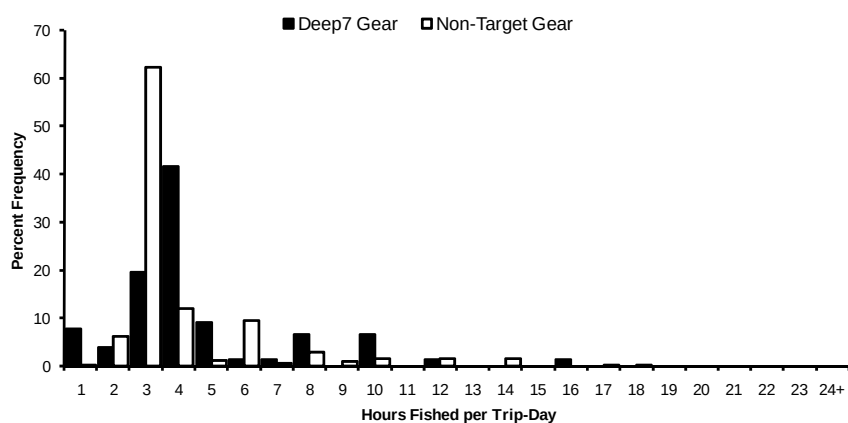


Figure 3.2.7.- Distribution of hours fished per trip-day from the 2002-2005 FRS database for trips to: (A) Northwestern Hawaiian Islands (NWHI); (B) Main Hawaiian Islands (MHI) commercial; and, (C) MHI Charter. From these observations, the cutoff of maximum hours in a day was determined to be 16 hours.

(A) 2 Gears Fishing MHI

Number of trip-area records	p(HLD)	p(HLO)	p(HLS)	p(NET)	p(SPR)
1016	0.63	0.37			
340	0.47		0.53		
68	0.32			0.68	
7	0.47				0.53
438		0.43	0.57		
5		0.48		0.52	
8		0.59			0.41
18			0.45	0.55	
19			0.38		0.62
21				0.55	0.45

(B) 3 Gears Fishing MHI

Number of trip-area records	p(HLD)	p(HLO)	p(HLS)
59	0.39	0.26	0.35

(C) 2 Gears Fishing

Number of trip-area records	p(HLD)	p(HLO)
213	0.76	0.24

Table 3.2.2.- Proportion hours fished by multiple gears in same area for: (A) MHI – 2 gears fishing; (B) MHI – 3 gears fishing; and (C) NWHI – 2 gears fishing. Refer to Table 2.4.16 for assigned gear categories.

3.3 Procedures for Missing/Problem Area, Port & License Variables

Table 3.3.1.- Total bottomfish records relative to those missing information on either trip_id and/or trip-distance. Area and port codes are components of trip-distance computation.

Year	Total DB records	Missing License	Missing Trip-Distance	Missing/Problem Area	Missing/Problem Port_Land
1948	19468	145	9991	1	9982
1949	20354	297	144	53	89
1950	17439	98	69	36	2
1951	14219	102	26	1	1
1952	15873	522	92	90	1
1953	17434	4630	309	270	5
1954	13895	216	2	0	0
1955	15293	962	0	0	0
1956	15927	1387	16	16	0
1957	12866	2303	2	1	0
1958	15792	4614	33	0	0
1959	12751	2384	14	1	0
1960	11461	1077	0	0	0
1961	9147	962	5	3	0
1962	11022	2813	303	299	0
1963	12250	1493	0	0	0
1964	9820	1125	63	36	0
1965	8918	1219	11	11	0
1966	10469	1596	27	0	0
1967	9180	1061	0	0	0
1968	9994	1272	8	0	0
1969	9634	208	3	0	0
1970	9228	229	0	0	0
1971	13452	982	1	1	0
1972	13544	416	3	1	0
1973	14771	330	8	0	6
1974	15262	686	14	8	6
1975	18618	186	14	0	14
1976	22297	52	24	0	23
1977	26115	87	112	5	46
1978	19997	48	7	1	0
1979	24214	101	10	4	6
1980	33919	346	1	1	0
1981	38330	1321	28	14	14
1982	36818	1407	37	2	31
1983	40498	1629	259	137	57
1984	38989	1920	200	162	19
1985	36163	988	816	682	9
1986	31637	293	415	404	0
1987	31582	943	658	656	0
1988	38815	672	309	256	0
1989	37069	338	452	452	0
1990	29527	153	65	65	0
1991	26049	37	146	46	0
1992	30256	90	274	41	12
1993	25648	228	100	16	28
1994	28720	45	70	63	7
1995	30560	3	69	60	0
1996	30212	0	119	44	75
1997	32614	10	55	55	24
1998	28353	0	35	29	0
1999	26565	0	22	22	0
2000	27296	0	14	10	0
2001	23051	0	23	23	0
2002	19724	0	27	27	0
2003	17476	0	50	27	23
2004	18142	0	62	58	4
2005	15233	0	145	37	108
Totals	1243950	44026	15762	4227	10592

Table 3.3.2.- Remaining problem records after the various correction procedures, and associated catches (pounds) for the 12 target bottomfish species. These records were removed from the bottomfish database.

Year	Problem Records	Species Codes												Deep7 Catch	Bottomfish Catch
		15	16	17	19	20	21	22	58	97	114	200	205		
1948	9990	35571	144972	14522	155493	60039	33370	32530	15279	217	0	0	0	286982	491993
1949	52	25	876	132	168	246	149	373	0	0	0	0	0	847	1969
1950	54	146	422	238	108	439	894	892	0	0	0	0	0	2278	3139
1951	3	0	0	0	0	12	0	0	0	0	0	0	0	0	12
1952	90	505	549	12	303	1487	164	0	0	4	0	0	0	988	3024
1953	262	1232	679	79	3041	153	309	400	86	104	0	0	0	5251	6083
1954	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1955	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1956	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1957	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1958	33	0	152	0	0	107	0	0	0	0	0	0	0	0	259
1959	11	102	221	342	0	92	65	391	0	4	0	0	0	904	1217
1960	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1961	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1962	303	876	4321	1303	9198	1639	1151	2567	157	100	0	0	0	15352	21312
1963	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1964	48	912	2124	492	493	103	739	9603	0	95	0	0	0	12334	14561
1965	11	1603	1338	13	76	397	841	138	0	0	0	0	0	2671	4406
1966	7	219	79	0	0	0	20	0	0	0	0	0	0	239	318
1967	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1968	7	0	50	0	181	10	0	0	0	0	0	0	0	181	241
1969	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1970	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1971	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1972	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1973	6	0	0	0	0	0	0	0	0	0	24	0	0	0	24
1974	8	0	0	0	0	0	0	0	0	0	348	0	0	0	348
1975	11	0	0	0	0	0	0	0	0	0	466	0	0	0	466
1976	24	0	0	0	0	0	0	0	0	0	1070	0	0	0	1070
1977	110	415	212	0	9	331	159	0	0	30	121	0	0	613	1277
1978	6	0	0	0	0	0	0	0	0	0	14	0	0	0	14
1979	9	0	0	0	0	203	0	0	0	0	4	0	0	0	207
1980	1	0	0	0	0	36	0	0	0	0	0	0	0	0	36
1981	28	0	38	0	0	0	0	0	10	0	336	0	35	10	419
1982	37	0	15	0	0	12	0	0	0	0	21	0	0	0	48
1983	223	116	0	19	194	534	38	48	20	0	925	0	14	435	1908
1984	143	1862	114	118	750	6353	1340	5663	15	3	329	42	205	9751	16794
1985	794	4563	1017	652	13083	4890	3233	6151	6	255	2710	2695	3	27943	39258
1986	398	1703	1711	111	1011	46	571	499	3	64	271	1754	205	3962	7949
1987	654	3270	324	60	9302	6	260	1540	0	2	200	5500	0	14434	20464
1988	292	247	166	112	128	1304	869	510	562	44	289	78	81	2472	4390
1989	435	121	0	79	534	74	1188	1596	9	9	14	0	0	3536	3624
1990	61	248	0	0	743	5596	294	86	88	4	30	869	0	1463	7958
1991	137	158	0	294	1069	189	190	137	16	27	6	121	0	1891	2207
1992	274	2101	7	245	5385	3615	354	111	268	225	4	2282	0	8689	14597
1993	99	35	0	99	621	209	197	426	13	40	5	16	0	1431	1661
1994	68	37	3	1199	671	11	812	251	76	373	98	123	206	3419	3860
1995	58	63	0	105	177	7	330	158	73	98	26	232	685	1004	1954
1996	44	2117	0	258	1498	375	774	152	0	121	0	899	262	4920	6456
1997	48	0	0	542	0	0	371	0	19	24	6	0	0	956	962
1998	35	14	0	108	102	43	0	0	0	134	0	64	230	358	695
1999	19	0	0	170	6	0	0	0	0	13	0	0	84	189	273
2000	10	0	0	72	0	0	0	0	0	0	0	0	106	72	178
2001	22	0	0	95	0	644	28	0	0	77	0	166	0	200	1010
2002	25	0	0	125	112	149	0	0	6	31	0	0	115	274	538
2003	26	196	0	31	42	120	65	63	0	12	0	617	56	409	1202
2004	38	0	0	87	104	200	10	0	0	2	31	36	0	203	470
2005	36	25	41	0	246	31	36	431	82	0	17	0	0	820	908
Totals	15,074	58,482	159,431	21,714	204,848	89,702	48,821	64,716	16,788	2,112	7,364	15,494	2,287	417,481	691,759

3.4 Species & Gear-specific Catch and Effort Database

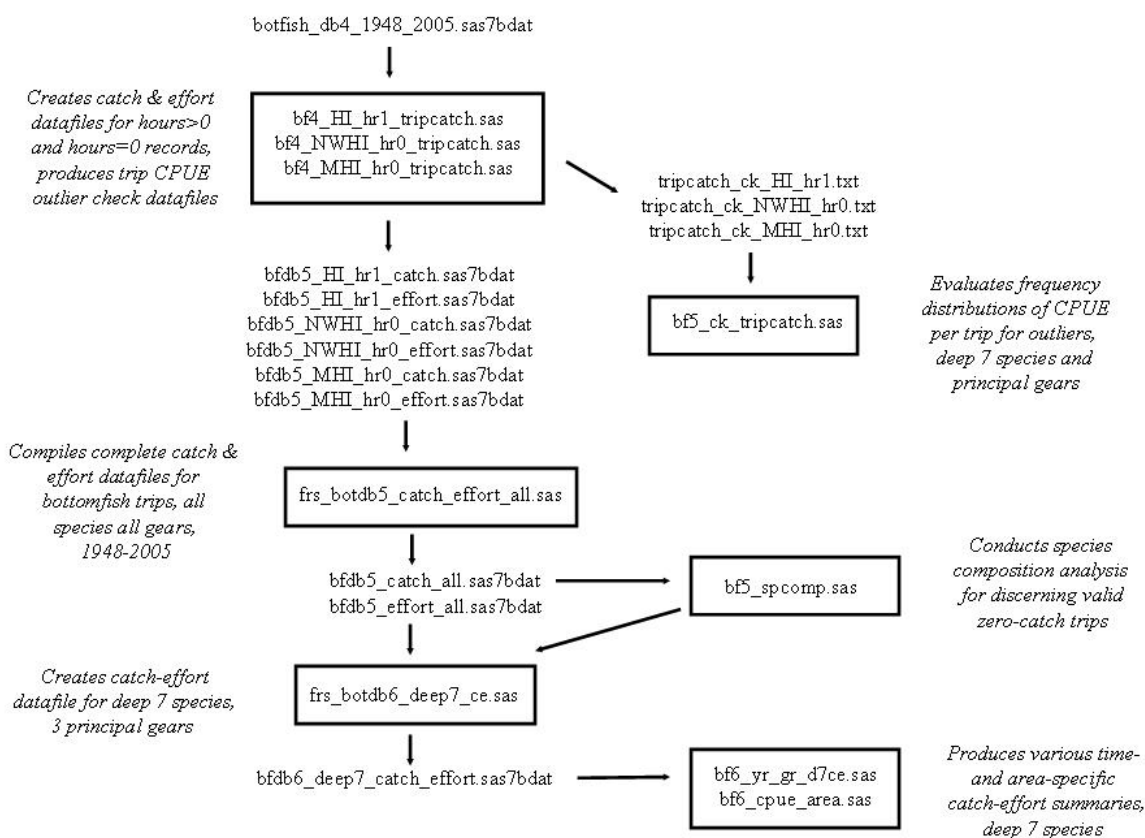


Figure 3.4.0.- Flow of procedures to compute catch-and-effort data files for the bottomfish complex.

3.4.1 Estimation of Fishing Effort (Trip Day)

Table 3.4.1.- Distribution of Deep7 bottomfish catches (pounds per trip-day) of the Deep7 species from the Northwestern Hawaiian Islands (NWHI) and main Hawaiian Islands (MHI) for the period 1948-2005. Data with (hours=1,uncorrected) evaluated from the 2002-2005 period, Data with (hours=0, and either uncorrected or corrected) was from 2002-2005 period, but trip days were estimated from the reported area-fished and port-of-landing.

Catch (lbs trip-day ⁻¹)	NWHI		MHI		NWHI		NWHI		MHI		MHI	
	hrs=1, uncorrected		hrs=1, uncorrected		hrs= 0, lic = ok		hours = 0, lic = no		hours = 0, lic = ok		hours = 0, lic = no	
	freq	%freq	freq	%freq	freq	%freq	freq	%freq	freq	%freq	freq	%freq
250	1188	69.5%	9194	98.2%	1259	57.1%	77	64.7%	142941	96.8%	3388	92.5%
500	335	19.6%	139	1.5%	577	26.2%	29	24.4%	3479	2.4%	194	5.3%
750	123	7.2%	21	0.2%	238	10.8%	11	9.2%	785	0.5%	53	1.4%
1000	44	2.6%	7	0.1%	78	3.5%	1	0.8%	251	0.2%	14	0.4%
1250	8	0.5%	2	0.0%	30	1.4%	0	0.0%	84	0.1%	6	0.2%
1500	9	0.5%	1	0.0%	16	0.7%	1	0.8%	38	0.0%	1	0.0%
1750	1	0.1%	1	0.0%	2	0.1%	0	0.0%	33	0.0%	3	0.1%
2000	1	0.1%	0	0.0%	3	0.1%	0	0.0%	8	0.0%	1	0.0%
2250	0	0.0%	0	0.0%	0	0.0%	0	0.0%	6	0.0%	0	0.0%
2500	0	0.0%	1	0.0%	0	0.0%	0	0.0%	5	0.0%	0	0.0%
2750	0	0.0%	0	0.0%	1	0.0%	0	0.0%	2	0.0%	0	0.0%
3000	0	0.0%	0	0.0%	0	0.0%	0	0.0%	1	0.0%	0	0.0%
3250	0	0.0%	0	0.0%	0	0.0%	0	0.0%	2	0.0%	0	0.0%
3500	0	0.0%	0	0.0%	0	0.0%	0	0.0%	1	0.0%	0	0.0%
3750	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	1	0.0%
4000	0	0.0%	0	0.0%	0	0.0%	0	0.0%	2	0.0%	0	0.0%
4250	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
4500	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
4750	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
5000	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
5250	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
5500	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
5750	0	0.0%	0	0.0%	0	0.0%	0	0.0%	1	0.0%	0	0.0%
Totals	1,709	100.0%	9,366	100.0%	2,204	100.0%	119	100.0%	147,639	100.0%	3,661	100.0%

Table 3.4.2.- Time series of Deep7 bottomfish 1948-2005 catch (yield in weight, Yw, in pounds) and fishing effort (trip days) for the combined gear category HLD (gears 3 and 5).

Year	Effort	Yw 15	Yw 17	Yw 19	Yw 21	Yw 22	Yw 58	Yw 97	Yw Deep7	YwD7/f
1948	1577.0	49017	16080	68956	16285	18280	1520	68	171783	108.93
1949	3477.6	68509	32358	237843	47637	78215	5552	270	473862	136.26
1950	3174.1	80269	31513	222239	21164	80397	3035	2028	443819	139.83
1951	2410.2	67859	40254	254113	29864	55021	6920	3035	459476	190.64
1952	2670.4	96954	30487	307423	47289	44973	6345	2853	538994	201.84
1953	1980.4	74045	32031	219300	40298	56502	3033	2688	429877	217.07
1954	1436.6	40134	27710	134865	25980	65788	1793	2558	300265	209.01
1955	1422.1	64739	26556	159979	29476	58405	1800	5213	347590	244.42
1956	1588.0	42347	29393	119196	44568	86769	8378	3423	335662	211.37
1957	1498.3	31215	17663	116948	29760	55049	2705	2202	257040	171.55
1958	1358.1	26456	15743	106784	20433	62933	3056	1713	238476	175.60
1959	1442.6	31406	20836	87066	23060	49257	1168	1402	215638	149.48
1960	902.4	21854	15626	85869	16513	40302	1127	914	183107	202.91
1961	892.3	20296	18581	90752	12191	42741	1425	679	187557	210.20
1962	1105.4	21689	16651	91697	21500	59387	1808	749	214586	194.13
1963	1367.4	32291	23482	114390	25997	52889	2218	2222	254856	186.38
1964	1086.6	23216	15314	104531	22229	47844	973	1514	216708	199.44
1965	1081.9	19584	13990	82667	18455	65700	1381	464	203323	187.93
1966	1412.4	32876	13060	96959	19550	69681	2289	1016	236843	167.69
1967	1436.7	23390	7009	109045	16038	58114	2604	664	218301	151.95
1968	1464.5	21115	5120	90226	22807	69850	4014	751	215348	147.05
1969	1293.9	21199	5574	89332	16594	48454	4606	1361	188414	145.62
1970	1167.0	40473	4446	52666	16664	38174	1329	1680	156599	134.19
1971	1612.0	49520	6953	82379	21815	47893	2089	2055	214316	132.95
1972	1824.8	36881	7104	117644	30205	50220	5024	1930	250833	137.46
1973	2144.3	50672	6453	131870	22035	42668	6489	2599	264930	123.55
1974	1909.0	39285	5759	110949	26360	40524	4241	1511	230538	120.76
1975	2408.6	59043	8786	157661	24192	65260	9886	1857	329094	136.63
1976	2791.7	56330	8923	118557	32386	88978	9327	1342	318635	114.14
1977	3087.3	45780	7104	142998	30628	66294	7784	1512	305187	98.85
1978	1816.9	72610	8506	171660	33775	60937	8437	3543	361285	198.85
1979	2851.8	64008	7735	202005	20884	48040	15480	3042	364046	127.65
1980	3420.6	54513	6395	226982	17694	33491	16531	2130	361157	105.58
1981	3781.7	30339	6679	188671	21949	54613	25713	1588	333334	88.14
1982	4457.8	44603	10107	197970	25200	61948	28271	1926	374483	84.01
1983	5163.0	43920	15537	220051	38257	92455	23111	3204	441698	85.55
1984	5864.9	57139	17770	347239	35627	109358	21103	4775	598876	102.11
1985	6497.9	94303	27848	289474	66515	232445	30073	7031	754187	116.07
1986	5899.2	100683	22870	297026	56072	235393	21667	5178	744788	126.25
1987	6003.8	71638	29557	401530	54389	194997	37406	4945	800466	133.33
1988	6603.4	33436	12024	351587	50112	159470	50390	2643	666265	100.90
1989	6802.1	67119	14578	371065	46937	163568	43959	2967	716995	105.41
1990	4963.6	74061	24886	221863	55147	130426	21738	5990	539075	108.61
1991	4557.7	65744	25024	215951	46699	128517	12909	6739	506141	111.05
1992	4919.9	53822	27677	251129	35338	87453	17832	5545	483716	98.32
1993	3934.0	66542	18529	295528	33175	101750	10266	7298	537022	136.51
1994	4781.4	79173	26365	322859	36676	101764	11098	8391	591107	123.63
1995	5218.7	69087	24627	277022	39648	122995	14474	6491	559563	107.22
1996	4769.8	54110	30997	224411	46238	97144	8884	8890	475444	99.68
1997	4974.7	73434	29286	259352	41611	126277	11903	8303	555141	111.59
1998	4655.9	84011	25181	226582	41762	109783	8406	8442	508823	109.29
1999	4071.7	72028	15618	208966	37824	159436	9114	6375	513433	126.10
2000	4672.9	41076	20826	206657	43003	168039	11116	5647	501037	107.22
2001	3588.4	36802	15262	157903	35499	131086	10387	5092	395619	110.25
2002	3434.8	37828	14583	144909	26329	124200	8367	4189	363840	105.93
2003	3243.7	45591	11836	136547	23917	123698	8037	3975	356845	110.01
2004	3208.9	44283	9275	133941	29212	155322	6024	4395	385661	120.18
2005	2790.4	49368	8681	103718	29892	107495	5301	6197	313442	112.33

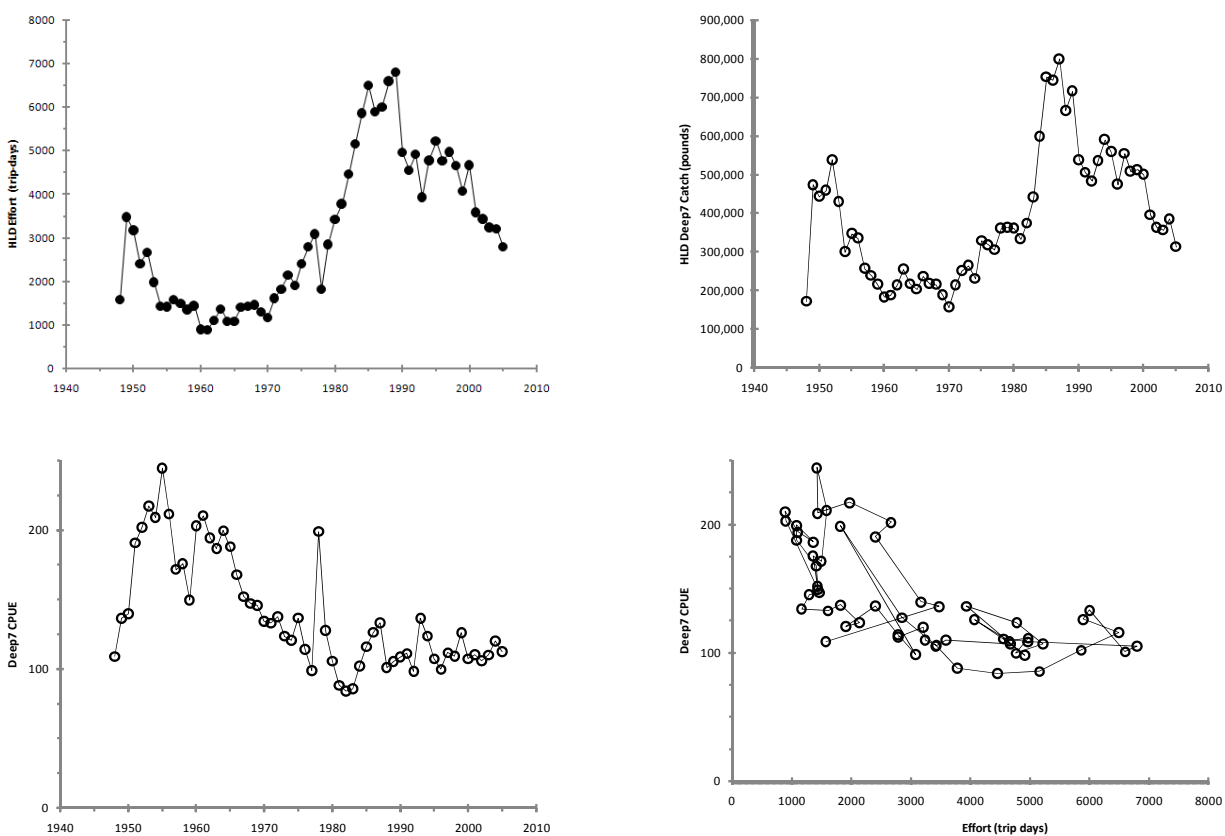


Figure 3.4.1.- Deep7 catch and effort data for HLD gear from 1948-2005: (A) time-series of effort (trip-days); (B) time-series of catch (pounds); (C) timeseries of catch-per-unit-effort (CPUE); and, (D) CPUE dependent on effort.

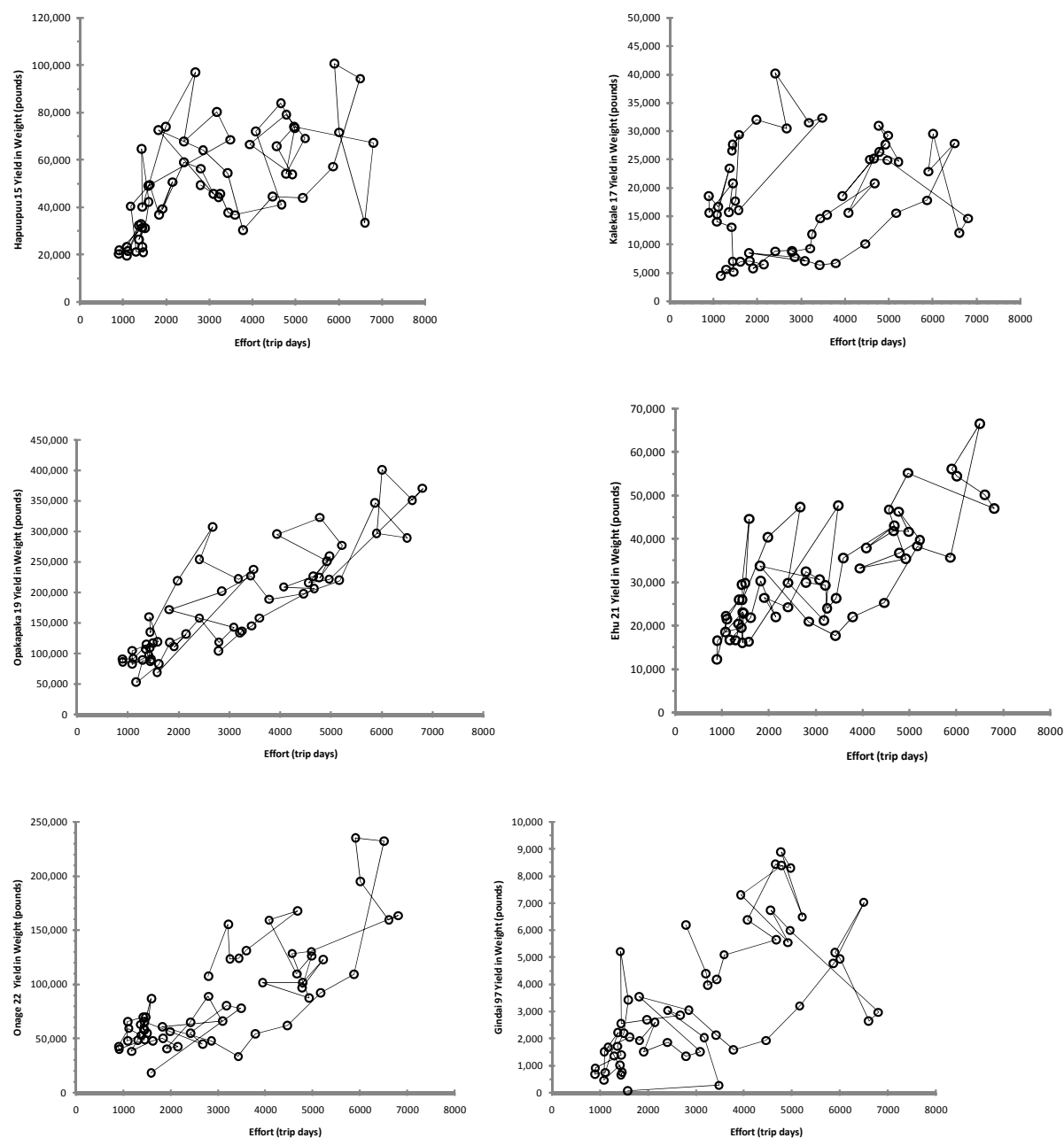
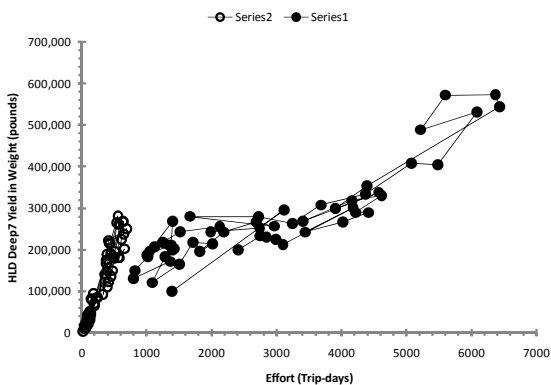
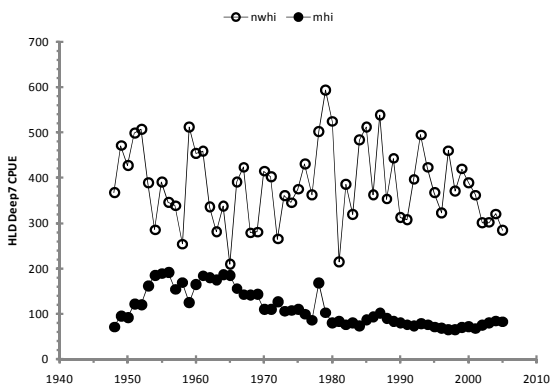
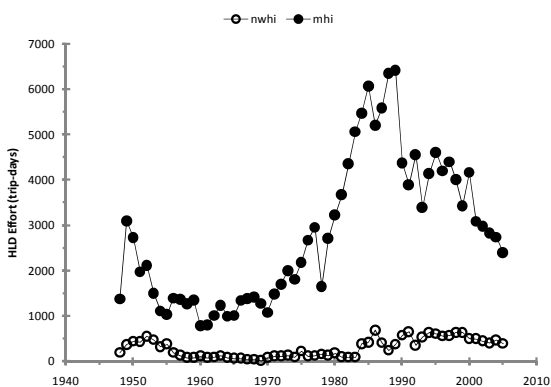


Figure 3.4.2.- Production curves for each species of the Deep7 complex.

Table 3.4.3- Deep7 bottomfish 1948-2005 catch and effort time-series for gear category HLS.

Year	Effort	Yw 15	Yw 17	Yw 19	Yw 21	Yw 22	Yw 58	Yw 97	Deep7	Ywlf
1948	255.5	178	1471	5561	2341	1227	417	0	11195	1.6
1949	475.0	1019	3767	15936	1968	1058	336	12	24096	0.7
1950	274.0	823	1537	9255	206	506	148	294	12769	0.5
1951	328.0	662	2601	5041	2095	1528	12	715	12654	0.0
1952	421.6	1397	5200	8005	9478	1285	101	744	26210	0.2
1953	563.1	1600	5474	9536	6950	1589	371	105	25625	0.7
1954	641.5	1795	5509	14892	7323	2087	148	172	31926	0.2
1955	887.7	3479	7169	19752	9632	6110	320	657	47119	0.4
1956	881.6	2405	5413	34171	5337	2740	501	258	50825	0.6
1957	535.8	1028	2975	9991	2541	1709	382	68	18694	0.7
1958	668.4	1756	3448	14148	2859	1525	309	338	24383	0.5
1959	163.6	336	735	929	192	97	11	18	2318	0.1
1960	212.7	513	570	1758	1444	550	50	164	5049	0.2
1961	186.5	230	505	1892	631	241	2	15	3516	0.0
1962	187.3	347	340	2523	937	631	81	40	4899	0.4
1963	315.4	427	733	5877	2054	1030	71	82	10274	0.2
1964	70.4	36	205	335	344	96	13	3	1032	0.2
1965	20.0	20	7	210	97	52	1	0	387	0.1
1966	7.0	0	10	486	2	54	70	3	625	10.0
1967	3.0	0	0	19	8	0	0	0	27	0.0
1968	6.5	10	4	0	72	96	0	6	188	0.0
1969	7.0	0	23	14	8	0	116	2	163	16.6
1970	11.5	0	20	72	2	17	13	3	127	1.1
1971	26.3	49	11	24	50	40	29	8	211	1.1
1972	12.5	31	4	27	29	41	32	3	167	2.6
1973	8.3	4	9	20	8	2	0	0	43	0.0
1974	15.2	26	36	159	52	2	40	32	347	2.6
1975	122.7	381	66	2121	267	852	654	89	4430	5.3
1976	133.1	825	188	1495	990	1227	203	138	5066	1.5
1977	179.4	673	173	2095	392	424	1166	175	5098	6.5
1978	267.0	935	1065	4549	1502	1414	1731	147	11343	6.5
1979	122.9	322	144	939	153	22	814	167	2561	6.6
1980	73.9	39	30	178	56	33	436	115	887	5.9
1981	72.9	8	33	391	56	70	303	94	955	4.2
1982	113.7	135	44	174	156	38	387	79	1013	3.4
1983	178.8	320	151	1187	293	722	379	212	3264	2.1
1984	48.2	10	26	256	46	458	175	80	1051	3.6
1985	91.8	18	110	461	37	1225	776	4	2631	8.5
1986	110.6	78	167	775	38	0	552	14	1624	5.0
1987	47.6	37	82	109	57	6	307	5	603	6.4
1988	50.9	102	107	453	55	107	164	0	988	3.2
1989	1.3	13	0	0	0	0	0	0	13	0.0
1990	2.5	26	0	273	0	0	0	0	299	0.0
1991	3.2	51	0	0	0	0	0	0	51	0.0
1993	2.0	30	0	0	0	0	0	0	30	0.0
1994	4.5	36	0	0	0	0	0	0	36	0.0
1995	14.0	10	6	17	0	0	169	0	202	12.1
1996	22.2	24	3	35	0	0	151	0	213	6.8
1997	43.7	29	32	64	13	5	435	0	578	10.0
1998	30.8	5	13	47	2	0	239	0	306	7.8
1999	47.9	0	16	188	2	0	352	0	558	7.3
2000	36.7	6	27	79	73	0	273	0	458	7.4
2001	27.8	22	21	124	56	15	65	1	304	2.3
2002	50.6	34	24	86	147	71	273	4	639	5.4
2003	73.4	130	14	683	85	32	507	32	1483	6.9
2004	81.5	52	38	654	135	81	617	47	1624	7.6
2005	101.6	124	36	900	116	184	908	57	2325	8.9

Figure 3.4.3.- Regional separation of NWHI and MHI catch and effort for the Deep7 bottomfish species.



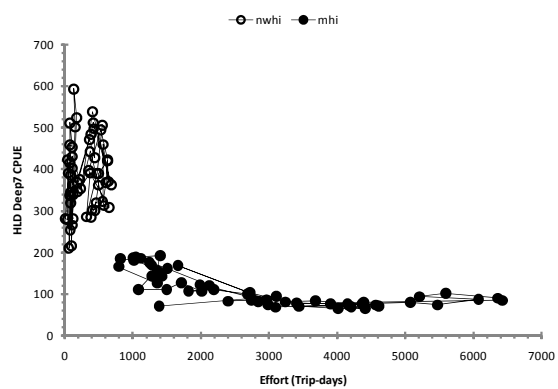
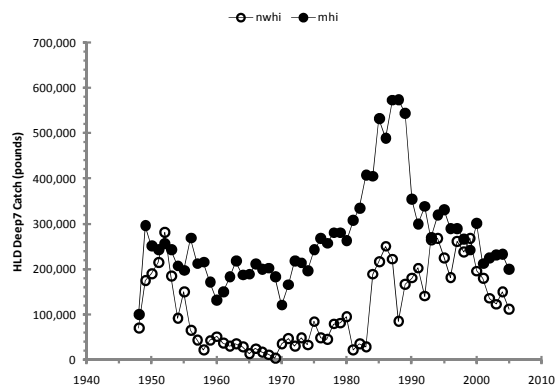


Figure 3.4.4.- Regional separation of NWHI and MHI catch and effort for Hapuupuu (species code: 15)

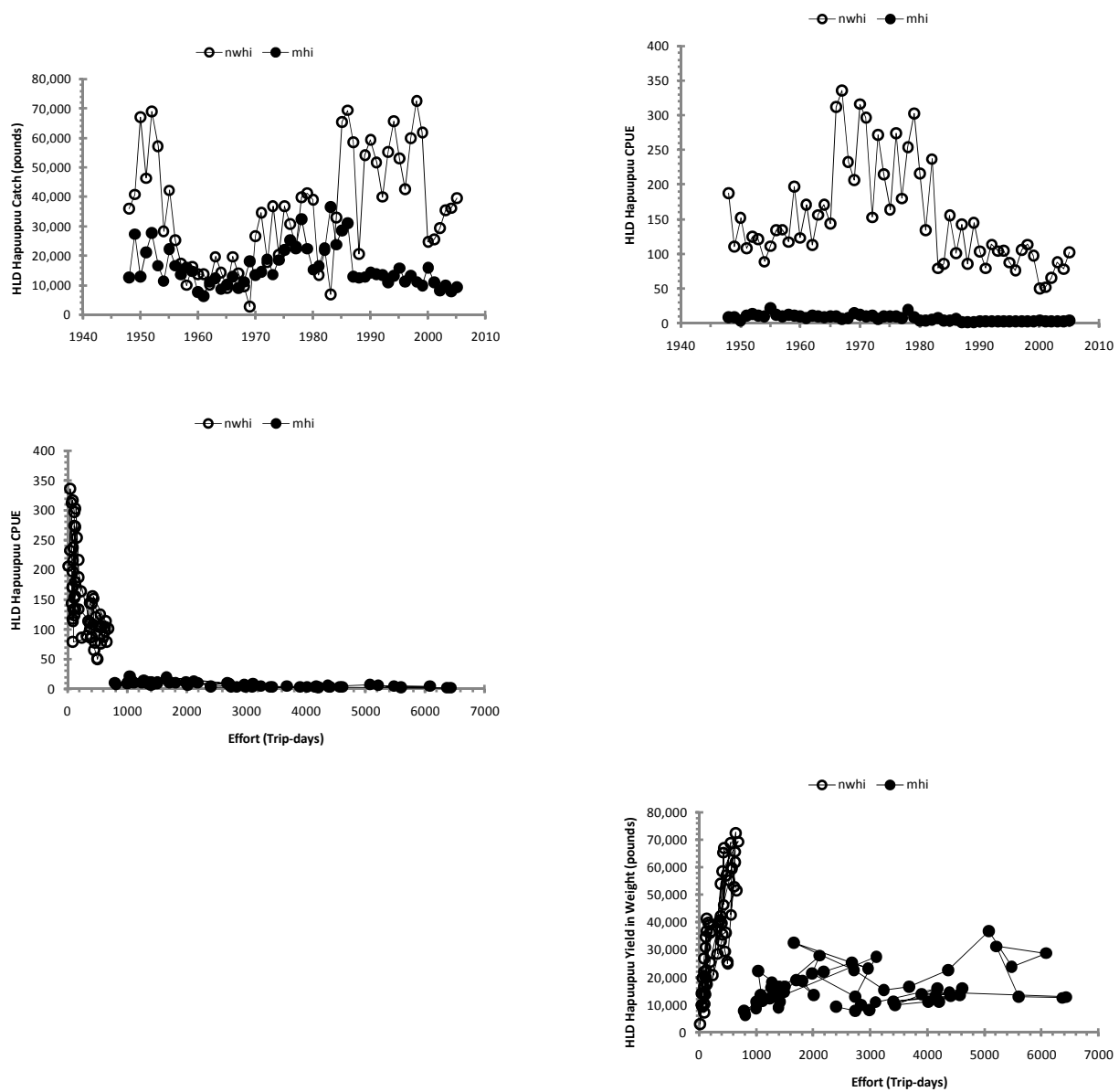
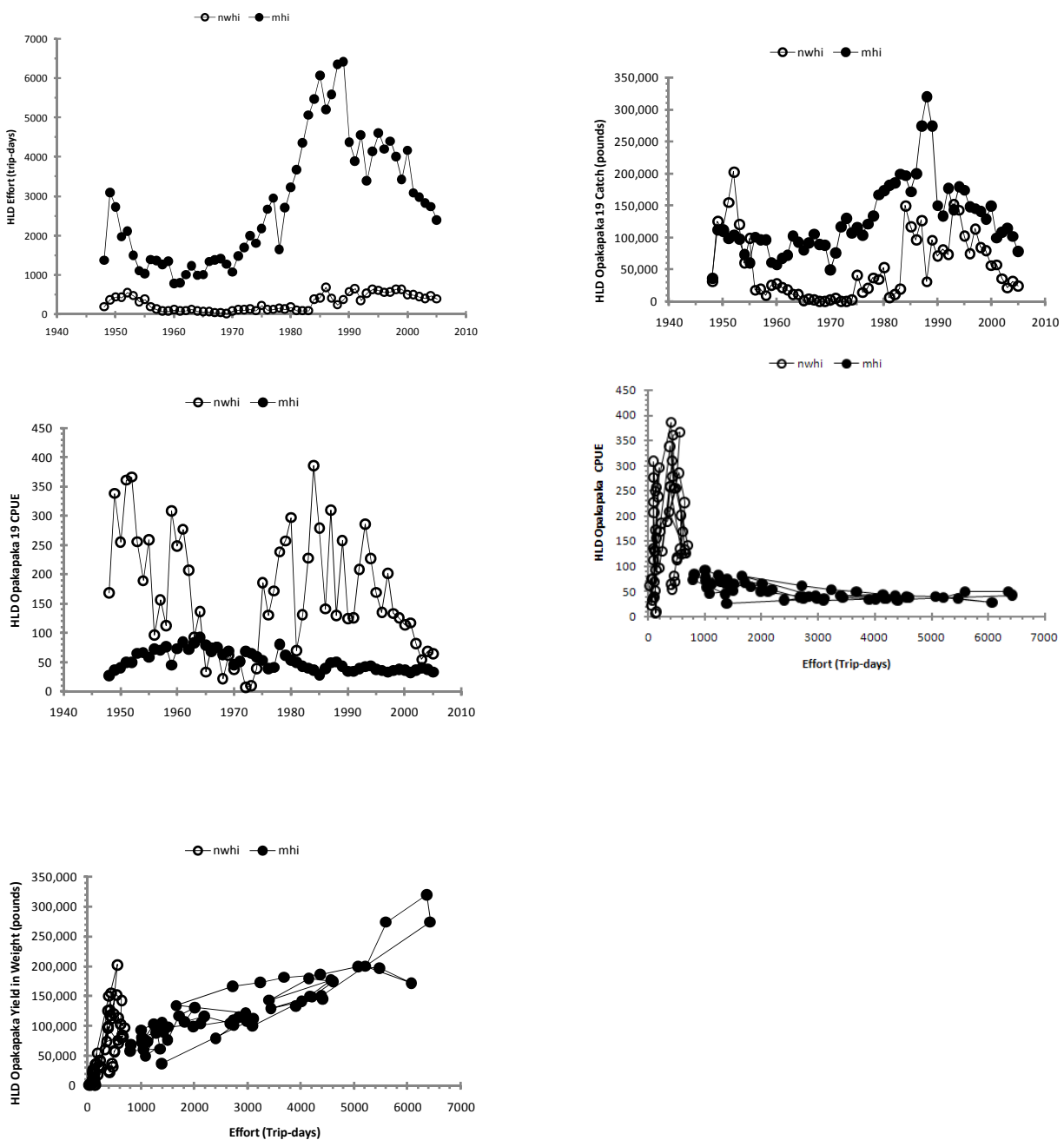
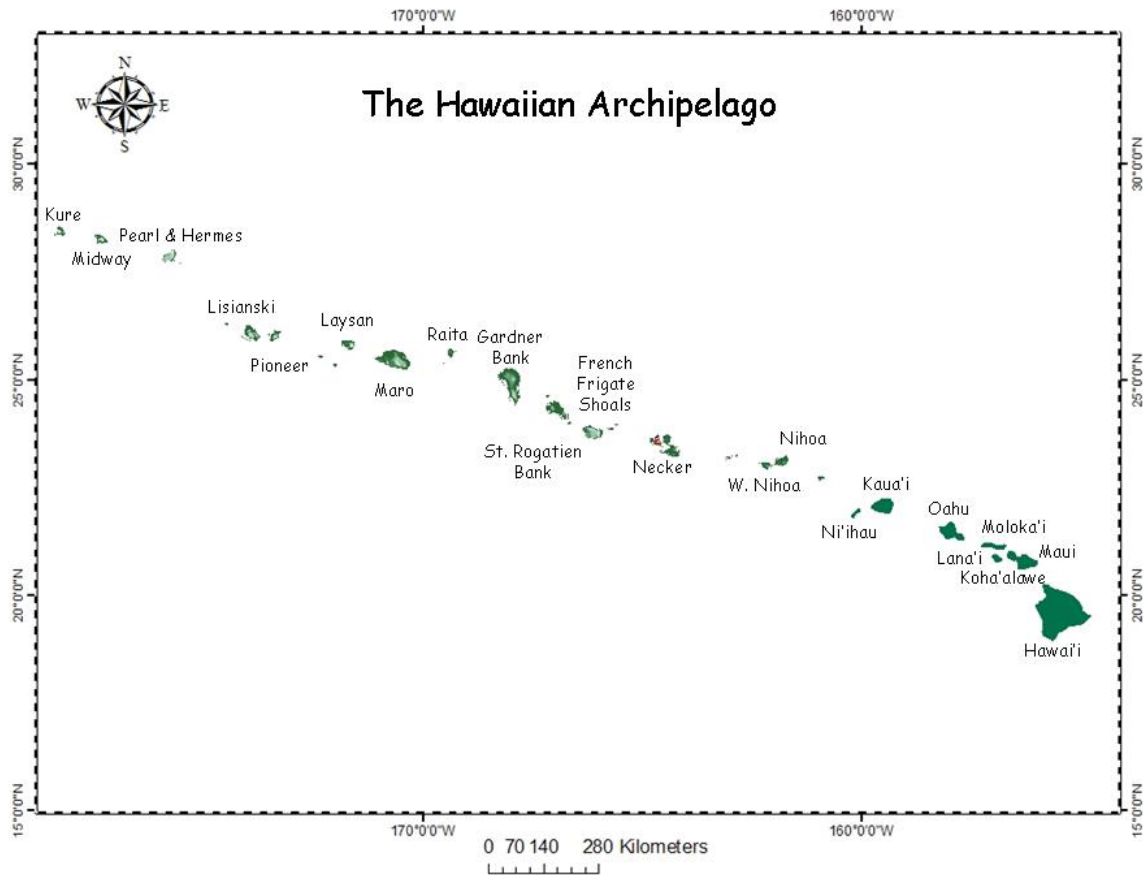


Figure 3.4.5.- Regional separation of NWHI and MHI catch and effort (HLD) for Opakapaka (species code: 19)



4.0 Spatial-Temporal Distributions of Bottomfish Effort and Catches

Figure 4.0.1.- Hawaiian Archipelago where bottomfish have been captured from 1948-2005.



4.1 Temporal Distribution Effort (Trip-Days) and Areas Fished

Table 4.1.1.- Distribution of areas fished (grid cells in DLNR map) and maximum number of fishing trip-days per area for 5-years stanzas between 1948-2005.

Period	Max Trips	Areas Fished	Total Trips
1948 - 1955	1,988	121	18,148
1955 - 1960	852	77	6,789
1961 - 1965	681	78	5,533
1966 - 1970	1,108	74	6,775
1971 - 1975	1,255	86	9,899
1976 - 1980	1,488	122	13,968
1981 - 1985	2,193	125	25,765
1986 - 1990	2,858	150	30,272
1991 - 1995	2,043	199	23,412
1996 - 2000	1,989	159	23,145
2001 - 2005	1,830	159	16,266

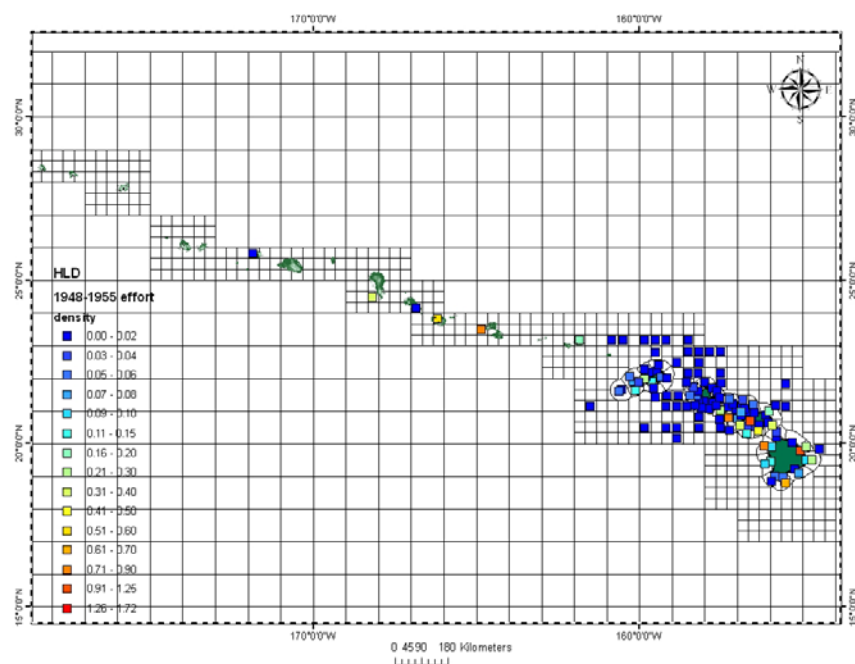
Average **1,662**

The spatial GIS mapping of nominal fishing effort (trip-days), the data for each map area cell were scaled the the average maximum trips for the period 1948-2005 (**Table 3.4.1**). Note that the same cells received up to 2,858 unique trip-days during any five-year stanza, as did the maximum trip-days fished. The maximum effort occurred during the 1986-1990 period. The maximum area fished occurred during the 1991-1995 period.

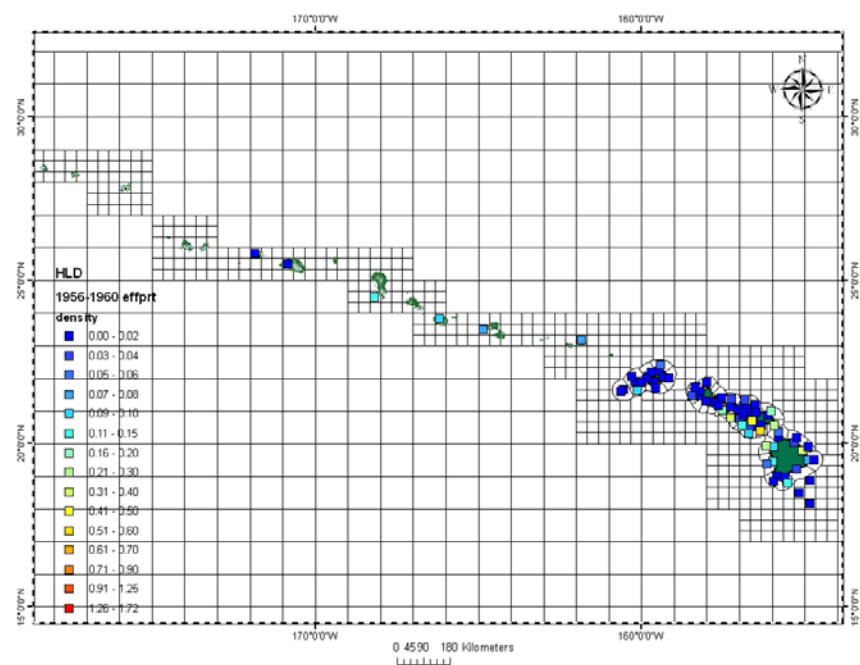
4.2 Spatial Mapping of Estimated Trip-Days by Bottomfish Fleet and Gear

Figure 4.2.1.- Spatial distribution of nominal bottomfish fishing effort (HLD trip-days) for 5-year stanzas from 1948-2005.

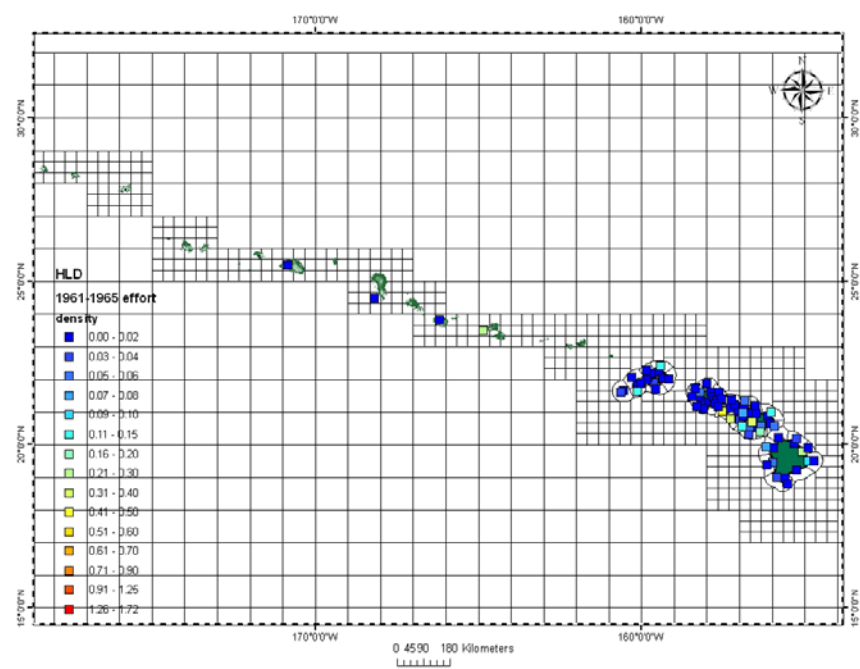
(A) 1948-1955



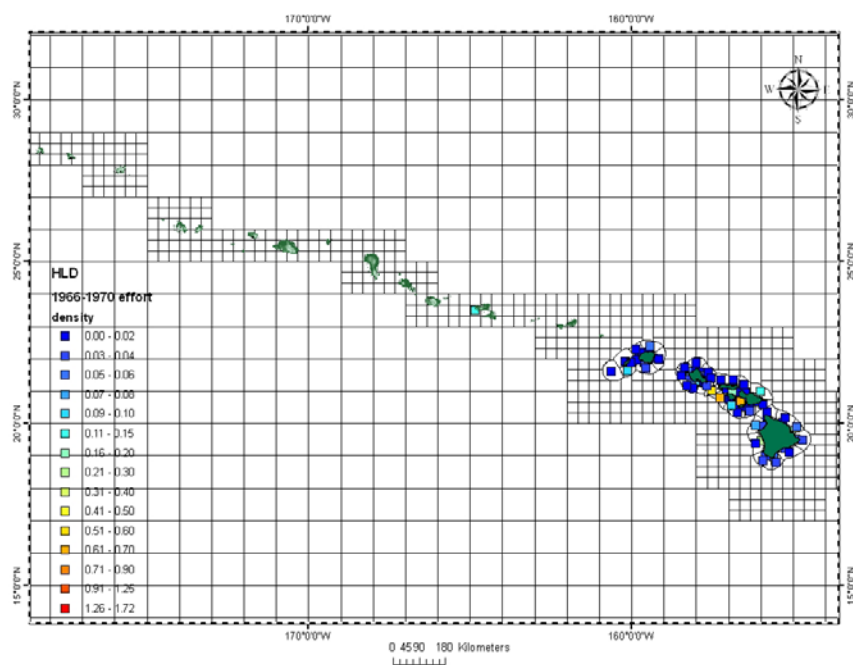
(B) 1956-1960



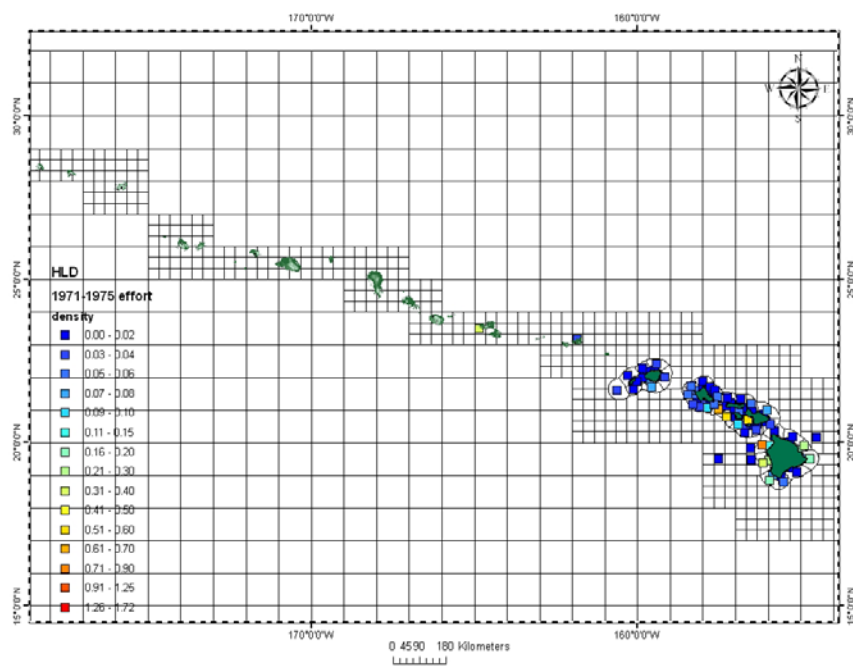
(C) 1961-1965



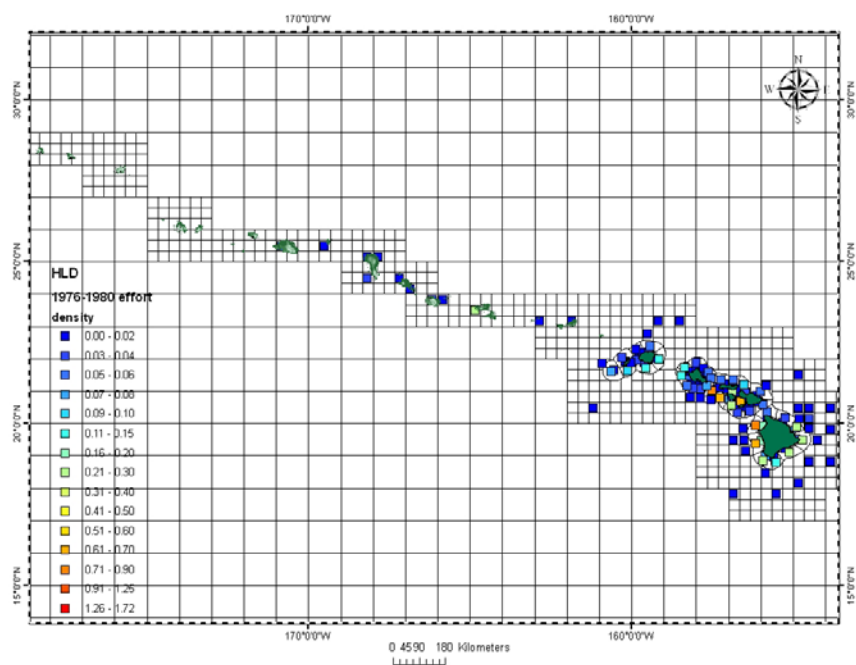
(D) 1966-1970



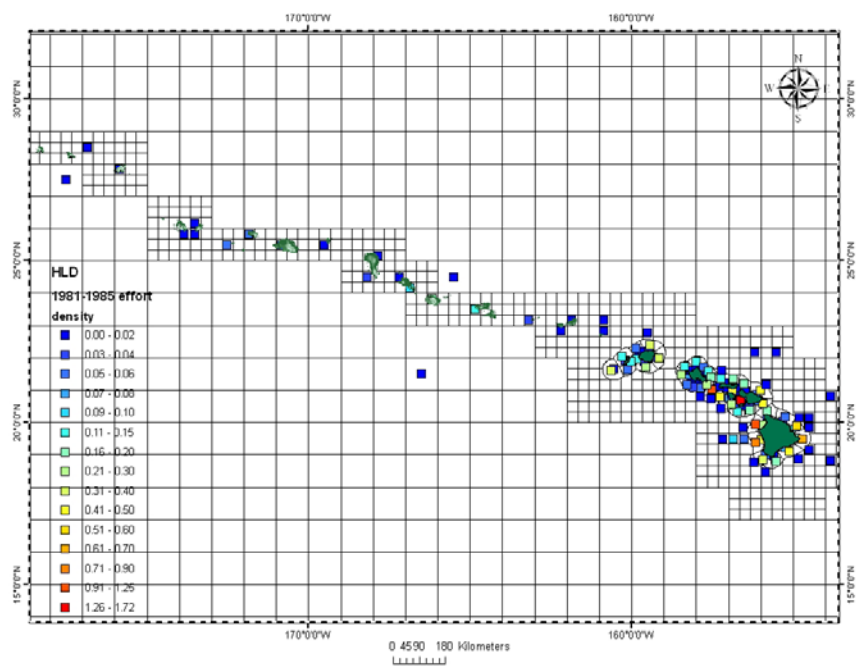
(E) 1971-1975



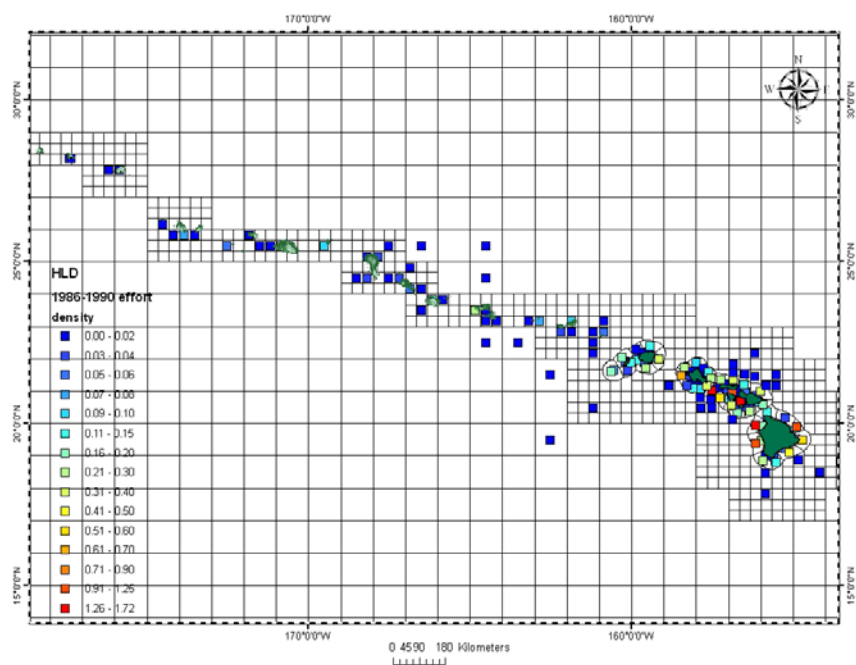
(F) 1976-1980



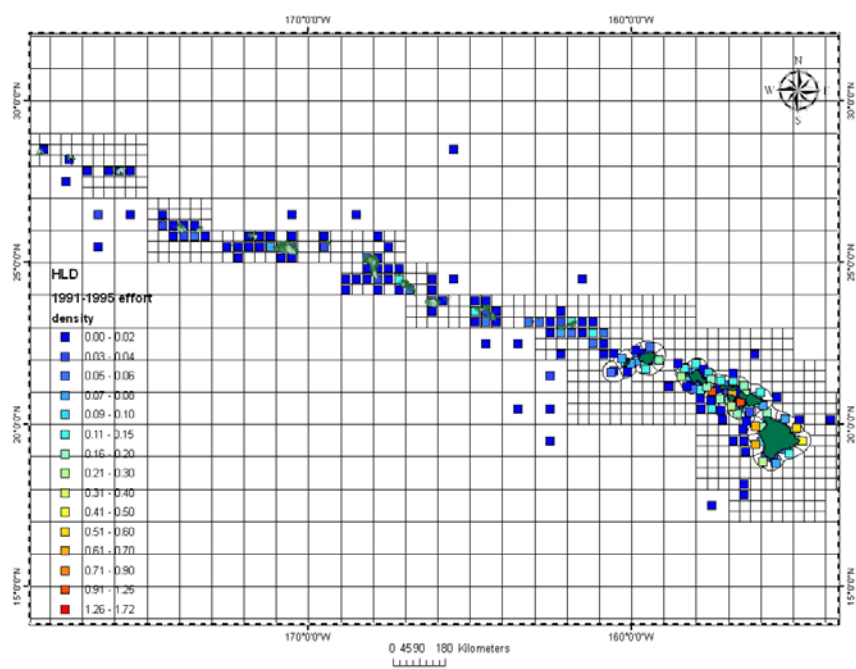
(G) 1981-1985



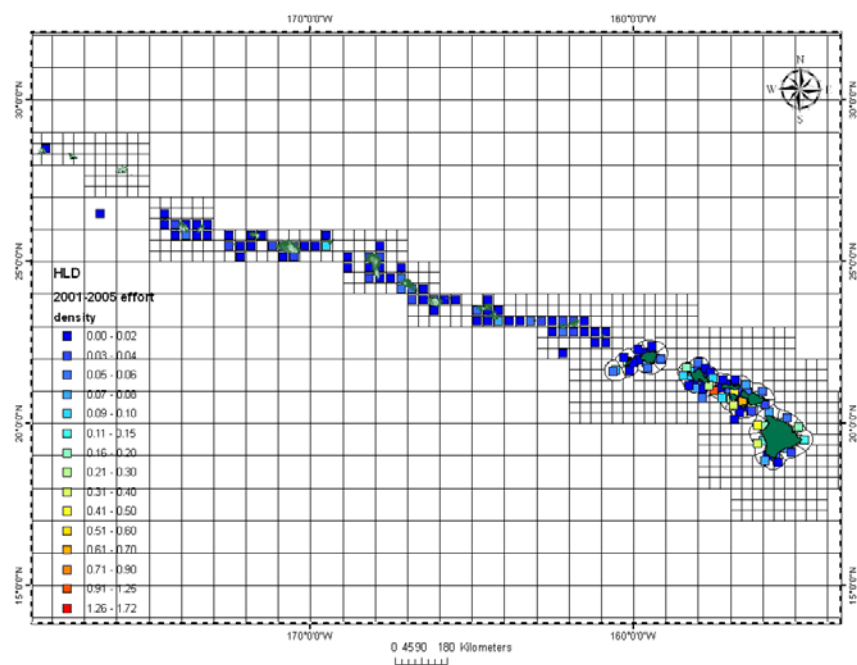
(H) 1986-1990



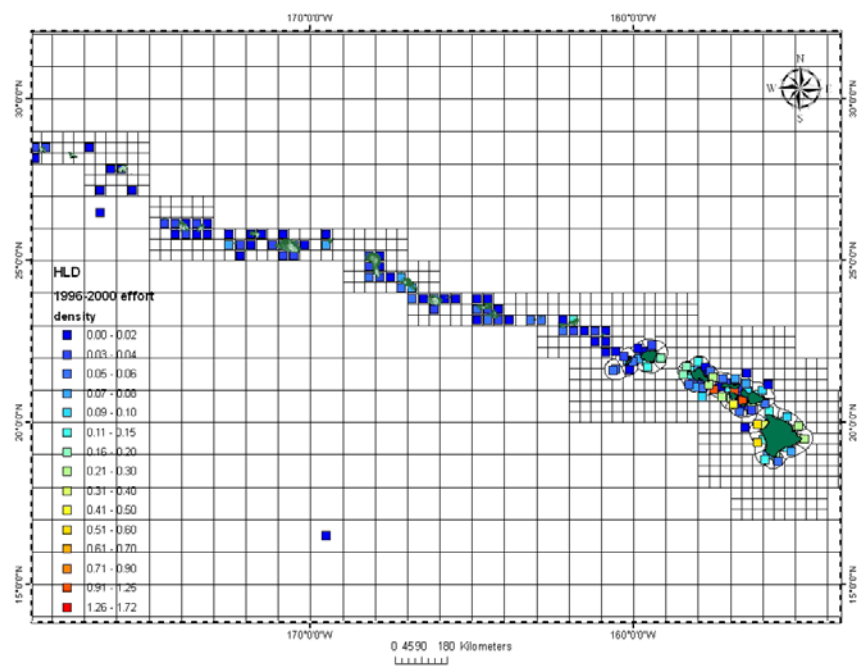
(I) 1991-1995



(J) 1996-2000



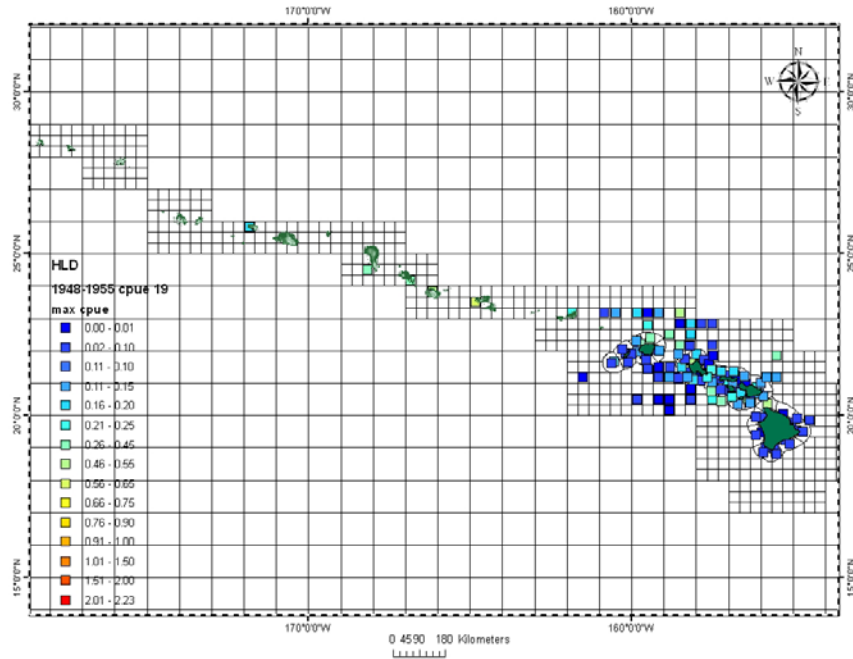
(K) 2001-2005



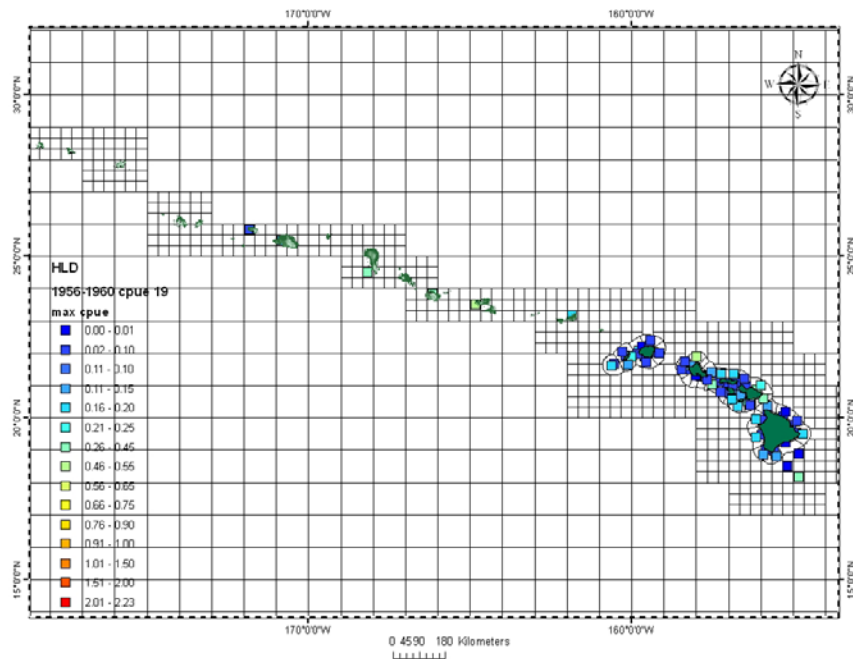
4.3 Spatial Distribution of CPUE by Indicator Bottomfish Species

Figure 4.3.1.- Spatial distribution of opakapaka (species 19) catch-per-unit-effort (pounds per fishing day) for deep handlines (gear 3) for 5-year stanzas from 1948-2005.

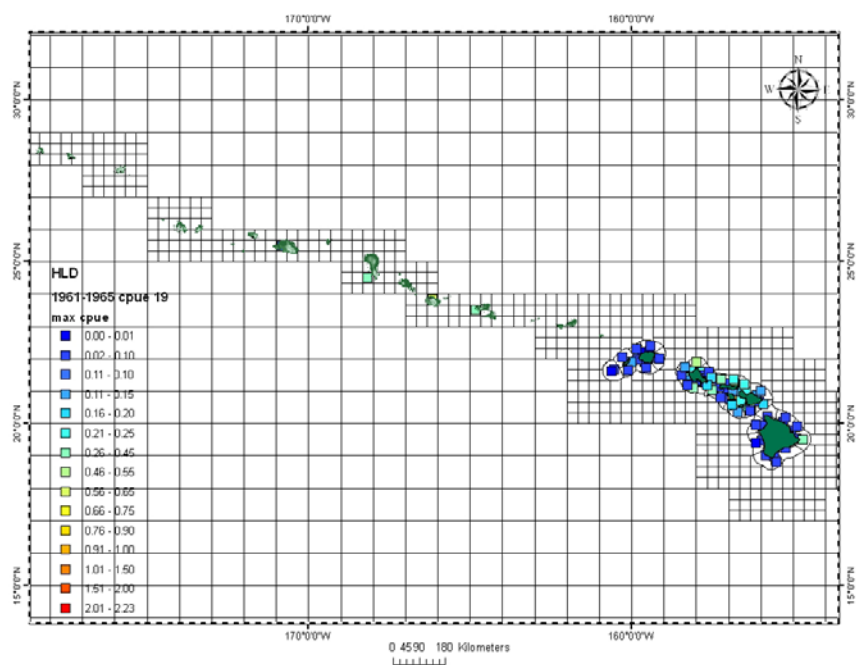
(A) 1948-1955



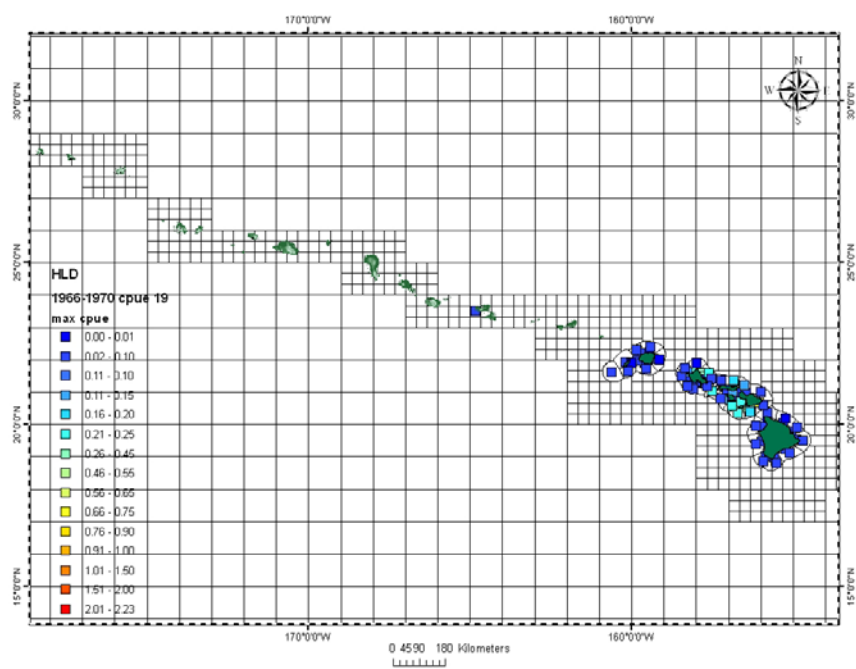
(B) 1956-1960



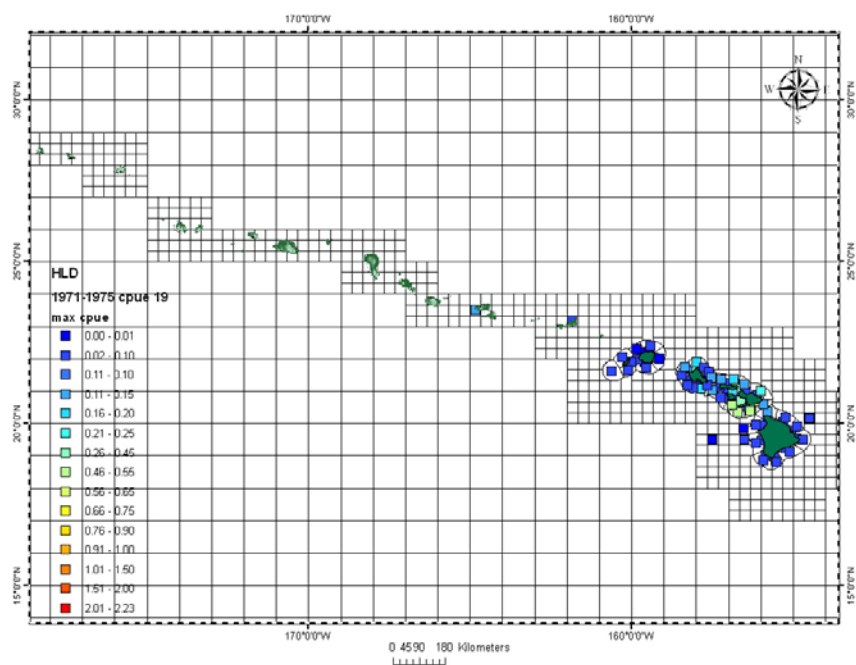
(C) 1961-1965



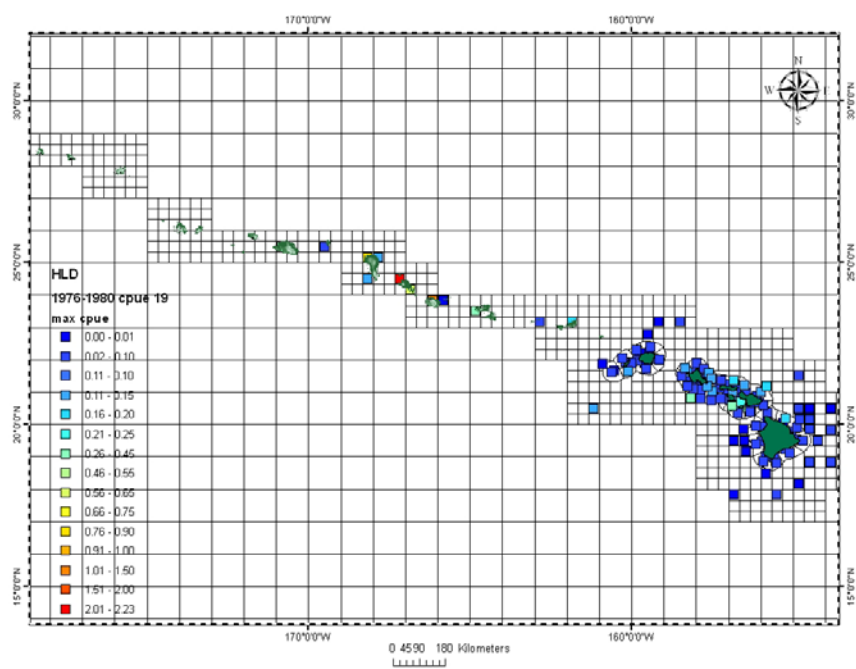
(D) 1966-1970



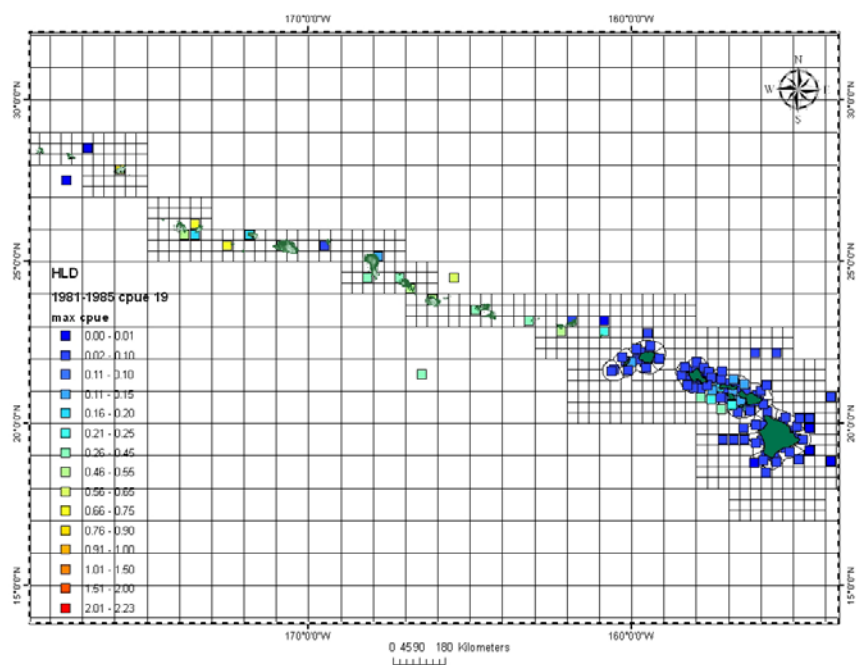
(E) 1971-1975



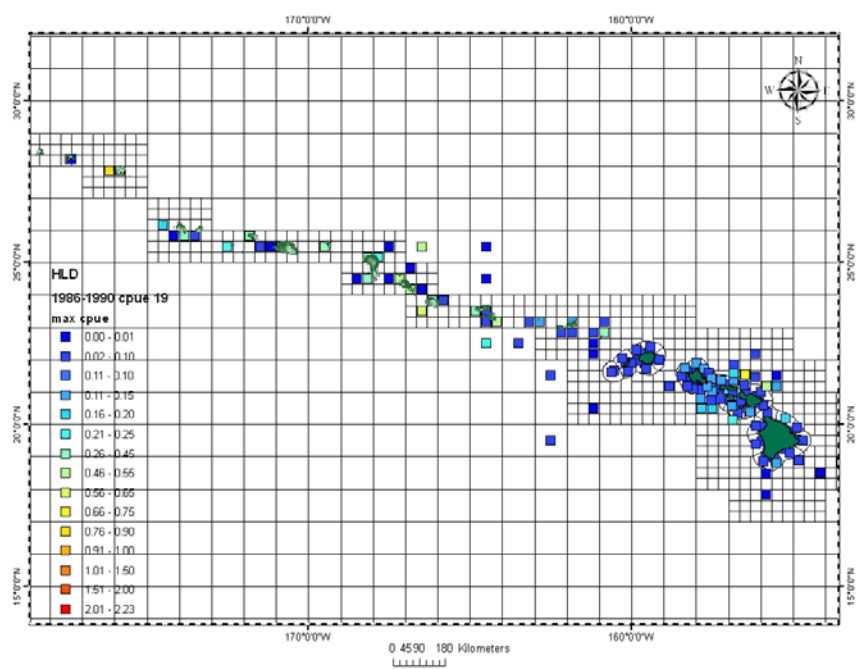
(F) 1976-1980



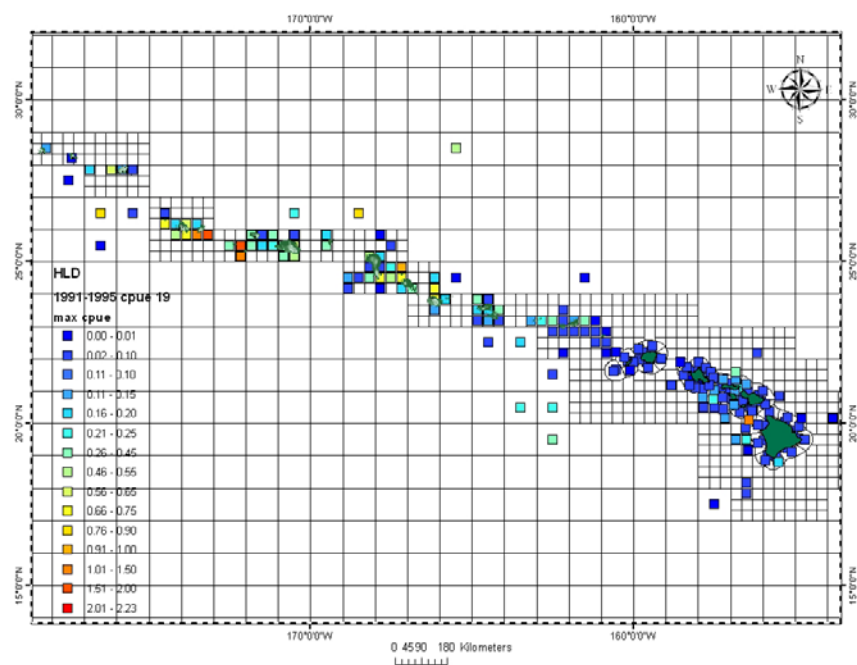
(G) 1981-1985



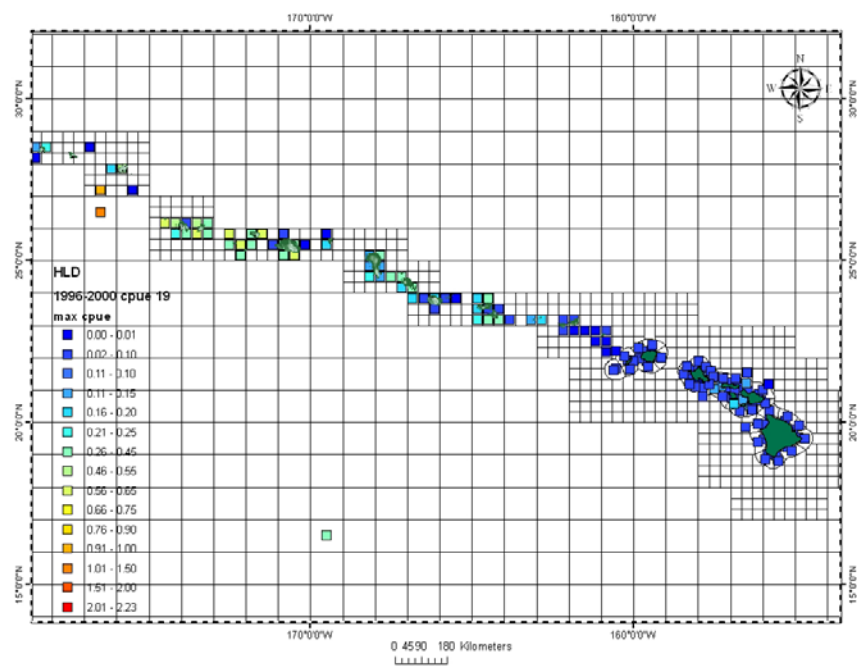
(H) 1986-1990



(I) 1991-1995



(J) 1996-2000



(K) 2001-2005

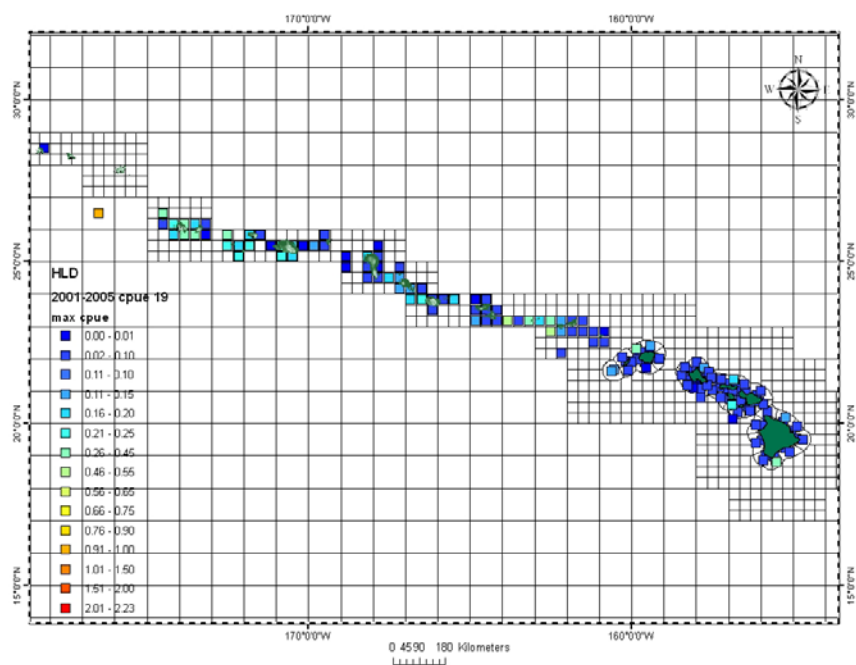
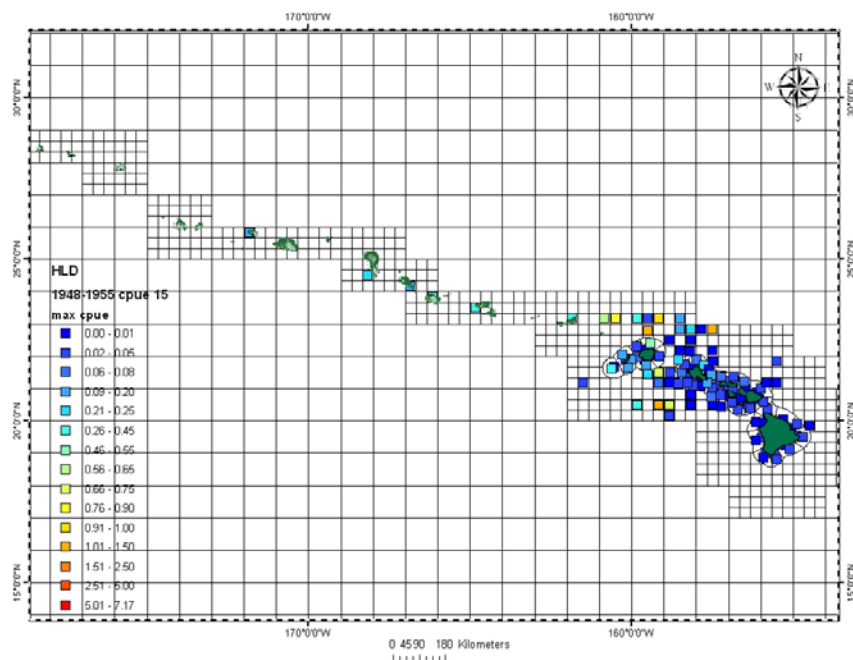
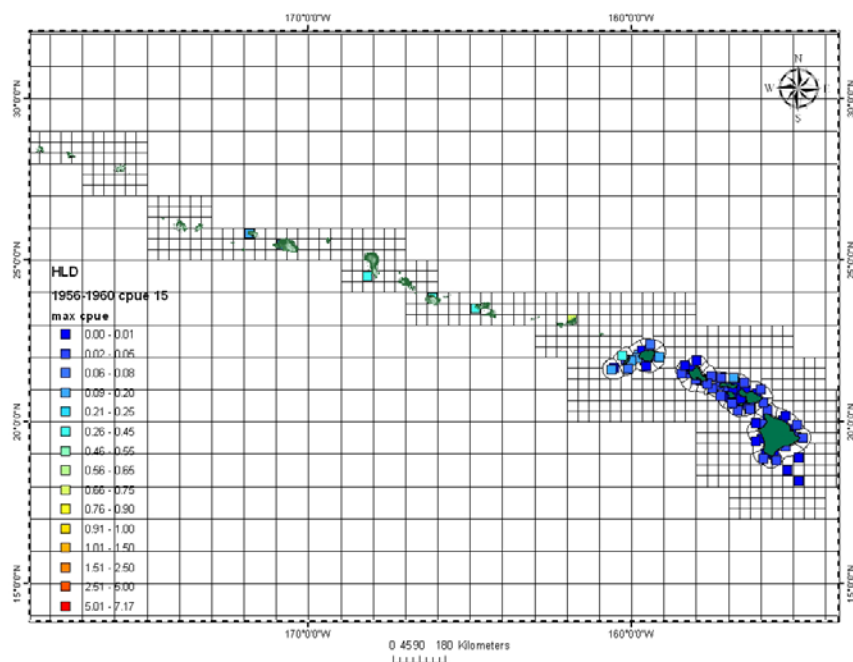


Figure 4.3.2.- Spatial distribution of hapuupuu (species 15) catch-per-unit-effort (pounds per fishing day) for deep handlines (gear 3) for 5-year stanzas from 1948-2005.

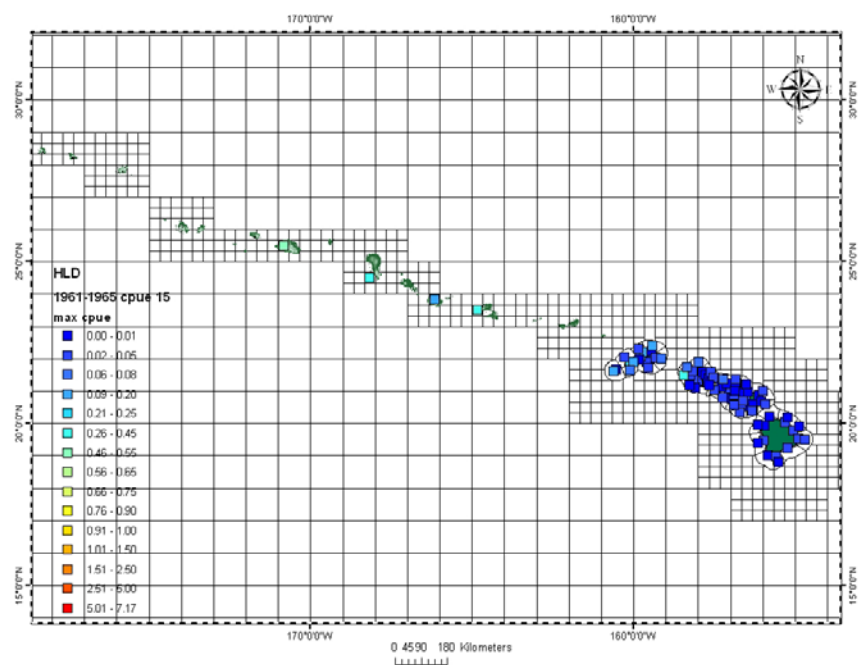
(A) 1948-1955



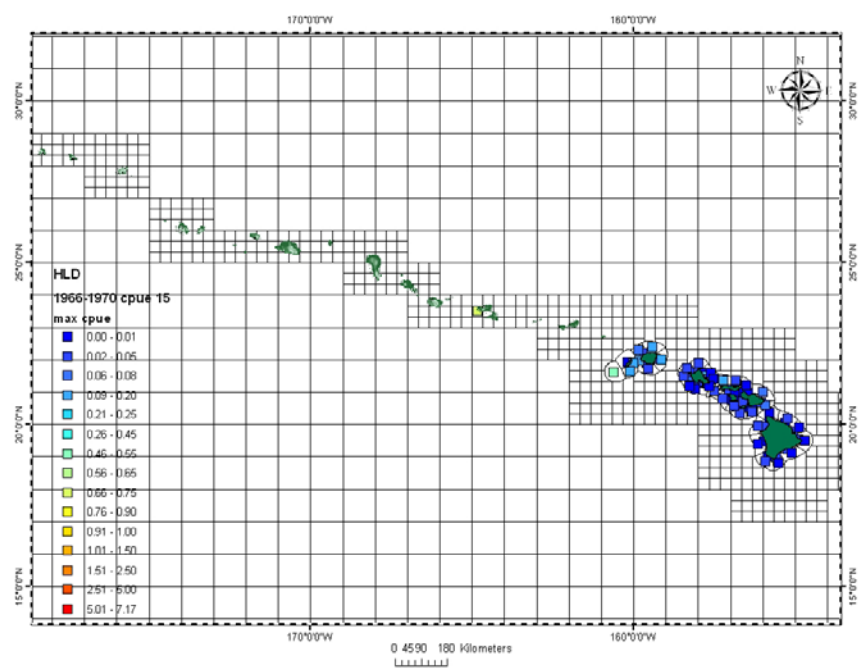
(B) 1956-1960



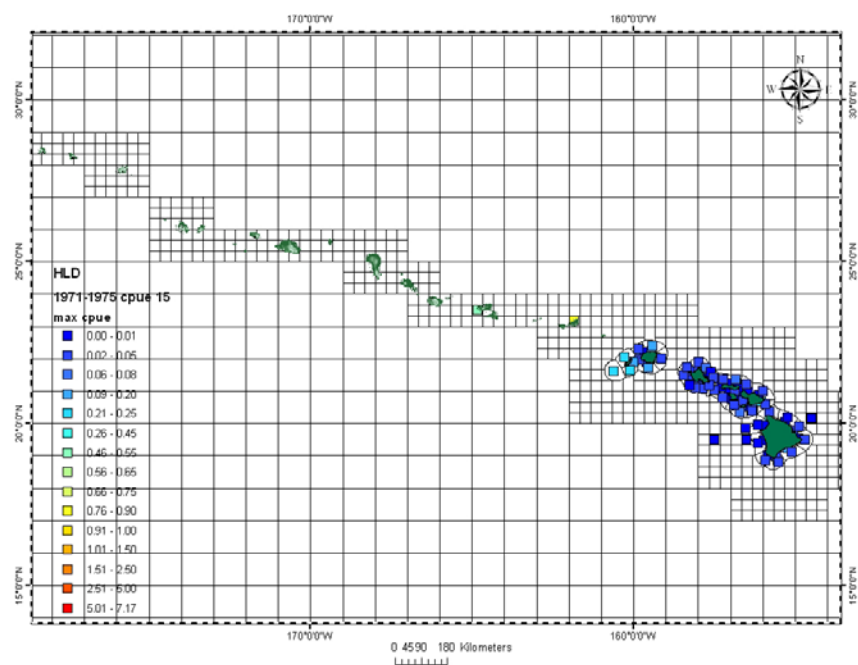
(C) 1961-1965



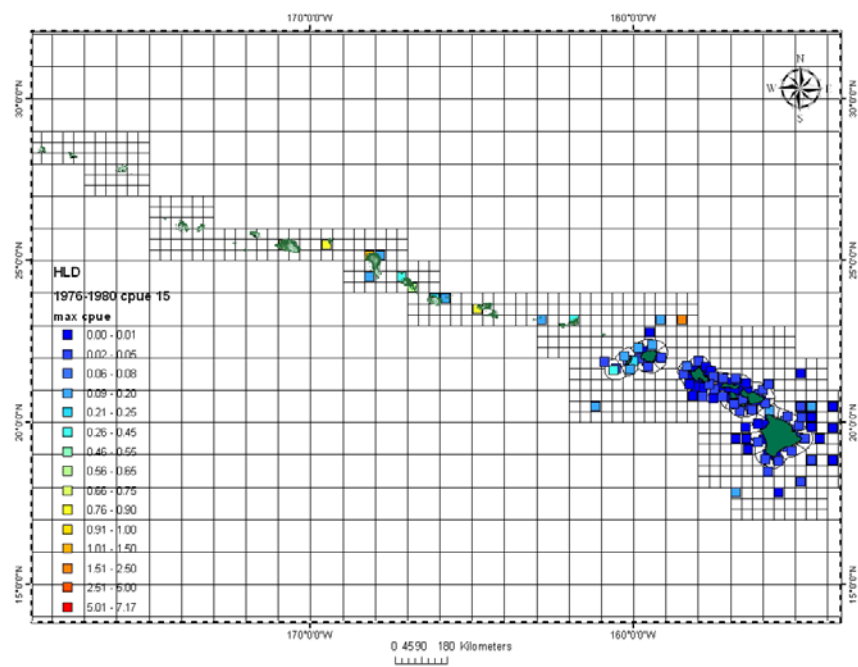
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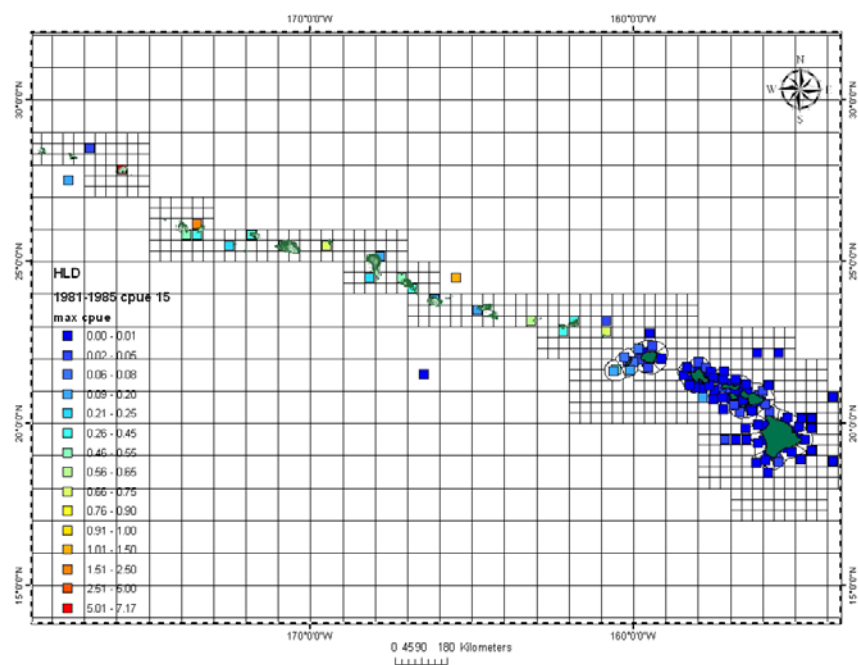
(E) 1971-1975



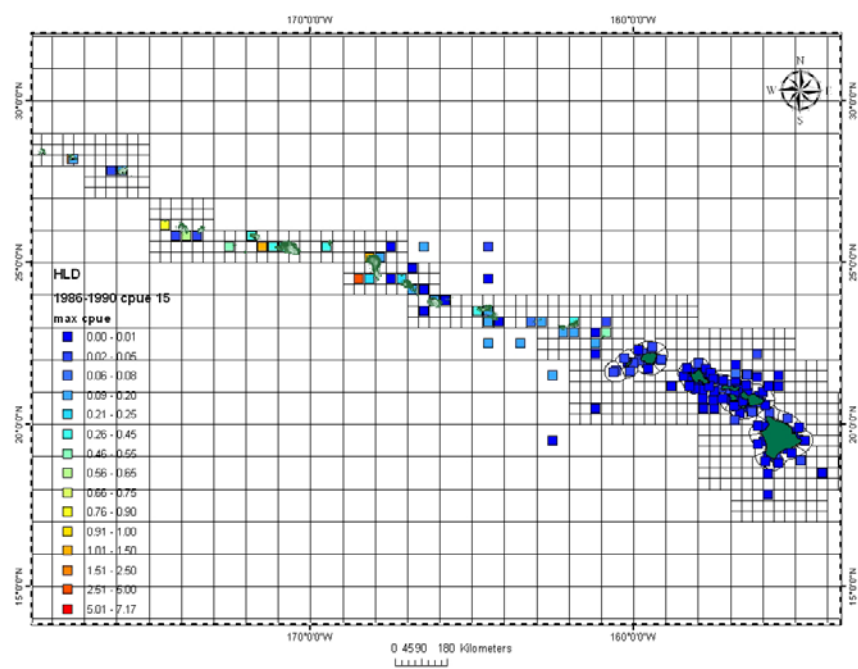
(F) 1976-1980



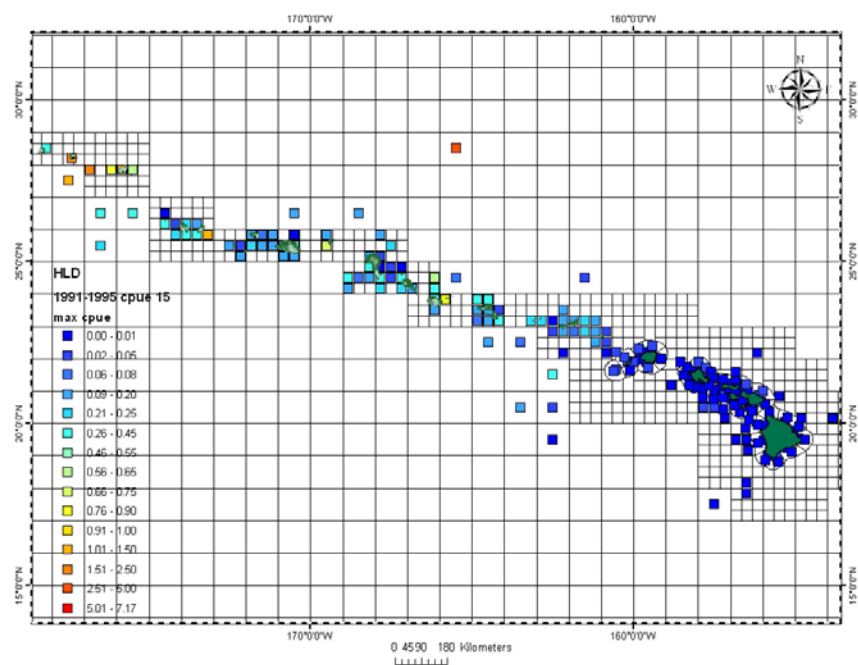
(G) 1981-1985



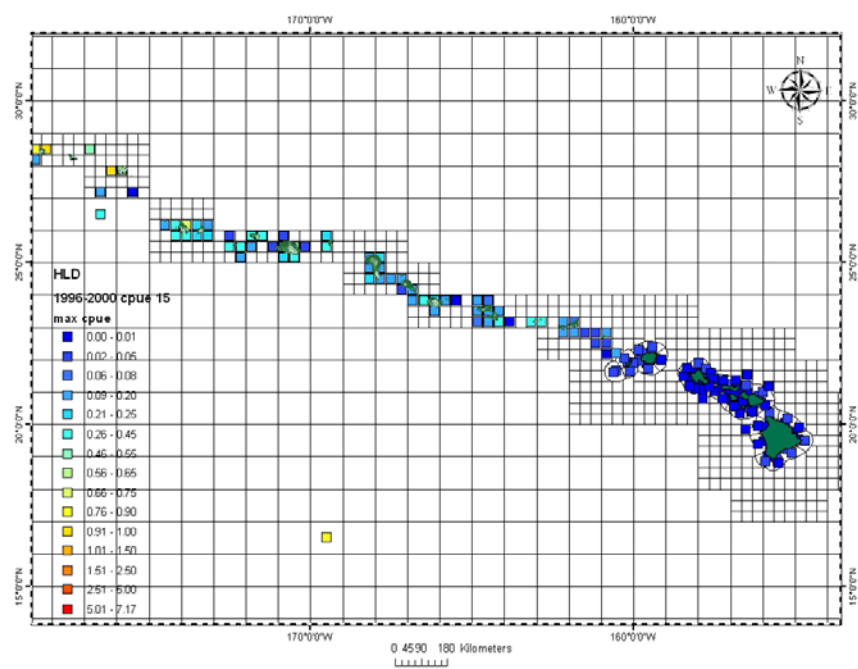
(H) 1986-1990



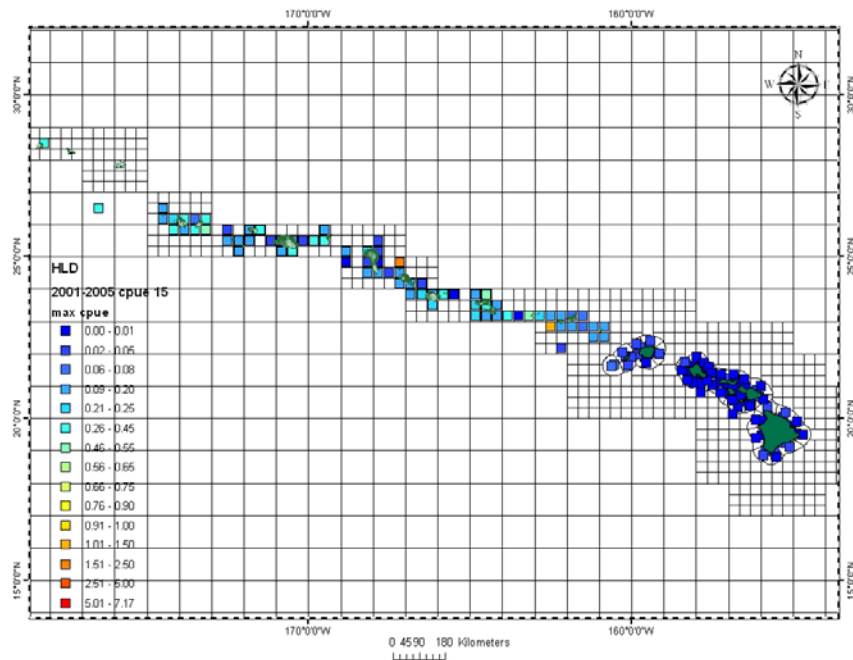
(I) 1991-1995



(J) 1996-2000



(K) 2001-2005



4.4 Summary of Recommendations

- Need gear-specific correction of CPUE by species so that the entire data series can be standardized between the years 1948 to 2005.
- Potential to improve data time-series estimation by including estimates of depth by fishing area, and by computing the fishable area for bottomfish within a given reported fished area.
- Link the species, gear, effort and catch data to more current size-structured data from the United Fishing dealer auction sources.
- Generate survey sampling design for complimentary fishery-independent studies by using this corrected and refined spatial database on historical time-series of bottomfish catches and effort, along with the auxillary data on bathymetry, etc.

5.0 References

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6.0 Appendices

6.1 Appendix A: SAS program code for database processing and development

6.1.1 SAS program 'frs_db1.sas'.

```
/*program frs_db1.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*year-specific processing*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';
/*
*/
/*initial processing*/
/*
*/
/*1948-2001*/
data a1948;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1948.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
/*proc contents data=a1;
quit;*/
/*proc print;
  var year month day area gear hours net_height species lbs value;
run;*/
data a1949;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1949.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1950;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1950.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
```

```
run;
data a1951;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1951.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1952;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1952.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1953;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1953.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1954;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1954.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1955;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1955.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
```

```
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1956;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1956.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1957;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1957.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1958;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1958.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1959;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1959.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
```

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lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1960;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1960.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1961;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1961.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1962;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1962.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1963;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1963.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1964;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1964.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
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@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1965;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1965.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1966;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1966.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1967;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1967.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1968;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1968.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
```

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port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1969;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1969.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1970;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1970.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1971;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1971.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1972;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1972.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1973;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1973.txt' lrecl=350;
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input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
  subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
  @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
  89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1974;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1974.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
  subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
  @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
  89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1975;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1975.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
  subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
  @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
  89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1976;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1976.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
  subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
  @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
  89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1977;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1977.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
  subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
  @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
  89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
```

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formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1978;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1978.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1979;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1979.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1980;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1980.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1981;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1981.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
```

```
run;
data a1982;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1982.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1983;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1983.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1984;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1984.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1985;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1985.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
    255
    lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
    f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1986;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1986.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
```

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wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1987;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1987.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1988;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1988.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1989;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1989.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1990;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1990.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
```

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lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1991;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1991.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1992;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1992.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1993;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1993.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1994;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1994.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1995;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1995.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
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@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1996;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1996.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1997;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1997.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1998;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1998.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-
255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a1999;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_1999.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
```

```
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a2000;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2000.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data a2001;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2001.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 num_sold 251-255
lbs_sold 256-263 @264 value 10.2 buyer$ 274-303 lost 304-308 release 309-313 damage 314-318
f_spec_gr$ 319 f_spec_ar$ 320 f_season$ 321 f_pieces$ 322 f_lbs$ 323 f_ctchlim$ 324;
run;
data b1;
set a1948 a1949 a1950 a1951 a1952 a1953 a1954 a1955 a1956 a1957 a1958 a1959 a1960 a1961
a1962 a1963 a1964 a1965 a1966 a1967 a1968 a1969 a1970 a1971 a1972 a1973 a1974 a1975
a1976 a1977 a1978 a1979 a1980 a1981 a1982 a1983 a1984 a1985 a1986 a1987 a1988 a1989
a1990 a1991 a1992 a1993 a1994 a1995 a1996 a1997 a1998 a1999 a2000 a2001;
avg_price=-9.0;
from_file='UN';
format num_sold 8.2 lbs_sold 9.2 avg_price 7.4;
run;
/*2002-2005*/
data a2002;
infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2002.txt' lrecl=350;
input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
@58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
89-90
wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
143
lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
port_dprtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 @251 num_sold
8.2
@259 lbs_sold 9.2 @268 avg_price 7.4 lost 275-279 release 280-284 damage 285-289 @290 value
8.2
f_spec_gr$ 298 f_spec_ar$ 299 f_season$ 300 f_pieces$ 301 f_lbs$ 302 f_ctchlim$ 303
from_file$ 304-305;
run;
/*proc freq;
tables from_file;
quit;*/
/*proc contents data=a2002;
quit;
proc print;
```

```
var year month day area gear hours net_height species lbs num_sold lbs_sold avg_price value;
run;*/
data a2003;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2003.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 @251 num_sold
    8.2
    @259 lbs_sold 9.2 @268 avg_price 7.4 lost 275-279 release 280-284 damage 285-289 @290 value
    8.2
    f_spec_gr$ 298 f_spec_ar$ 299 f_season$ 300 f_pieces$ 301 f_lbs$ 302 f_ctchlim$ 303
    from_file$ 304-305;
run;
data a2004;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2004.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 @251 num_sold
    8.2
    @259 lbs_sold 9.2 @268 avg_price 7.4 lost 275-279 release 280-284 damage 285-289 @290 value
    8.2
    f_spec_gr$ 298 f_spec_ar$ 299 f_season$ 300 f_pieces$ 301 f_lbs$ 302 f_ctchlim$ 303
    from_file$ 304-305;
run;
data a2005;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\FRS_2005.txt' lrecl=350;
  input license 1-6 trip_end 7-14 year 15-18 month 19-20 day 21-22 bouy$ 23-24 area 25-29
    subarea$ 30 gear 31-32 sets 33-35 @36 hours 7.2 no_gears 43-47 hooks 48-52 net_length 53-57
    @58 net_height 5.2 bank$ 63-80 bank_quad$ 81-82 depth_beg 83-85 depth_end 86-88 wind_speed
    89-90
    wind_dir$ 91-92 wave_ht 93-94 cur_speed$ 95 cur_dir$ 96-97 nocatch$ 98 port_land 99-101
    logbook 102-107 num_stolen 108-112 crew 113-115 chartered$ 116 r_month 117-118 r_year 119-122
    formtype$ 123-130 timelink 131-138 f_area$ 139 f_buoy$ 140 f_gear$ 141 f_ar_gr$ 142 f_by_pr$
    143
    lname$ 144-167 fname$ 168-185 vessel$ 186-205 uscg$ 206-212 ha_no$ 213-220 port_land2 221-223
    port_dprrtr 224-226 trip_beg 227-234 species 235-237 caught 238-242 @243 lbs 8.1 @251 num_sold
    8.2
    @259 lbs_sold 9.2 @268 avg_price 7.4 lost 275-279 release 280-284 damage 285-289 @290 value
    8.2
    f_spec_gr$ 298 f_spec_ar$ 299 f_season$ 300 f_pieces$ 301 f_lbs$ 302 f_ctchlim$ 303;
  from_file='UN';
run;
data b2;
  set a2002 a2003 a2004 a2005;
  buyer='XXXXXXXXXXXXXXXXXXXXXXXXXXXX';
  format value 10.2;
run;
data b3;
  set b1 b2;
run;
data frs.fish_db1_1948_2005;
  set b3;
run;
```


6.1.2 SAS program 'frs_botdb1.sas'.

```
/*program frs_botdb1.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*creates bottomfish database for initial descriptive analysis*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.fish_db1_1948_2005;
quit;*/
data a1;
  set frs.fish_db1_1948_2005;
  if 15<=species<=17 or 19<=species<=22 or species=58 or species=97
    or species=114 or species=200 or species=205 then bf=1;
  else bf=0;
  if species=15 or species=17 or species=19 or species=20 or 21<=species<=22
    or species=58 or species=97 or species=114 then fine9=1;
  else fine9=0;
  if species=15 or species=17 or species=19 or 21<=species<=22
    or species=58 or species=97 then deep7=1;
  else deep7=0;
  if bf>0 or gear=3 or gear=5 then bft=1;
  else bft=0;
run;
proc means data=a1 nway noprint;
  class license year month day;
  var bft;
  output out=b1 (drop=_type_ _freq_) sum=bfsum;
quit;
data b2;
  set b1;
  if bfsum>0 then bf_trip=1;
  else bf_trip=0;
  keep license year month day bf_trip;
run;
proc sort data=a1 out=b3;
  by license year month day;
quit;
data b4;
  merge b2 b3;
  by license year month day;
  if bf_trip=1;
run;
data frs.botfish_db1_1948_2005;
  set b4;
run;
```

6.1.3 SAS program 'frs_botdb2.sas'.

```
/*program frs_botdb2.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*creates refined bottomfish dataset DB2*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.botfish_db1_1948_2005;
quit;*/
data bf1;
  set frs.botfish_db1_1948_2005;
  drop nocatch num_stolen num_sold value damage avg_price f_area f_buoy
    f_gear f_ar_gr f_by_pr f_spec_gr f_spec_ar f_season f_pieces f_lbs
    f_ctchlim from_file r_month r_year timelink wind_speed wind_dir wave_ht
    cur_speed cur_dir port_land2 port_dprtr logbook bf_trip crew formtype
    buyer lname fname vessel uscg ha_no release lost caught;
run;
/*basic variable cleanup, designate missing values, resolve problem catches*/
data bf2;
  set bf1;
  if license=0 then license=-9;
  else license=license;
  if trip_end=. then trip_end=-9;
  else trip_end=trip_end;
  if bouy=' ' then bouy='-9';
  else bouy=bouy;
  if subarea=' ' then subarea='Z';
  else subarea=subarea;
  if bank=' ' then bank='-9';
  else bank=bank;
  if bank_quad=' ' then bank_quad='-9';
  else bank_quad=bank_quad;
  if trip_beg=. then trip_beg=-9;
  else trip_beg=trip_beg;
  if lbs=0 and lbs_sold>0 then lbs=lbs_sold;
  else lbs=lbs;
  drop lbs_sold;
run;
proc sort;
  by license year month day port_land;
quit;
/*refine designation of bf trip-records*/
proc means data=bf2 nway noprint;
  class license year month day port_land;
  var bft;
  output out=bf3 (drop=_type_ _freq_) sum=bfsum;
quit;
data bf4;
  set bf3;
  if bfsum>0 then bf_trip=1;
  else bf_trip=0;
  keep license year month day port_land bf_trip;
run;
data bf5;
```

```
merge bf2 bf4;
by license year month day port_land;
if bf_trip=1;
run;
/*assign gear_cat*/
data bf6;
set bf5;
if gear=3 then gear_cat=3;
else if gear=4 or gear=9 then gear_cat=4;
else if gear=5 then gear_cat=5;
else if gear=6 or gear=10 or gear=35 or gear=61 or gear=62
or gear=63 or gear=2 or gear=8 or gear=1 or gear=92 or gear=93
then gear_cat=6;
else if gear=11 or gear=52 or gear=24 or gear=51 or gear=53 or gear=54
then gear_cat=11;
else if gear=13 or gear=14 then gear_cat=13;
else if gear=20 or gear=22 or gear=23 or gear=21 or gear=26 or gear=27 or
gear=28
or gear=31 or gear=33 or gear=25 or gear=30 or gear=40 then gear_cat=20;
else gear_cat=99;
run;
/*proc freq data=bf6;
tables gear_cat;
quit;*/
data frs.botfish_db2_1948_2005;
set bf6;
run;
```

6.2 Appendix B: SAS program code for dataset analysis procedures

6.2.1 SAS program 'frs1_bfspecies_year.sas'.

```
/*program frs1_bfspecies_year.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*catchw time-series, species, region */
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.fish_db1_1948_2005;
quit;*/
data a1;
  set frs.fish_db1_1948_2005;
  if 15<=species<=17 or 19<=species<=22 or species=58 or species=97
    or species=114 or 140<=species<=141 or species=200 or species=202
    or species=205 or species=208 then bf=1;
  else bf=0;
  if species=15 or species=17 or species=19 or species=20 or 21<=species<=22
    or species=58 or species=97 or species=114 then fine9=1;
  else fine9=0;
  if species=15 or species=17 or species=19 or 21<=species<=22
    or species=58 or species=97 then deep7=1;
  else deep7=0;
run;
data a2;
  set a1;
  if fine9=1;
run;
/*catchw by region by year, fine9*/
data a3;
  set a2;
  if area<55 or area=999 or 2919<=area<=9999 or area=99999 then region=0;
  else if 100<=area<=128 then region=1;
  else if 300<=area<=333 then region=2;
  else if 400<=area<=429 then region=3;
  else if 500<=area<=528 then region=4;
  else if 129<=area<=258 or 334<=area<=390 or 431<=area<=475
    or 530<=area<=855 then region=5;
  else if 874<=area<=875 or 1001<=area<=2339 or 15217<=area<=17828 then
    region=6;
  else region=-9;
run;
/*proc freq data=a3;
  tables region;
quit;
data ck1;
  set a3;
  if region=-9;
run;
proc print;
quit;*/
proc means data=a3 noprint nway;
  class species year region;
  var lbs;
```

```
    output out=ar1 (drop=_type_ _freq_) sum=catch_wt;
quit;
/*proc print;
quit;*/
data ar2;
    set ar1;
    if region=0 then delete;
run;
proc transpose out=ar3;
    by species year;
    var catch_wt;
    id region;
quit;
proc print noobs;
    var species year _1 _2 _3 _4 _5 _6;
quit;
proc means data=ar1 nway noprint;
    class year region;
    var catch_wt;
    output out=ar4 (drop=_type_ _freq_) sum=totcatch;
quit;
data ar5;
    set ar4;
    if region=0 then delete;
run;
proc transpose out=ar3;
    by year;
    var totcatch;
    id region;
quit;
proc print noobs;
    var year _1 _2 _3 _4 _5 _6;
quit;
```

6.2.2 SAS program 'frs1_gear_bfspp.sas'.

```
/*program frs1_gear_bfspp.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*desc of gear-bfspecies associations*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.fish_db1_1948_2005;
quit;*/
data a1;
  set frs.fish_db1_1948_2005;
  if 15<=species<=17 or 19<=species<=22 or species=58 or species=97
    or species=114 or 140<=species<=141 or species=200 or species=202
    or species=205 or species=208 then bf=1;
  else bf=0;
  if species=15 or species=17 or species=19 or 21<=species<=22
    or species=58 or species=97 then deep7=1;
  else deep7=0;
run;
/*trip-records by gear*/
proc means data=a1 nway noprint;
  class license year month day gear;
  var bf deep7;
  output out=a2 (drop=_type_ _freq_) sum=bfsum dp7sum;
quit;
data a3;
  set a2;
  if bfsum>0 then bf_trip=1;
  else bf_trip=0;
  if dp7sum>0 then dp7_trip=1;
  else dp7_trip=0;
run;
proc means data=a3 nway noprint;
  class gear;
  var bf_trip dp7_trip;
  output out=a4 (drop=_type_ _freq_) n=tot_trips sum=bf_trips dp7_trips;
quit;
/*proc print;
quit;*/
data g1;
  infile 'c:\sgs\HI_main8_FD_DB\HDAR-
  FishermanReportingSystem\fisher_db\gear_list.prn';
  input gear 1-2 gear_desc$ 12-66 dummy 80;
run;
proc sort;
  by gear;
quit;
data a5;
  merge g1 a4;
  by gear;
  if bf_trips=. then delete;
  bf_pct=(bf_trips/tot_trips)*100;
  dp7_pct=(dp7_trips/tot_trips)*100;
  format bf_pct dp7_pct 6.2;
```

```
run;
proc sort;
  by descending bf_trips;
quit;
proc print noobs;
  var gear gear_desc tot_trips bf_trips bf_pct dp7_trips dp7_pct;
quit;
/*catch by bottomfish species by gear*/
data bg1;
  set a1;
  if bf=1;
  if 3<=gear<=5 or 9<=gear<=11 or gear=31 or gear=52 then pr_gear=1;
  else pr_gear=0;
run;
proc means nway noprint;
  class species gear;
  var lbs;
  id pr_gear;
  output out=bg2(drop=_type_ _freq_) sum=catch_wt;
quit;
/*proc print;
quit;*/
data bg3;
  set bg2;
  if pr_gear=1;
run;
proc sort;
  by species gear;
quit;
/*proc print;
quit;*/
proc transpose out=bg4;
  by species;
  var catch_wt;
  id gear;
quit;
proc print noobs;
  var species _3 _4 _5 _9 _10 _11 _31 _52;
quit;
/*proc print;
quit;*/
proc means data=bg2 nway noprint;
  class gear;
  var catch_wt;
  output out=bg5 (drop=_type_ _freq_) sum=totcatch;
quit;
proc print noobs;
quit;
proc means data=bg1 nway noprint;
  class gear species;
  var lbs;
  output out=bg6 (drop=_type_ _freq_) sum=catch_wt;
quit;
/*proc print;
quit;*/
proc transpose data=bg6 out=bg7;
  by gear;
```

```
var catch_wt;  
id species;  
quit;  
proc print noobs;  
var gear _15 _16 _17 _19 _20 _21 _22 _58 _97 _114  
_140 _141 _200 _202 _205 _208;  
quit;
```


6.2.3 SAS program 'bf1_dbvariables.sas'

```
/*program bf1_dbvariables.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*processing analysis of bottomfish species database variables*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.botfish_db1_1948_2005;
quit;*/
data bf1;
  set frs.botfish_db1_1948_2005;
run;
/*basic variable cleanup, designate missing values*/
data bf2;
  set bf1;
  if license=0 then license=-9;
  else license=license;
  if trip_end=. then trip_end=-9;
  else trip_end=trip_end;
  if bouy=' ' then bouy='-9';
  else bouy=bouy;
  if subarea=' ' then subarea='Z';
  else subarea=subarea;
  if bank=' ' then bank='-9';
  else bank=bank;
  if bank_quad=' ' then bank_quad='-9';
  else bank_quad=bank_quad;
  if wind_dir=' ' then wind_dir='-9';
  else wind_dir=wind_dir;
  if cur_speed=' ' then cur_speed='Z';
  else cur_speed=cur_speed;
  if cur_dir=' ' then cur_dir='-9';
  else cur_dir=cur_dir;
  if formtype=' ' then formtype='-9';
  else formtype=formtype;
  if timelink=. then timelink=-9;
  else timelink=timelink;
  if lname=' ' then lname='-9';
  else lname=lname;
  if fname=' ' then fname='-9';
  else fname=fname;
  if vessel=' ' then vessel='-9';
  else vessel=vessel;
  if uscg=' ' then uscg='-9';
  else uscg=uscg;
  if ha_no=' ' then ha_no='-9';
  else ha_no=ha_no;
  if trip_beg=. then trip_beg=-9;
  else trip_beg=trip_beg;
  if from_file=' ' then from_file='-9';
  else from_file=from_file;
  if avg_price=. then avg_price=-9;
  else avg_price=avg_price;
  drop bf_trip crew f_gear buyer;
```

```
run;
/*proc freq data=bf2;
  tables license year month day bouy area subarea gear sets hooks;
  tables bank bank_quad wind_dir cur_speed cur_dir nocatch port_land;
  tables logbook chartered r_month r_year formtype f_area f_buoy;
  tables f_ar_gr f_by_pr lname fname vessel uscg ha_no port_land2
    port_dprtr species;
  tables f_spec_gr f_spec_ar f_season f_pieces f_lbs f_ctchlim from_file;
quit;
proc univariate data=bf2 plot nobyplot;
  var trip_end hours no_gears net_length net_height;
  var depth_beg depth_end wind_speed wave_ht;
  var num_stolen timelink;
  var trip_beg caught lbs num_sold lbs_sold value;
  var lost release damage avg_price;
quit;*/
/*produce percent missing tables by year, variable classes*/
data msv1;
  set bf2;
  if license=-9 then lic_cnt=0;
  else lic_cnt=1;
  if logbook=0 then logbk_cnt=0;
  else logbk_cnt=1;
  if chartered='F' then chart_cnt=0;
  else chart_cnt=1;
  if formtype='-9' then form_cnt=0;
  else form_cnt=1;
  if lname='-9' then lname_cnt=0;
  else lname_cnt=1;
  if fname='-9' then fname_cnt=0;
  else fname_cnt=1;
  if vessel='-9' then ves_cnt=0;
  else ves_cnt=1;
  if uscg='-9' then uscg_cnt=0;
  else uscg_cnt=1;
  if ha_no='-9' then hano_cnt=0;
  else hano_cnt=1;
  if trip_end=-9 then trpend_cnt=0;
  else trpend_cnt=1;
  if year=-9 then year_cnt=0;
  else year_cnt=1;
  if month=-9 then mnth_cnt=0;
  else mnth_cnt=1;
  if day=-9 then day_cnt=0;
  else day_cnt=1;
  if r_month=0 then rmon_cnt=0;
  else rmon_cnt=1;
  if r_year=0 then ryr_cnt=0;
  else ryr_cnt=1;
  if timelink=-9 then tlnk_cnt=0;
  else tlnk_cnt=1;
  if trip_beg=-9 then trpbeg_cnt=0;
  else trpbeg_cnt=1;
  if bouy='-9' then by_cnt=0;
  else by_cnt=1;
  if area<25 or area=9999 or area=99999 then area_cnt=0;
  else area_cnt=1;
```

```
if subarea='Z' then sbar_cnt=0;
else sbar_cnt=1;
if bank='-9' then bnk_cnt=0;
else bnk_cnt=1;
if bank_quad='-9' then bkqd_cnt=0;
else bkqd_cnt=1;
if port_land=0 then prtln_cnt=0;
else prtln_cnt=1;
if port_land2=0 then prtln2_cnt=0;
else prtln2_cnt=1;
if port_dprtr=0 then prtdpt_cnt=0;
else prtdpt_cnt=1;
if gear=0 then gr_cnt=0;
else gr_cnt=1;
if sets=0 then set_cnt=0;
else set_cnt=1;
if hours=0 then hr_cnt=0;
else hr_cnt=1;
if no_gears=0 then ngr_cnt=0;
else ngr_cnt=1;
if hooks=0 then hk_cnt=0;
else hk_cnt=1;
if net_length=0 then netlt_cnt=0;
else netlt_cnt=1;
if net_height=0 then netht_cnt=0;
else netht_cnt=1;
if nocatch='F' then noctch_cnt=0;
else noctch_cnt=1;
if num_stolen=0 then nstol_cnt=0;
else nstol_cnt=1;
if species=0 or species=999 then spp_cnt=0;
else spp_cnt=1;
if caught=0 then ctchn_cnt=0;
else ctchn_cnt=1;
if lbs=0 then ctchw_cnt=0;
else ctchw_cnt=1;
if num_sold=0 then nmsold_cnt=0;
else nmsold_cnt=1;
if lbs_sold=0 then wtsold_cnt=0;
else wtsold_cnt=1;
if value=0 then val_cnt=0;
else val_cnt=1;
if lost=0 then lost_cnt=0;
else lost_cnt=1;
if release=0 then rels_cnt=0;
else rels_cnt=1;
if damage=0 then dam_cnt=0;
else dam_cnt=1;
if avg_price<=0 then avprc_cnt=0;
else avprc_cnt=1;
if depth_beg=0 then dpbeg_cnt=0;
else dpbeg_cnt=1;
if depth_end=0 then dpend_cnt=0;
else dpend_cnt=1;
if wind_speed=0 then wspd_cnt=0;
else wspd_cnt=1;
if wind_dir='-9' then wdir_cnt=0;
```

```
    else wdir_cnt=1;
    if wave_ht=0 then wvht_cnt=0;
    else wvht_cnt=1;
    if cur_speed='Z' then crsp_cnt=0;
    else crsp_cnt=1;
    if cur_dir='-9' then crdr_cnt=0;
    else crdr_cnt=1;
    if from_file='-9' or from_file='UN' then frfile_cnt=0;
    else frfile_cnt=1;
    if f_area='F' then flgar_cnt=0;
    else flgar_cnt=1;
    if f_buoy='F' then flgby_cnt=0;
    else flgby_cnt=1;
    if f_ar_gr='F' then flgargr_cnt=0;
    else flgargr_cnt=1;
    if f_by_pr='F' then flgbypr_cnt=0;
    else flgbypr_cnt=1;
    if f_spec_gr='F' or f_spec_gr='0' then flgspgr_cnt=0;
    else flgspgr_cnt=1;
    if f_spec_ar='F' or f_spec_ar='0' then flgspar_cnt=0;
    else flgspar_cnt=1;
    if f_season='F' then flgsn_cnt=0;
    else flgsn_cnt=1;
    if f_pieces='F' then flgpc_cnt=0;
    else flgpc_cnt=1;
    if f_lbs='F' then flglb_cnt=0;
    else flglb_cnt=1;
    if f_ctchlim='F' then flgclim_cnt=0;
    else flgclim_cnt=1;
run;
/*trip var records*/
/*proc freq data=msv1;
    tables lic_cnt logbk_cnt chart_cnt form_cnt lname_cnt fname_cnt
           ves_cnt uscg_cnt hano_cnt;
quit;*/
proc means data=msv1 nway noprint;
    class year;
    var lic_cnt logbk_cnt chart_cnt form_cnt lname_cnt fname_cnt
        ves_cnt uscg_cnt hano_cnt;
    output out=msv_trp (drop=_type_ _freq_) n=n sum=license logbook
        chartered formtype lname fname vessel uscg ha_no;
quit;
data msv_trp2;
    set msv_trp;
    pct_lic=(license/n)*100;
    pct_log=(logbook/n)*100;
    pct_chart=(chartered/n)*100;
    pct_form=(formtype/n)*100;
    pct_lnm=(lname/n)*100;
    pct_fnm=(fname/n)*100;
    pct_ves=(vessel/n)*100;
    pct_uscg=(uscg/n)*100;
    pct_hano=(ha_no/n)*100;
run;
proc print noobs;
    var year n pct_lic pct_lnm pct_fnm pct_ves pct_uscg pct_hano pct_log
        pct_form pct_chart;
```

```
quit;
/*date var records*/
/*proc freq data=msv1;
  tables trpend_cnt year_cnt mnth_cnt day_cnt rmon_cnt ryr_cnt
    tlnk_cnt trpbeg_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var trpend_cnt year_cnt mnth_cnt day_cnt rmon_cnt ryr_cnt
    tlnk_cnt trpbeg_cnt;
  output out=msv_trp3 (drop=_type_ _freq_) n=n sum=trpend yr
    month day rmonth ryear timelink tripbeg;
quit;
data msv_trp4;
  set msv_trp3;
  pct_trpend=(trpend/n)*100;
  pct_year=(yr/n)*100;
  pct_mnth=(month/n)*100;
  pct_day=(day/n)*100;
  pct_rmnth=(rmonth/n)*100;
  pct_ryear=(ryear/n)*100;
  pct_tlnk=(timelink/n)*100;
  pct_trpbeg=(tripbeg/n)*100;
run;
proc print noobs;
  var year n pct_trpend pct_year pct_mnth pct_day pct_rmnth pct_ryear
    pct_tlnk
    pct_trpbeg;
quit;
/*location var records*/
/*proc freq data=msv1;
  tables by_cnt area_cnt sbar_cnt bnk_cnt bkqd_cnt prtln2_cnt
    prtln2_cnt prtdpt_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var by_cnt area_cnt sbar_cnt bnk_cnt bkqd_cnt prtln2_cnt
    prtln2_cnt prtdpt_cnt;
  output out=msv_trp5 (drop=_type_ _freq_) n=n sum=buoy area
    sbar bnk bkqd prtln2 prtln2 prtdpt;
quit;
data msv_trp6;
  set msv_trp5;
  pct_by=(buoy/n)*100;
  pct_area=(area/n)*100;
  pct_sbar=(sbar/n)*100;
  pct_bnk=(bnk/n)*100;
  pct_bkqd=(bkqd/n)*100;
  pct_prtln2=(prtln2/n)*100;
  pct_prtln2=(prtln2/n)*100;
  pct_prtdpt=(prtdpt/n)*100;
run;
proc print noobs;
  var year n pct_by pct_area pct_sbar pct_bnk pct_bkqd pct_prtln2 pct_prtln2
    pct_prtdpt;
quit;
/*gear var records*/
```

```
/*proc freq data=msv1;
  tables gr_cnt set_cnt hr_cnt ngr_cnt hk_cnt netlt_cnt
    netht_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var gr_cnt set_cnt hr_cnt ngr_cnt hk_cnt netlt_cnt
    netht_cnt;
  output out=msv_trp7 (drop=_type_ _freq_) n=n sum=gear set
    hr ngr hk netlt netht;
quit;
data msv_trp8;
  set msv_trp7;
  pct_gear=(gear/n)*100;
  pct_set=(set/n)*100;
  pct_hr=(hr/n)*100;
  pct_ngr=(ngr/n)*100;
  pct_hk=(hk/n)*100;
  pct_netlt=(netlt/n)*100;
  pct_netht=(netht/n)*100;
run;
proc print noobs;
  var year n pct_gear pct_set pct_hr pct_ngr pct_hk pct_netlt pct_netht;
quit;
/*catch var records*/
/*proc freq data=msv1;
  tables noctch_cnt nstol_cnt spp_cnt ctchn_cnt ctchw_cnt nmsold_cnt
    wtsold_cnt val_cnt lost_cnt rels_cnt dam_cnt avprc_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var noctch_cnt nstol_cnt spp_cnt ctchn_cnt ctchw_cnt nmsold_cnt
    wtsold_cnt val_cnt lost_cnt rels_cnt dam_cnt avprc_cnt;
  output out=msv_trp9 (drop=_type_ _freq_) n=n sum=noctch nstol
    spp ctchn ctchw nmsold wtsold val lost rels dam avprc;
quit;
data msv_trp10;
  set msv_trp9;
  pct_noctch=(noctch/n)*100;
  pct_nstol=(nstol/n)*100;
  pct_spp=(spp/n)*100;
  pct_ctchn=(ctchn/n)*100;
  pct_ctchw=(ctchw/n)*100;
  pct_nmsold=(nmsold/n)*100;
  pct_wtsold=(wtsold/n)*100;
  pct_val=(val/n)*100;
  pct_lost=(lost/n)*100;
  pct_rels=(rels/n)*100;
  pct_dam=(dam/n)*100;
  pct_avprc=(avprc/n)*100;
run;
proc print noobs;
  var year n pct_noctch pct_nstol pct_spp pct_ctchn pct_ctchw pct_nmsold
    pct_wtsold pct_val pct_lost pct_rels pct_dam pct_avprc;
quit;
/*depth-env var records*/
/*proc freq data=msv1;
```

```

tables dpbeg_cnt dpend_cnt wspd_cnt wdir_cnt wvht_cnt crsp_cnt
  crdr_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var dpbeg_cnt dpend_cnt wspd_cnt wdir_cnt wvht_cnt crsp_cnt
    crdr_cnt;
  output out=msv_trp11 (drop=_type_ _freq_) n=n sum=dpbeg dpend
    wspd wdir wvht crsp crdr;
quit;
data msv_trp12;
  set msv_trp11;
  pct_dpbeg=(dpbeg/n)*100;
  pct_dpend=(dpend/n)*100;
  pct_wspd=(wspd/n)*100;
  pct_wdir=(wdir/n)*100;
  pct_wvht=(wvht/n)*100;
  pct_crsp=(crsp/n)*100;
  pct_crdr=(crdr/n)*100;
run;
proc print noobs;
  var year n pct_dpbeg pct_dpend pct_wspd pct_wdir pct_wvht pct_crsp
    pct_crdr;
quit;
/*database-flag var records*/
/*proc freq data=msv1;
  tables frfile_cnt flgar_cnt flgby_cnt flgargr_cnt flgbypr_cnt flgspgr_cnt
    flgspar_cnt flgsn_cnt flgpc_cnt flglb_cnt flgclim_cnt;
quit;*/
proc means data=msv1 nway noprint;
  class year;
  var frfile_cnt flgar_cnt flgby_cnt flgargr_cnt flgbypr_cnt flgspgr_cnt
    flgspar_cnt flgsn_cnt flgpc_cnt flglb_cnt flgclim_cnt;
  output out=msv_trp11 (drop=_type_ _freq_) n=n sum=frfile flgar
    flgby flgargr flgbypr flgspgr flgspar flgsn flgpc flglb flgclim;
quit;
data msv_trp12;
  set msv_trp11;
  pct_frfile=(frfile/n)*100;
  pct_flgar=(flgar/n)*100;
  pct_flgby=(flgby/n)*100;
  pct_flgargr=(flgargr/n)*100;
  pct_flgbypr=(flgbypr/n)*100;
  pct_flgspgr=(flgspgr/n)*100;
  pct_flgspar=(flgspar/n)*100;
  pct_flgsn=(flgsn/n)*100;
  pct_flgpc=(flgpc/n)*100;
  pct_flgglb=(flglb/n)*100;
  pct_flgclim=(flgclim/n)*100;
run;
proc print noobs;
  var year n pct_frfile pct_flgar pct_flgby pct_flgargr pct_flgbypr
    pct_flgspgr
    pct_flgspar pct_flgsn pct_flgpc pct_flgglb pct_flgclim;
quit;

```

6.2.4 SAS program 'bf1_0205_desc.sas'

```
/*program bf1_0205_desc.sas*/
/*processes Hawaii fisherman reporting system database*/
/*MHI, 2002-2005*/
/*descriptive analysis of bottomfish and deep7 trip effort data*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.botfish_db1_1948_2005;
quit;*/
data bf1b;
  set frs.botfish_db1_1948_2005;
  if 2002<=year<=2005;
run;
/*basic variable cleanup, designate missing values*/
data bf2b;
  set bf1b;
  if license=0 then license=-9;
  else license=license;
  if trip_end=. then trip_end=-9;
  else trip_end=trip_end;
  if bouy=' ' then bouy='-9';
  else bouy=bouy;
  if subarea=' ' then subarea='Z';
  else subarea=subarea;
  if bank=' ' then bank='-9';
  else bank=bank;
  if bank_quad=' ' then bank_quad='-9';
  else bank_quad=bank_quad;
  if wind_dir=' ' then wind_dir='-9';
  else wind_dir=wind_dir;
  if cur_speed=' ' then cur_speed='Z';
  else cur_speed=cur_speed;
  if cur_dir=' ' then cur_dir='-9';
  else cur_dir=cur_dir;
  if formtype=' ' then formtype='-9';
  else formtype=formtype;
  if timelink=. then timelink=-9;
  else timelink=timelink;
  if lname=' ' then lname='-9';
  else lname=lname;
  if fname=' ' then fname='-9';
  else fname=fname;
  if vessel=' ' then vessel='-9';
  else vessel=vessel;
  if uscg=' ' then uscg='-9';
  else uscg=uscg;
  if ha_no=' ' then ha_no='-9';
  else ha_no=ha_no;
  if trip_beg=. then trip_beg=-9;
  else trip_beg=trip_beg;
  if from_file=' ' then from_file='-9';
  else from_file=from_file;
  if avg_price=. then avg_price=-9;
  else avg_price=avg_price;
```



```
drop bf_trip crew f_gear buyer;
run;
data dt1b;
  set bf2b;
  if license=-9 then delete;
  /*if hours=0 then delete;*/
  /*if hours>24;*/
  /*if 1<=area<=574;*/
  if area=0 or 1<=area<=574 or area=999 or area=9999 or area=99999 then
    delete;
  /*if chartered='T' then delete;*/
  if port_land=0 then delete;
  /*if (port_land=port_land2) or (port_land2=0) then n_ports=1;
  else n_ports=2;*/
  /*if no_gears=0 then delete;*/
  /*if sets=0 then delete;*/
  /*if hooks=0 then delete;*/
  /*if net_length=0 then delete;*/
  /*if net_height=0 then delete;*/
  /*if f_area='T';*/
  /*if f_buoy='T';*/
  /*if f_ar_gr='T';*/
  /*if f_spec_gr='T';*/
  /*if f_lbs='T';*/
  /*if f_ctchlim='T';*/
  /*if lbs=0 and nocatch='F';*/
  /*if nocatch='T';*/
  /*if release>0;*/
  /*if depth_beg>0;*/
run;
/*proc sort;
  by license year month day port_land;
quit;*/
/*proc freq;
  tables year*area;
quit;*/
/*proc univariate plot nobyplot;
  var depth_beg depth_end;
quit;*/
/*proc print;
  var license year month day trip_beg trip_end port_land
    area subarea gear no_gears hours species lbs caught
    nocatch lost release;
quit;*/
/*num of different landing ports per trip*/
proc means data=dt1b nway noprint;
  class license year month day port_land;
  var lbs;
  output out=ptck1 (drop=_type_ _freq_) n=nrecs;
quit;
proc means nway noprint;
  class license year month day;
  var nrecs;
  output out=ptck2 (drop=_type_ _freq_) n=n_port;
quit;
proc freq;
  tables n_port;
```

```
quit;
/*data ptck3;
  set ptck2;
  if n_port>1;
run;
proc sort;
  by license year month day;
quit;
data ptck4;
  merge dt1 ptck3;
  by license year month day;
  if n_port=. then delete;
run;
proc print;
  var license year month day trip_beg trip_end port_land
  area subarea gear no_gears hours species lbs caught;
quit;*/
/*num of different fishing areas per trip*/
proc means data=dt1b nway noprint;
  class license year month day port_land area subarea;
  var lbs;
  output out=ar1 (drop=_type_ _freq_) n=nrecs;
quit;
proc means nway noprint;
  class license year month day port_land;
  var nrecs;
  output out=ar2 (drop=_type_ _freq_) n=n_area;
quit;
proc means nway noprint;
  class n_area;
  var day;
  output out=ar3 (drop=_type_ _freq_) n=n_trips;
quit;
proc print noobs;
quit;
/*data arck1;
  set ar2;
  if n_area>3;
run;
proc sort;
  by license year month day;
quit;
data arck2;
  merge dt1 arck1;
  by license year month day;
  if n_area=. then delete;
run;
proc sort;
  by n_area;
quit;
proc print;
  var license year month day trip_beg trip_end port_land
  area n_area subarea gear no_gears hours species lbs caught;
quit;*/
/*num of different gears per trip*/
proc means data=dt1 nway noprint;
  class license year month day port_land gear;
```

```
var lbs;
output out=gr1 (drop=_type_ _freq_) n=nrecs;
quit;
proc means nway noprint;
class license year month day port_land;
var nrecs;
output out=gr2 (drop=_type_ _freq_) n=n_gear;
quit;
proc means nway noprint;
class n_gear;
var day;
output out=gr3 (drop=_type_ _freq_) n=n_trips;
quit;
proc print noobs;
quit;
/*check gear-area comp. of hours*/
/*proc means data=dt1 nway noprint;
class license year month day port_land gear no_gears area subarea hours;
var lbs;
output out=hr1 (drop=_type_ _freq_) n=nrecs;
quit;
proc means nway noprint;
class license year month day port_land gear no_gears area subarea;
var nrecs;
output out=hr2 (drop=_type_ _freq_) n=n_hours;
quit;
proc means nway noprint;
class n_hours;
var day;
output out=hr3 (drop=_type_ _freq_) n=n_trips;
quit;
proc print noobs;
quit;
data hrck1;
set hr2;
if n_hours>1;
keep license year month day port_land n_hours;
run;
proc sort;
by license year month day port_land;
quit;
data hrck2;
merge dt1 hrck1;
by license year month day port_land;
if n_hours=. then delete;
run;
proc print;
var license year month day trip_beg trip_end port_land
area subarea gear no_gears sets hooks net_length hours n_hours species
lbs caught;
quit;*/
/*release and lost by species*/
proc means data=dt1 nway noprint;
class species;
var release lost;
output out=hr1 (drop=_type_ _freq_) sum=num_release num_lost;
quit;
```

```
proc print noobs;
quit;
/*trip records & ports per month*/
proc means data=dt1b nway noprint;
  class license year month day port_land;
  var lbs;
  output out=trip1 (drop=_type_ _freq_) n=nrecs;
quit;
proc means nway noprint;
  class license year month;
  var nrecs;
  output out=trip2 (drop=_type_ _freq_) n=n_trips;
quit;
/*proc print;
quit;*/
proc means data=trip1 nway noprint;
  class license year month port_land;
  var nrecs;
  output out=port1 (drop=_type_ _freq_) n=n_recalls;
quit;
proc means nway noprint;
  class license year month;
  var n_recalls;
  output out=port2 (drop=_type_ _freq_) n=n_ports;
quit;
/*proc print;
quit;*/
data trip3;
  merge trip2 port2;
  by license year month;
run;
/*proc print;
quit;*/
/*proc freq;
  tables n_trips n_ports;
quit;*/
proc means nway noprint;
  class year;
  var n_trips n_ports;
  output out=trip4 (drop=_type_ _freq_) n=n mean=av_trips av_ports
    min=min_trips min_ports max=max_trips max_ports median=med_trips
    med_ports;
quit;
proc print noobs;
  var year n min_trips av_trips med_trips max_trips min_ports av_ports
    med_ports max_ports;
quit;
```

6.2.5 SAS program 'bf1_problem_catch.sas'

```
/*program bf1_problem_catch.sas*/
/*processes Hawaii fisherman reporting system database*/
/*1948-2005*/
/*processing analysis of bottomfish species database variables*/
/*analyzes lbs=0 records*/
options nonumber nodate;
libname frs 'c:\sgs\HI_main8_FD_DB\HDAR-FishermanReportingSystem\fisher_db';

/*proc contents data=frs.botfish_db1_1948_2005;
quit;*/
/*data bf1;
  set frs.botfish_db1_1948_2005;
  if lbs=0;
  keep year lbs lbs_sold caught num_sold nocatch num_stolen lost
  release damage;
run;*/
/*proc freq data=bf1;
  tables nocatch;
quit;*/
/*missing lbs records by year*/
data cm1;
  set bf1;
  if lbs_sold>0 then sold=1;
  else sold=0;
  if caught>0 then num_catch=1;
  else num_catch=0;
  if num_sold>0 then n_sold=1;
  else n_sold=0;
  if nocatch='T' then none=1;
  else none=0;
  if num_stolen>0 then stolen=1;
  else stolen=0;
  if lost>0 then n_lost=1;
  else n_lost=0;
  if release>0 then n_rel=1;
  else n_rel=0;
  if damage>0 then n_dam=1;
  else n_dam=0;
run;
proc means nway noprint;
  class year;
  var sold num_catch n_sold none stolen n_lost n_rel n_dam;
  output out=cm2 (drop=_type_ _freq_) n=totrecs sum=lbs_sold caught
  num_sold nocatch num_stolen lost release damage;
quit;
proc print noobs;
quit;
```